



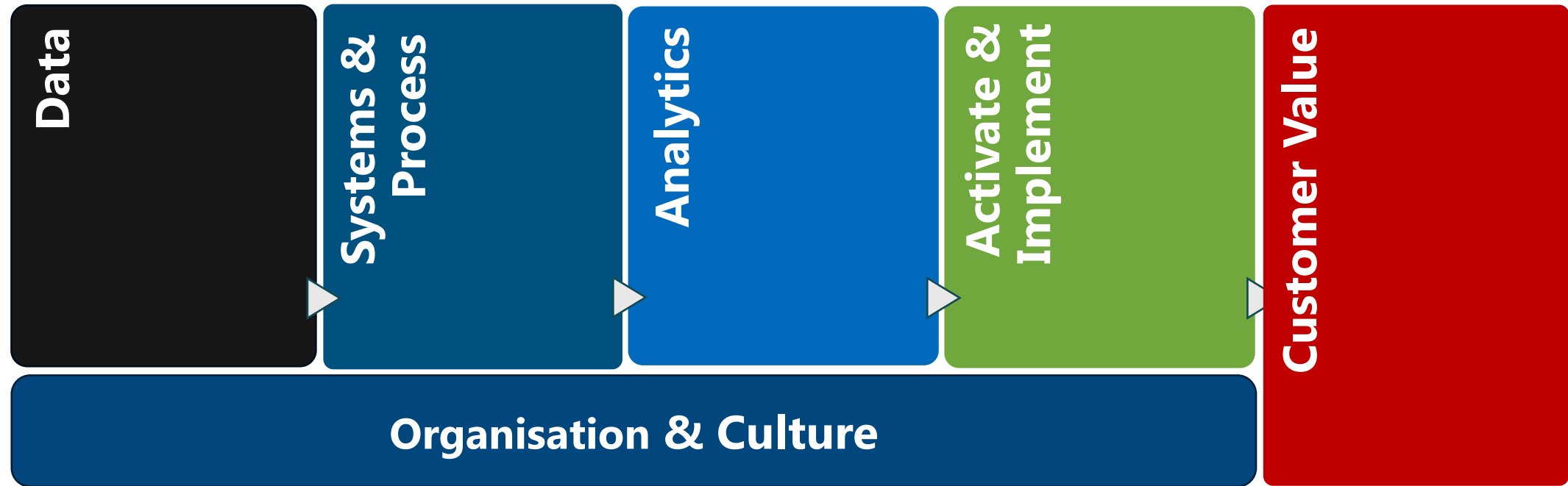
POSTMASTER DIGITALE INNOVATIE, AI & CHANGE

Analytics: AI, ML en DL



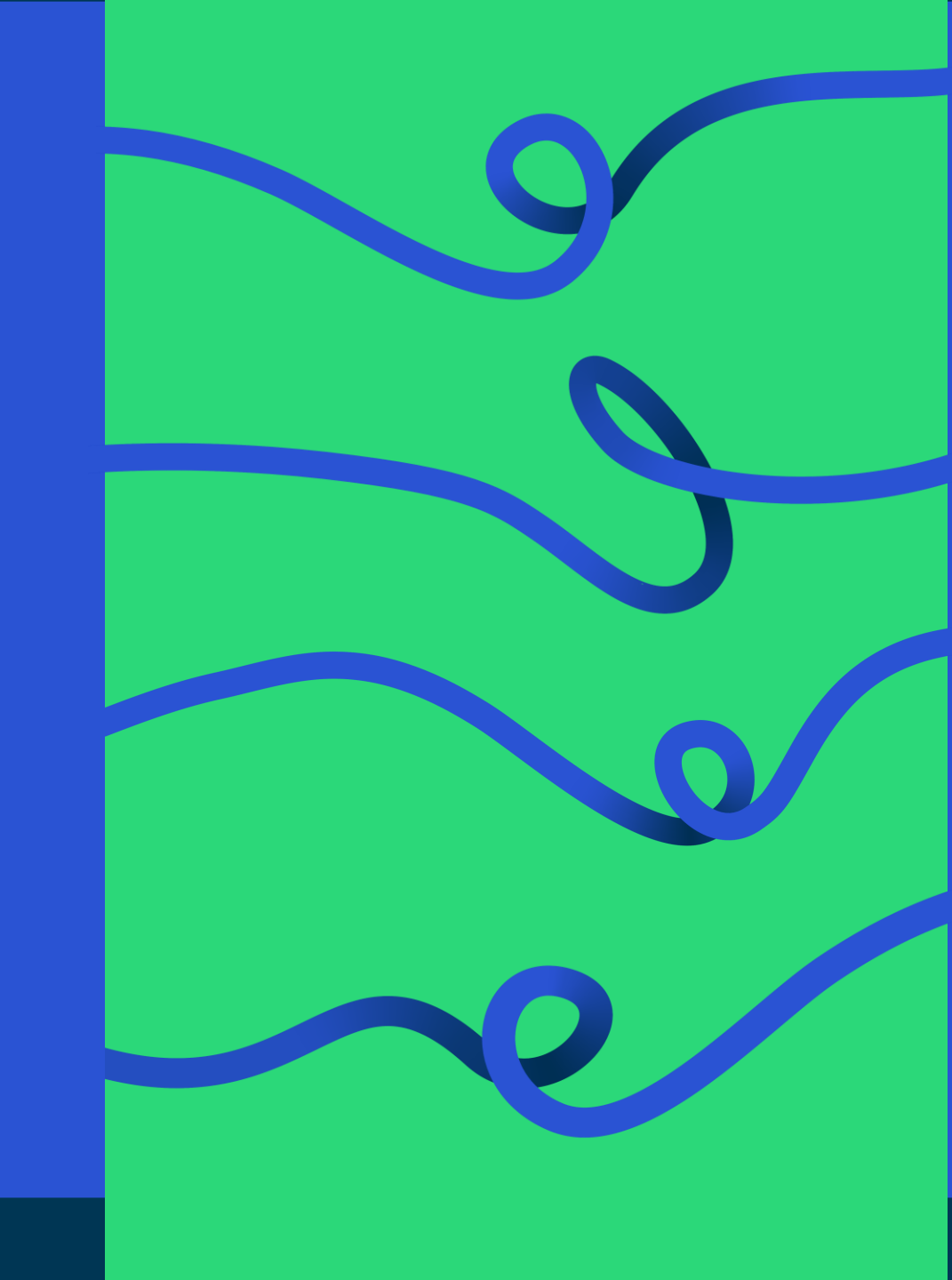
Waarde creatie met 'Data Driven' strategie

5 Building Blocks



The background features a dark space filled with vibrant, multi-colored particle trails in shades of blue, green, orange, and red, resembling a complex data network or a nebula. A large, solid blue circle is positioned in the center-right, containing the word 'Analytics' in white. The right side of the image transitions into a light gray area, and a yellow curved shape is visible at the bottom right corner.

Analytics



AGENDA

- A Intro & descriptive analytics (10 min)
- B Predictive analytics
 - a. supervised (45 min)
 - b. unsupervised (55 min)
 - c. reinforced (15 min)
- C Prescriptive analytics (35 min)

**Wat moet
een restaurant manager
van koken weten?**

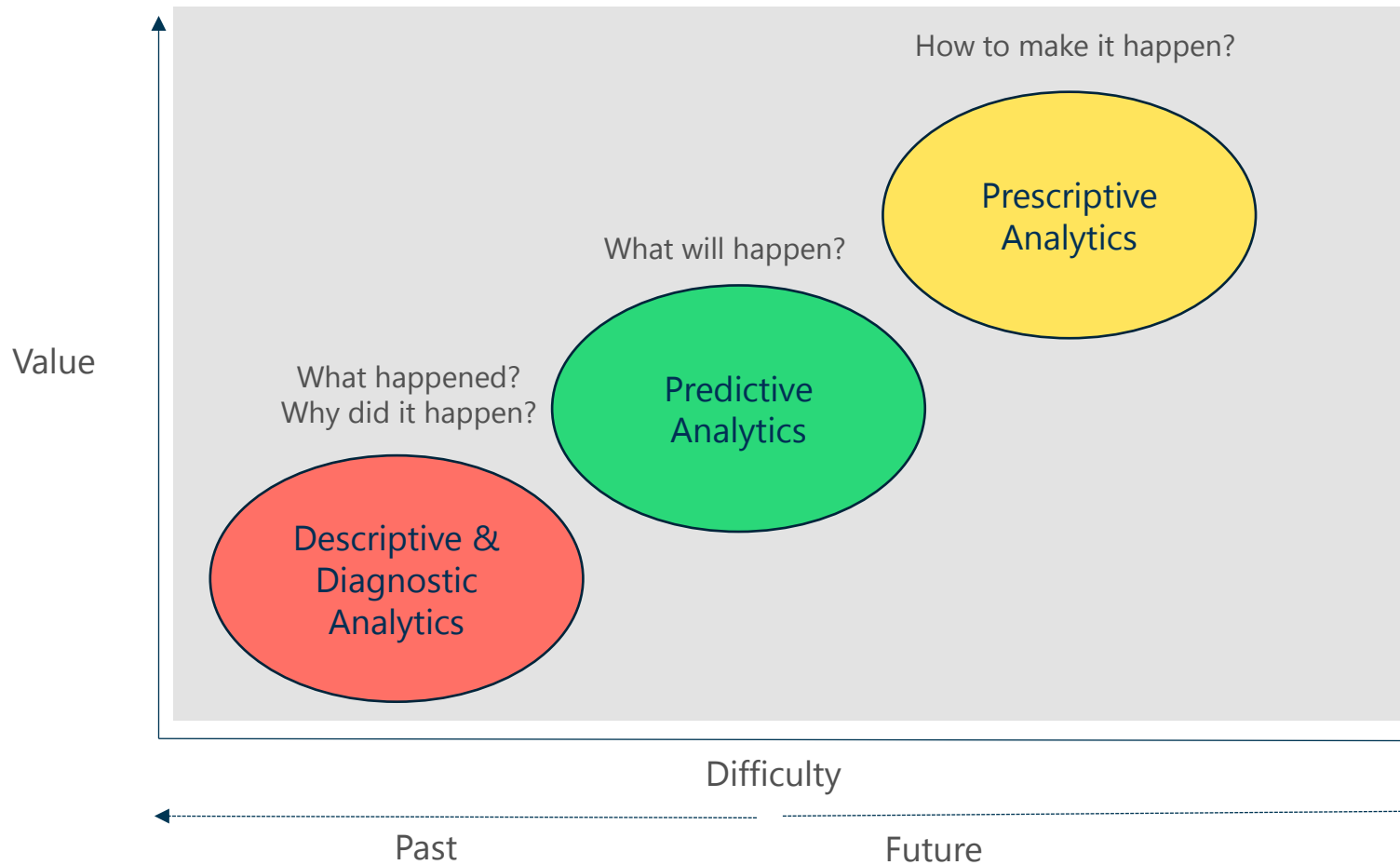


Een blik
onder
de motorkap



Development Data Driven Models

Levels Data Analytics





Descriptive & Diagnostic Analytics

- Data summarization (frequencies, cross tabulations)
- Data quality assessment (mean, standard deviation, t-test)
- Finding correlations (bivariate, partial, distances)

Aim to explain:

Management reports,
dashboards, basket analysis

Dashboards

Contacten sales

[Afspraak inplannen](#)[Registreer \(showroom\)bezoek](#)[Registreer telefoon/e-mail contact](#)[Laat collega terugbellen](#) (483/169) Daniëlle Dijkman - Keller (0)Laat contacten zien van dealer: 

- ☐ Louwman Heemskerk
(85/25 | 0)
- ☐ Louwman Welzorg Auto op Maat
(398/144 | 5)

Laat contacten zien van gebruikers met functie(s): 

- ☐ Verkoop personenauto's nieuw (0)
- ☐ Verkoop personen auto's gebruikt (0)
- ☐ Verkoop bedrijfswagens nieuw (0)
- ☐ Verkoop bedrijfswagens gebruikt (0)
- ☐ Accountmanagers (0)

Laat contacten zien van: 

- ☒ **Niet toegewezen (5)**
- ☐ Antonio Dzido (1/1)
- ☐ Arthur Bot (4)
- ☐ Brain Schippers (0)
- ☐ Dennis Douma (32)
- ☐ Desirée Rooijackers (0)
- ☐ Gordon Kocken (33/2)
- ☐ Hans de Vos (49/24)
- ☐ Inname (17/2)
- ☐ Jan Visscher (3)
- ☐ Jeroen de heus (0)
- ☐ Jesse Habraken (3)

[Toon meer](#) 

Momenteel staan er geen openstaande contacten op jouw naam en maken we op basis van jouw profielinstellingen op dat je zelf actief geen Sales contacten opvolgt. Pas [hier](#) je profiel aan indien je wel actief Sales contacten wil opvolgen

Bron



Interesse in



Nieuw/Gebruikt

PW/BW



Zoek op naam, postcode, e-mail, telefoon of kenteken



Nieuwe contacten

Klantnaam	Datum aanvraag	Actie verkoper	Interesse in	Inruil	Bron	Behandelaar
  Louwman Welzorg Auto op Maat	19-04-2023	Volg contact op	-	-	Contactive...	Toewijzen
  Louwman Welzorg Auto op Maat	gisteren	Volg contact op	Renault Megane scenic 37-ZF-NZ	-	Contactive...	Toewijzen
  Louwman Welzorg Auto op Maat	gisteren	Volg contact op	 Citroën Berlingo TP-113-P	-	Offertegeb...	Toewijzen

Gemeente

Noordenveld

Postcode 4P

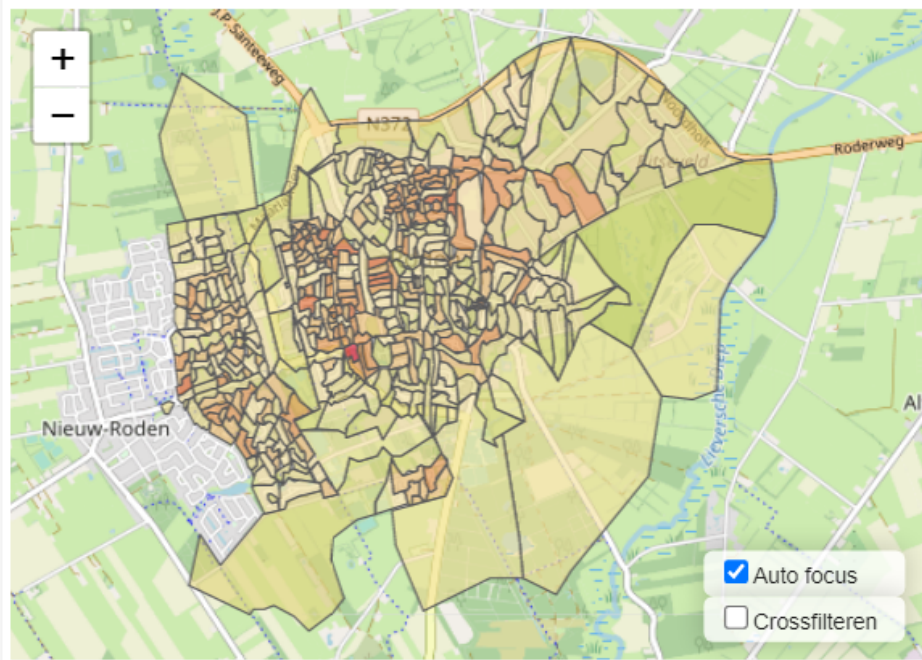
9301

Project

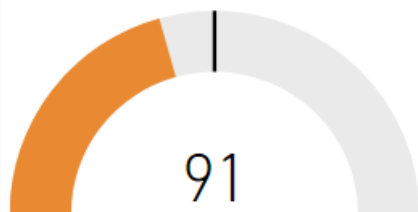
Alle



index wijkgerichte aanpak (Rood = 75 en Groen = 125)



Index wijkgerichte aanpak



Wijkgerichte aanpak leeswijzer

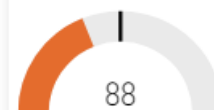
Deze Wijkgerichte Aanpak rapportage geeft een overzicht hoe een wijk ervoor staat op basis van data. Elke categorie krijgt een index score van 50 tot 150. Hierbij is de index score van het gemiddelde van de gemeente 100.

Veiligheid



Over de afgelopen 3 jaar is postcodegebied 9301 met een index van 87,8 van een bovengemiddeld slecht niveau. Dit is een gemiddelde van de index op overlast meldingen (78) en de index op misdrijf meldingen (97,6).

Er zijn in totaal 2488 meldingen gedaan waarvan 43% in de categorie overlast en 57% in de categorie misdrijf.

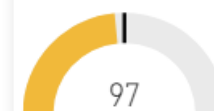


Woningkwaliteit



Met een index van 96,9 is de woningkwaliteit gemiddeld.

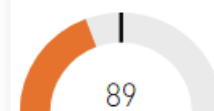
Ingezoomd op de vier indicatoren in het postcodegebied 9301 zijn energielabels hoger dan gemiddeld en zijn WOZ waardes lager dan gemiddeld.



Passende woonsituatie



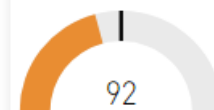
Met een index van 88,6 is het woonaanbod vaker niet passend dan gemiddeld. Van de vier voorgedefinieerde niet passende woonsituaties komt de woonsituatie 'huishoudens van 65 jaar of ouder in een grondgebonden huurwoning' het meeste voor in het postcodegebied 9301. De woonsituatie die het minst vaak voorkomt is 'scheefwoners die te goedkoop wonen'.



Wijksamenstelling



In het postcodegebied 9301 wonen bovengemiddeld veel 80 jaar of ouder huishoudens, -veel huishoudens met een Boven modaal inkomen en -veel huishoudens met een Middelbare opleiding.



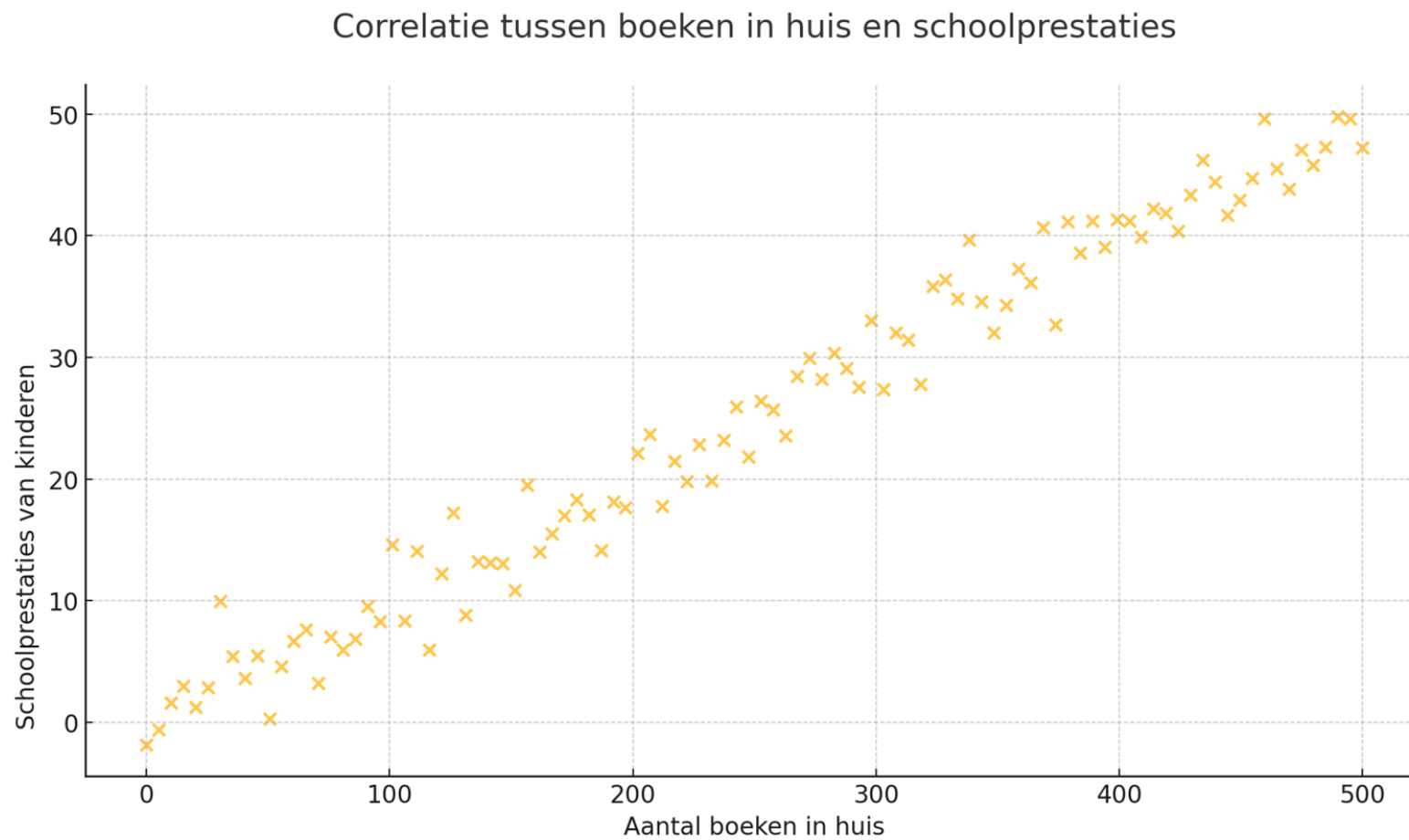
Descriptive analytics

Correlation coefficient

- Between -1 en 1
- If correlation score is $> -0,5$ and $< 0,5$:
None to a little relation between the samples
- If close to -1 or 1:
Strong relation between the samples



Big Data - Impact



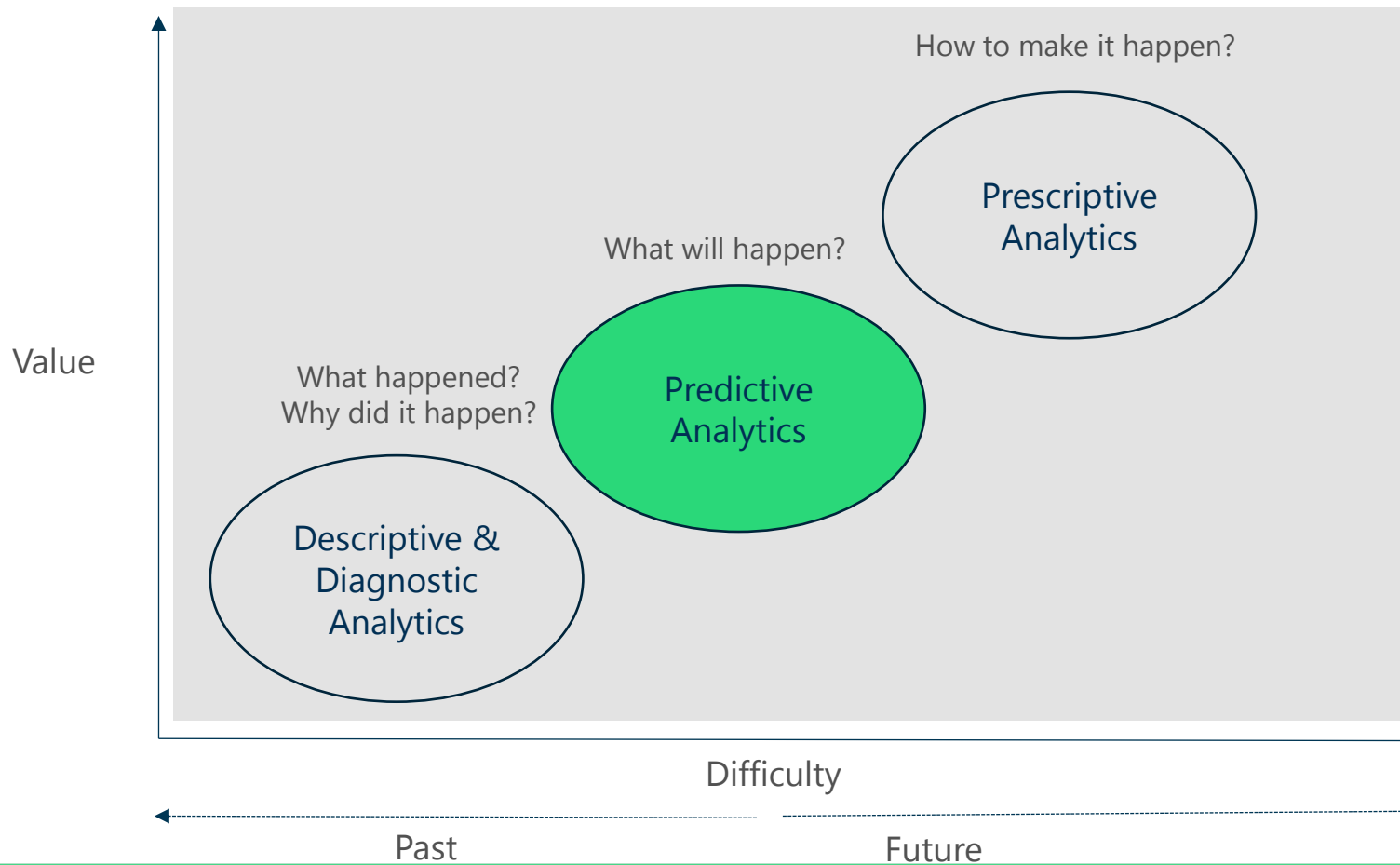
Correlation & Causality



Predictive Analytics

Development Data Driven Models

Levels Data Analytics

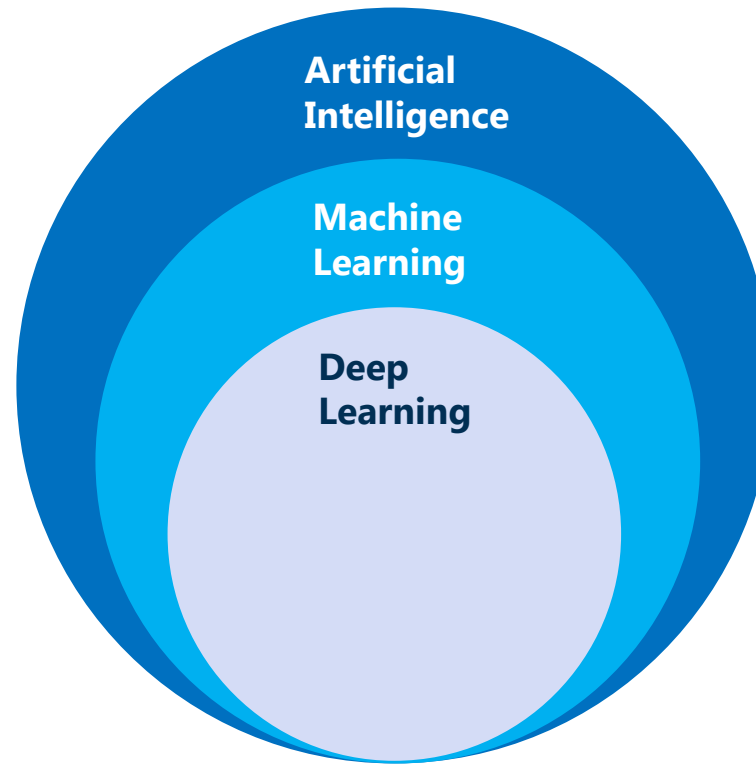
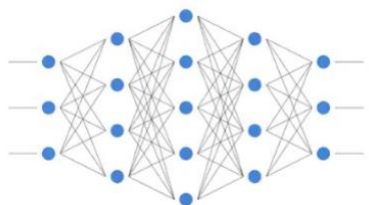


AI, ML & DL

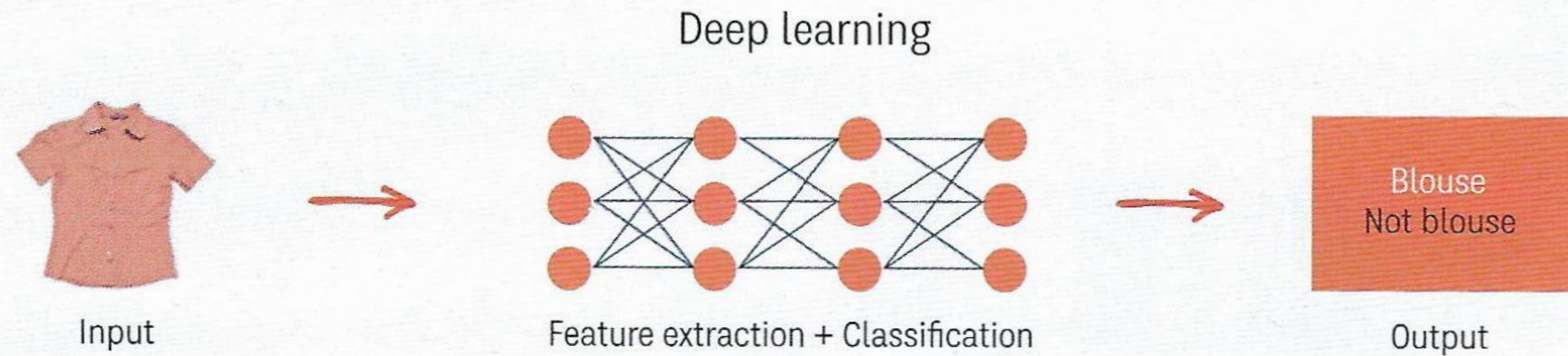
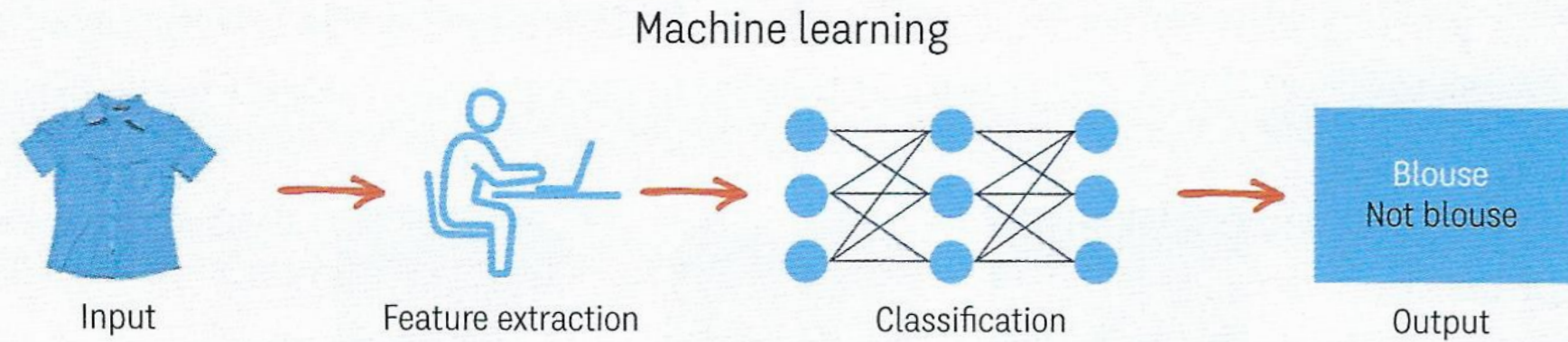
AI is an umbrella term for all automated repetitive decisions and actions made by machines, that mimic human decision-making and creation processes.

ML is the practice of using algorithms to parse data, learn from it through feedback, and then make a determination or prediction about something in the world.

DL is a subset of machine learning that uses neural networks



Machine Learning vs. Deep Learning

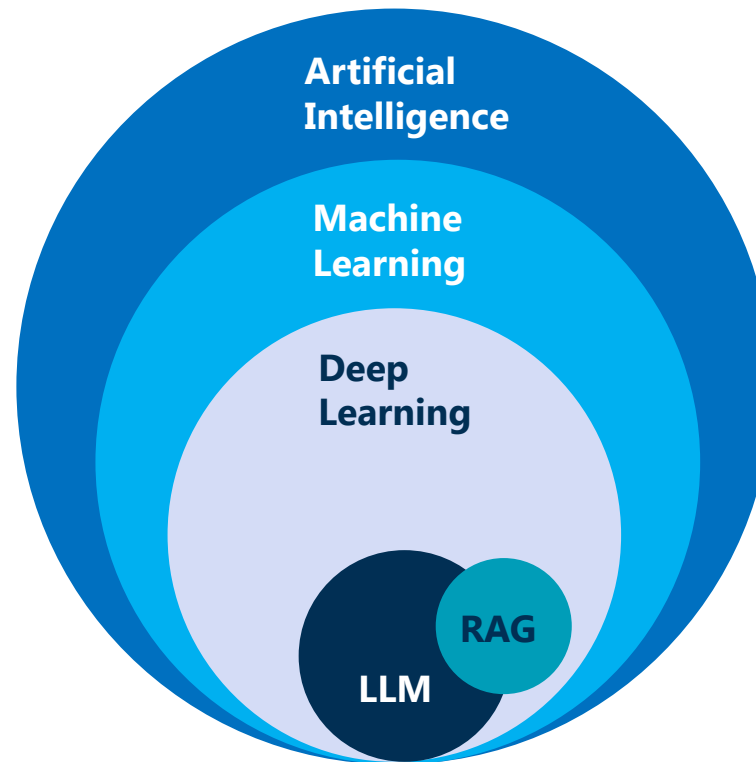
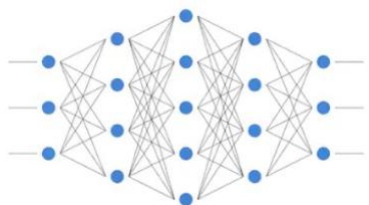


LLM & RAG

AI is an umbrella term for all automated repetitive decisions and actions made by machines, that mimic human decision-making and creation processes.

ML is the practice of using algorithms to parse data, learn from it through feedback, and then make a determination or prediction about something in the world.

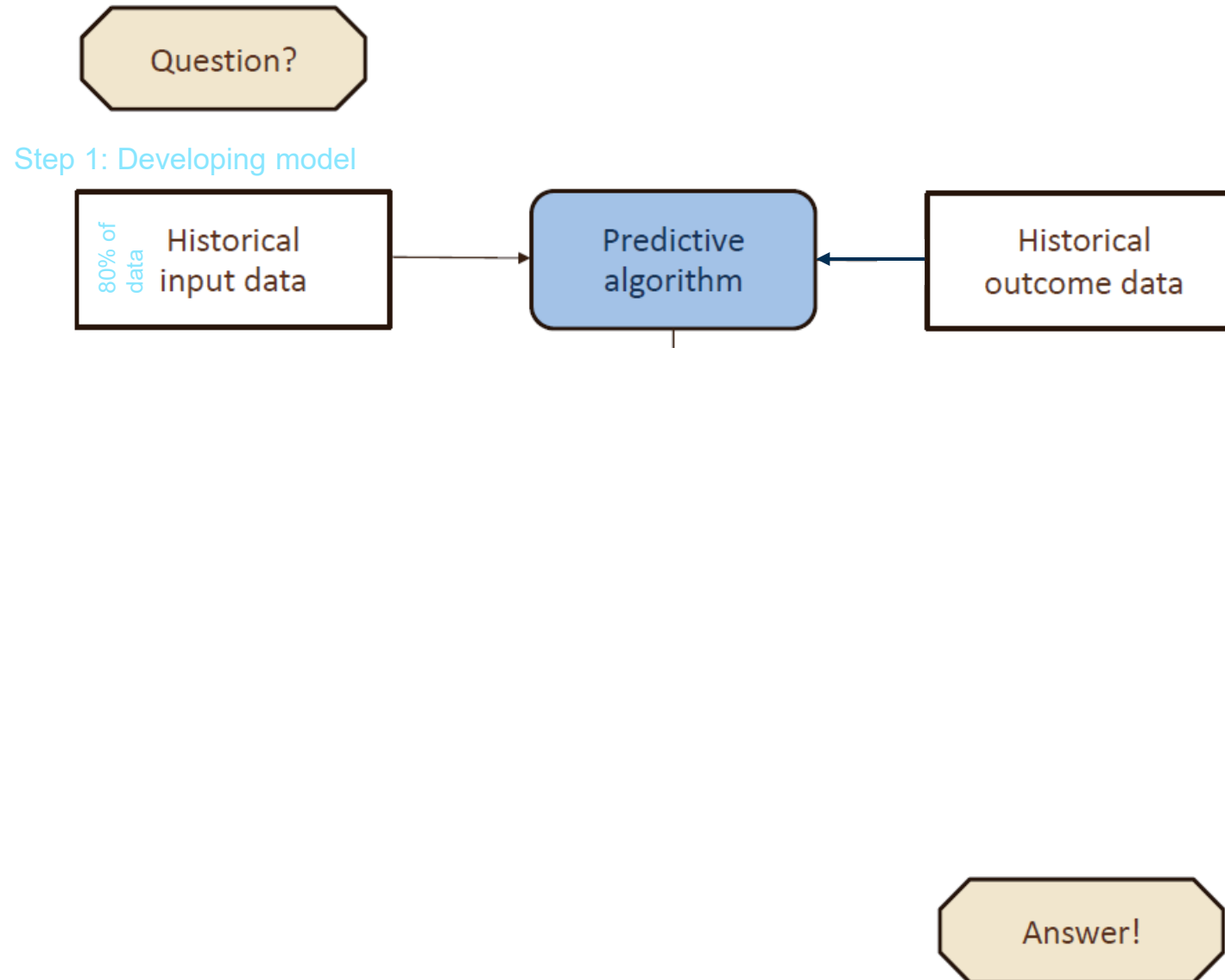
DL is a subset of machine learning that uses neural networks



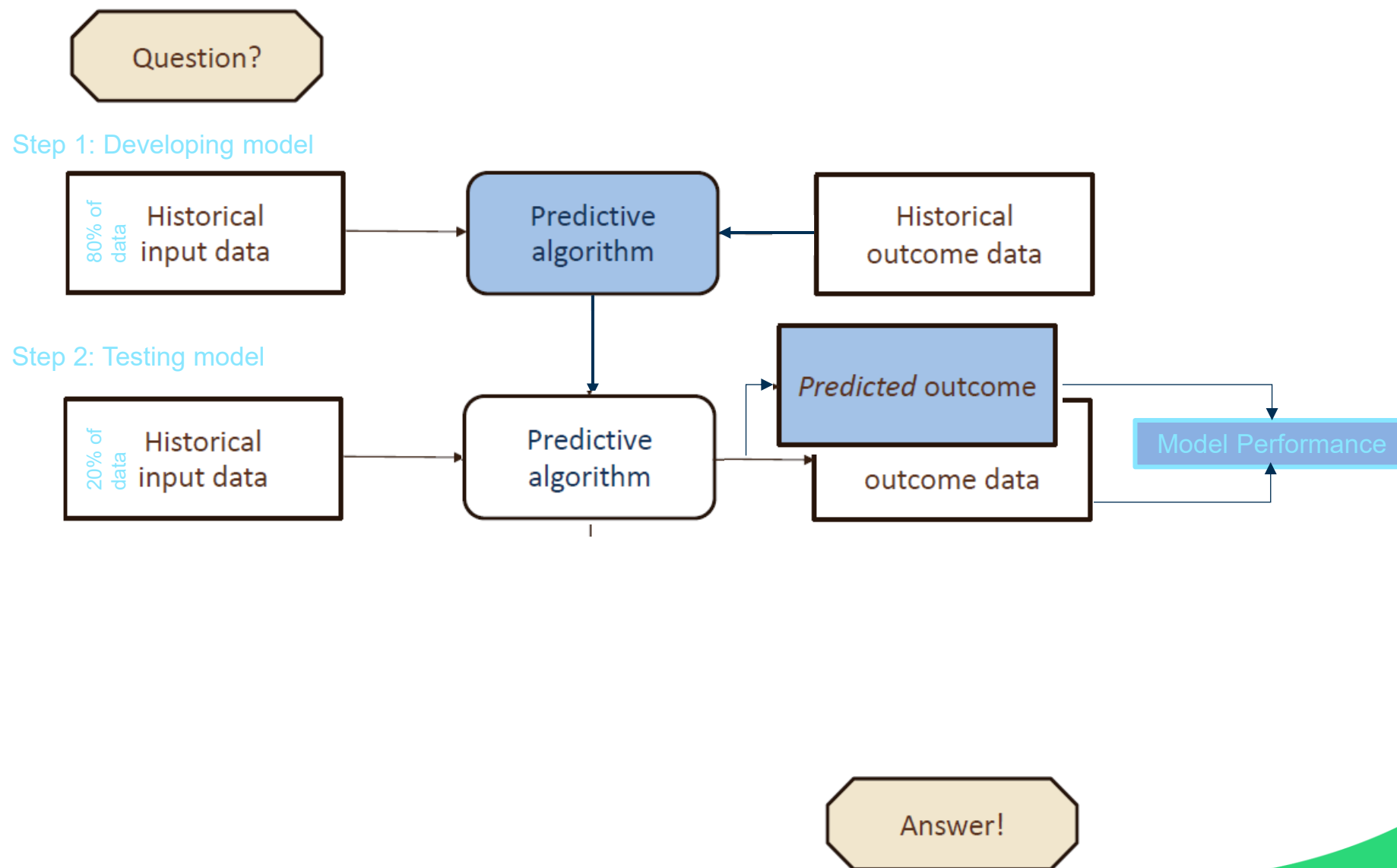
LLM means large language model, **trained** on vast amounts of data using machine learning techniques. The most well-known example is ChatGPT.

RAG (Retrieval-Augmented Generation) searches for relevant and trustworthy information in preselected external sources.

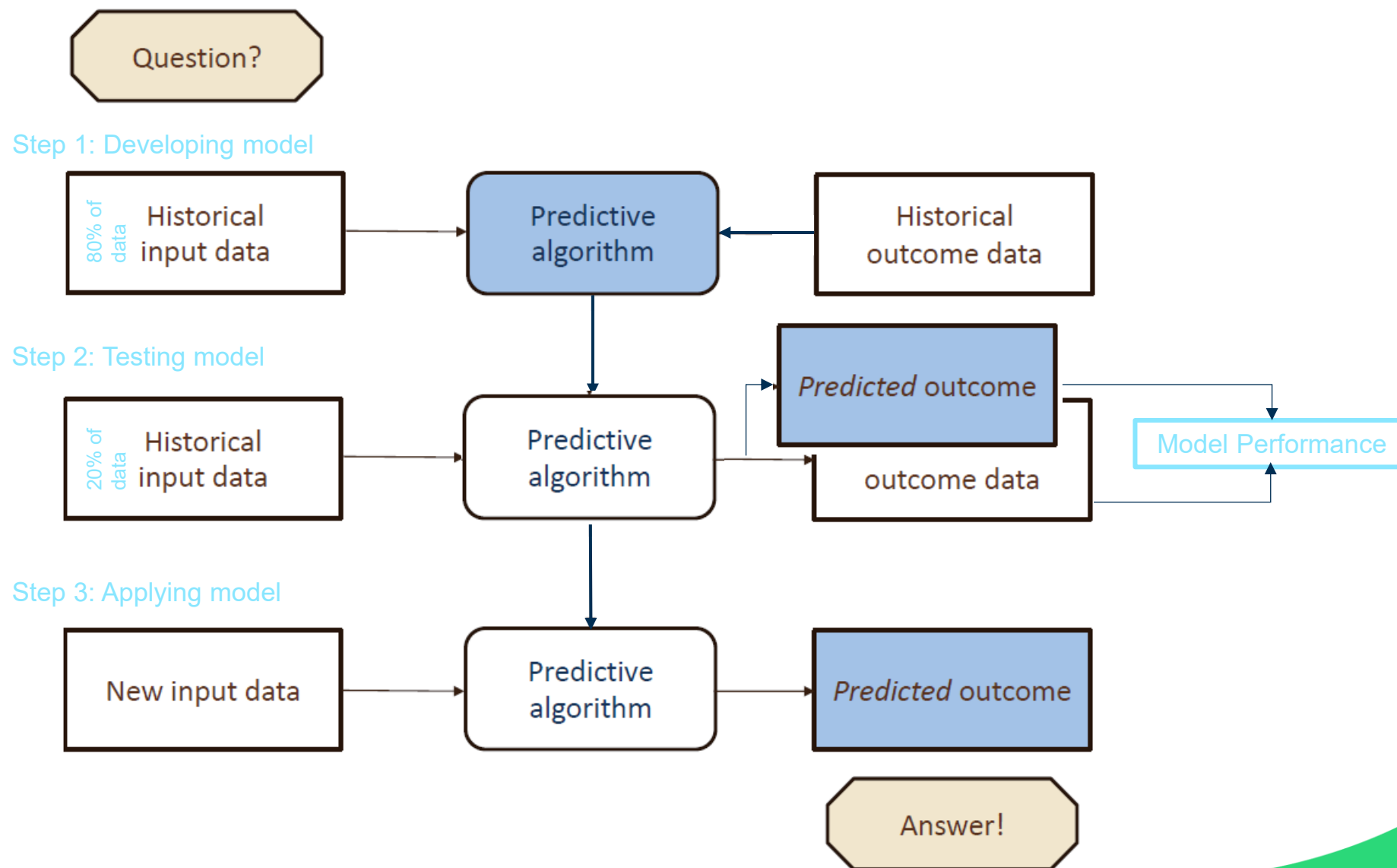
Data Analytics



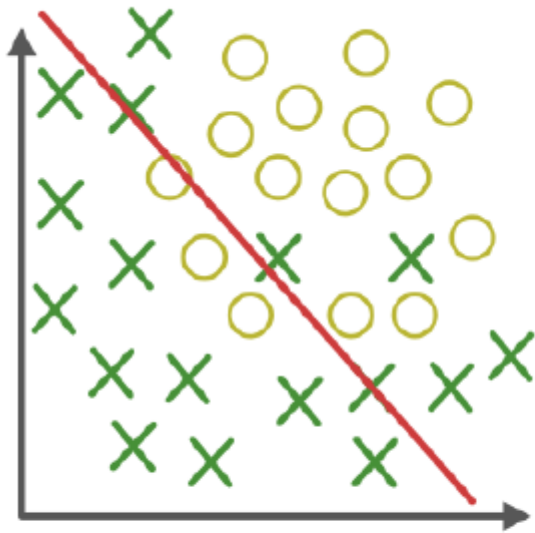
Data Analytics



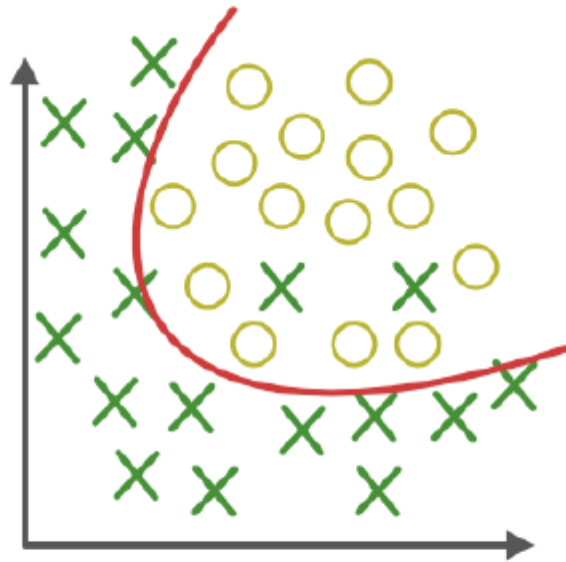
Data Analytics



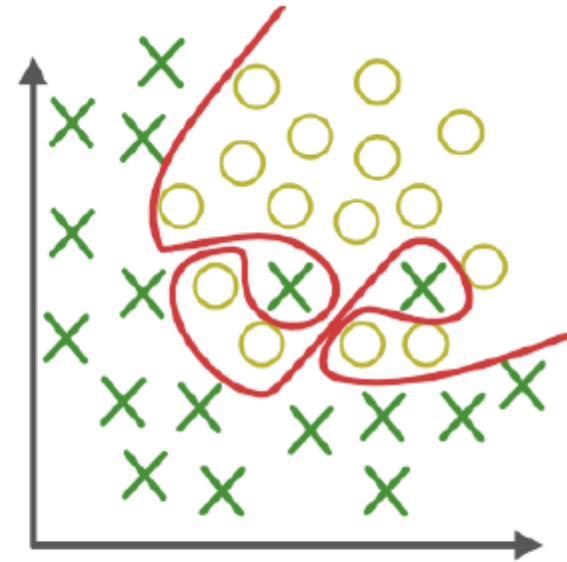
Impact fitting on model results; underfitting and overfitting



Underfitting



Appropriate fit



Overfitting



Predictive Analytics

**Predictive
Supervised**

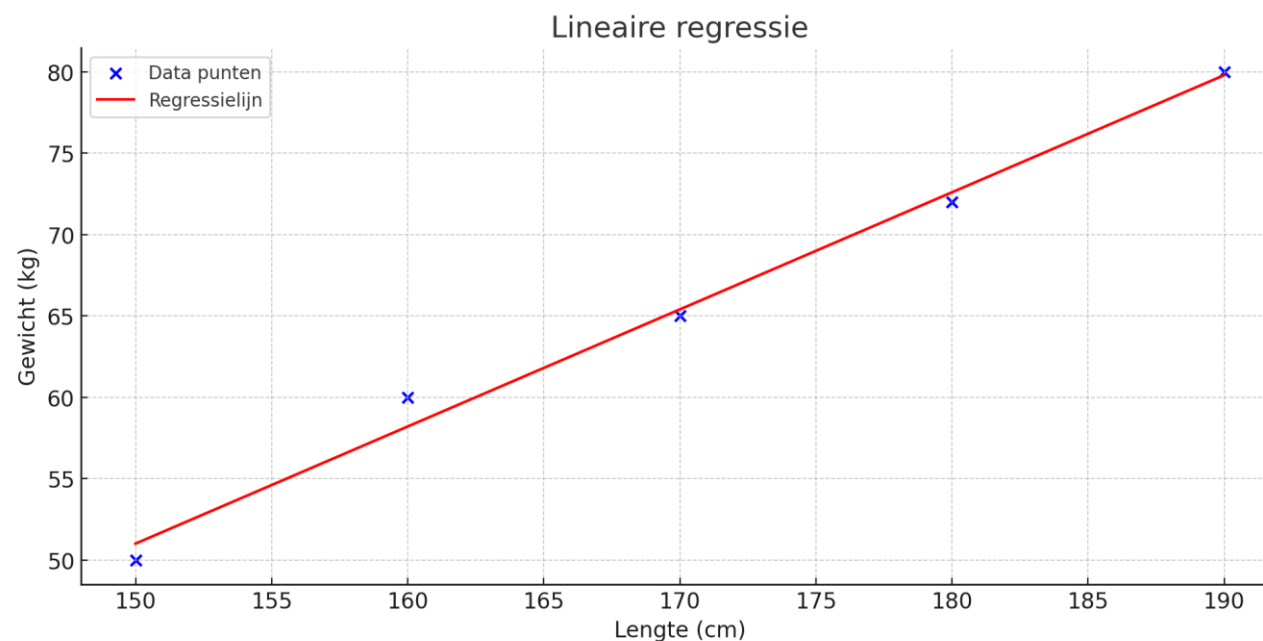
- A. Supervised learning**
Propensity, classification
- B. Unsupervised learning
Clustering, Principal Component Analysis
- C. Reinforcement learnings
Collaborative Filtering, recommendation
models based on nearest neighbourhood

Predictive Analytics - Supervised

A.1 Supervised learning; Propensity models (regression)

Predictive
Supervised
Propensity

Supervised learning models make true predictions about a customers (future) behaviour by learning from the past regarding **a clearly defined given output**.



Predictive Analytics - Supervised

Linear regression

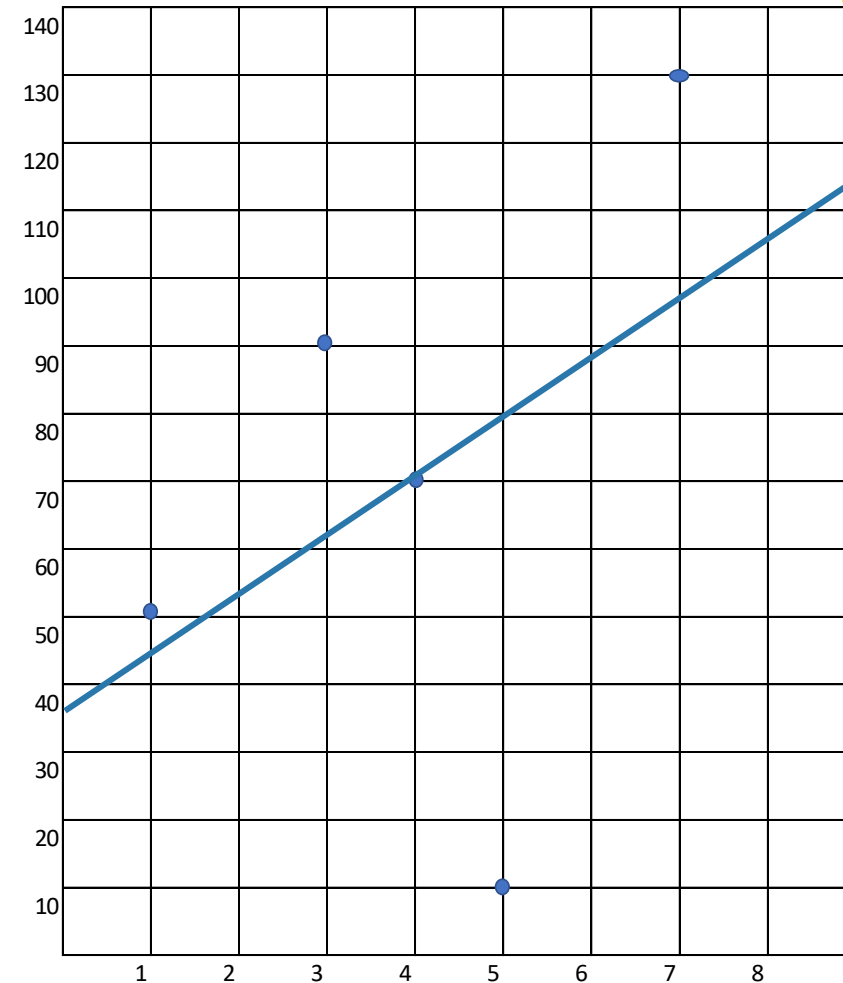
mean 1 (years)	4
standard deviation 1	2
mean 2: (salary x '000)	70
standard deviation 2	40
correlation coefficient r	0,4

$$Y = ax + b$$

$a = r \cdot (sd2 / sd1)$	$0,4 \cdot (40 / 2) = 8$
$b = g2 - a \cdot g1$	$70 - 8 \cdot 4 = 38$

$y = ax + b$	$y = 8x + 38$
--------------	---------------

Salary x '000



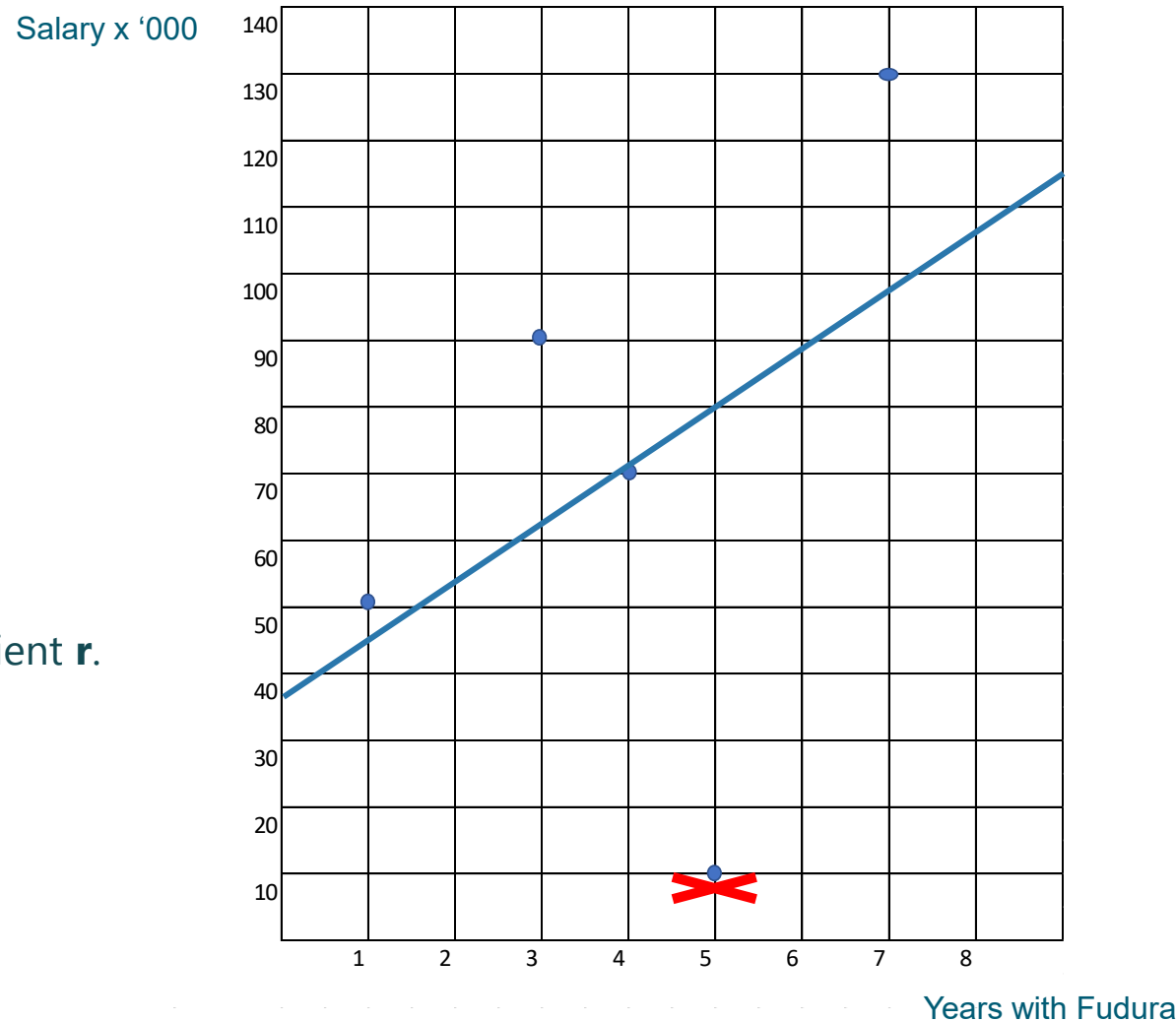
Years with Fudura

**Predictive
Supervised
Propensity**

Task - Regression

Impact outliers on model results

1. How will the new regression line show up?
Draw the new regression line in the map.
2. What will the impact on the regression?
Give an estimation of the correlation coefficient r .



Predictive Analytics - Supervised

Impact outliers on model results

Salary x '000

mean 1	4	3,75
standard deviation 1	2	2,165
mean 2	70	85
standard deviation 2	40	29,58
correlation coefficient r	0,4	0,92

$$Y = ax + b$$

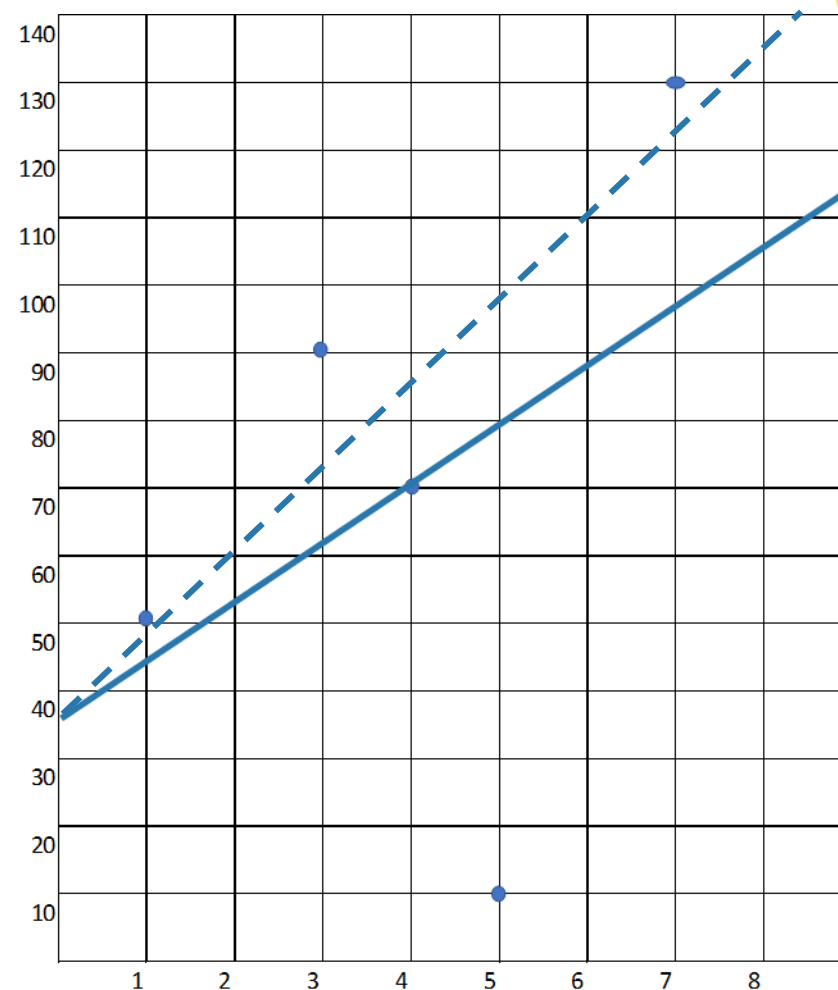
$a = r \cdot (sd2 / sd1)$	$0,4 \cdot (40 / 2) = 8$	$0,92 \cdot (29,58 / 2,165) = 12,5$
$b = g2 - a \cdot g1$	$70 - 8 \cdot 4 = 38$	$85 - (12,5 \cdot 3,75) = 38$

$y = ax + b$	$y = 8x + 38$
--------------	---------------

After elimination outlier 5,10 :

Regression equation: $y = 12,5x + 38$

Correlation coefficient 0,92



**Predictive
Supervised
Propensity**

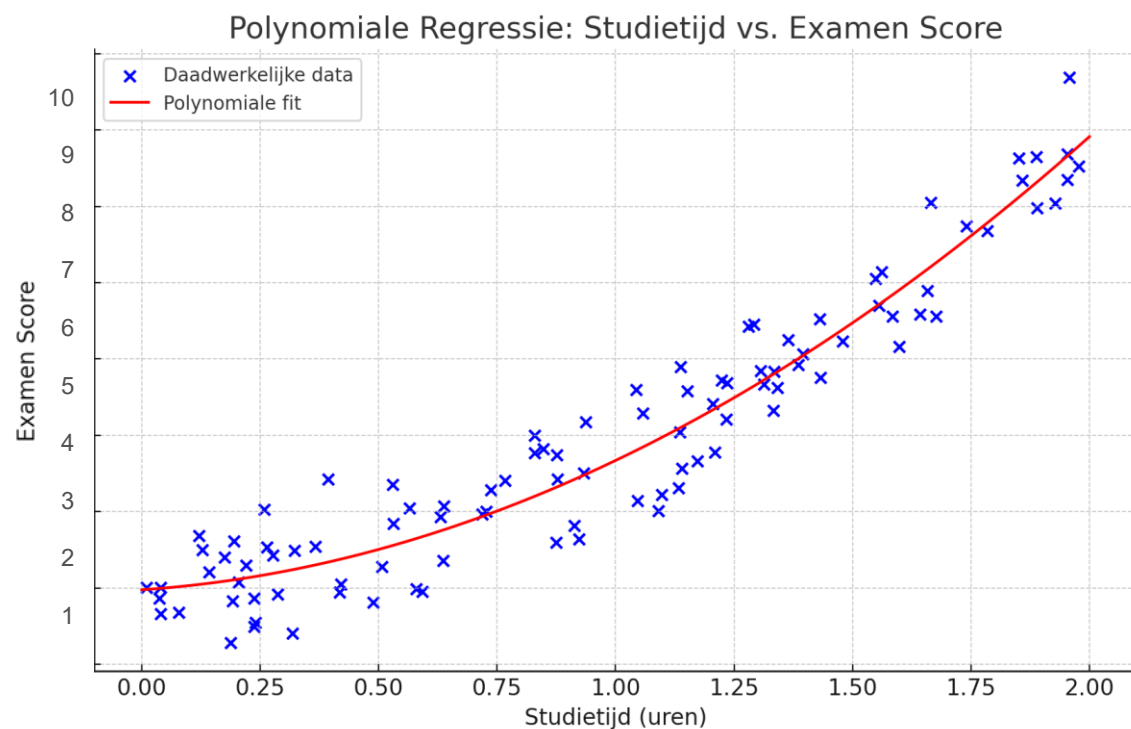
Years with Fudura

Predictive Analytics - Supervised

A.1 Supervised learning; Propensity models (regression)

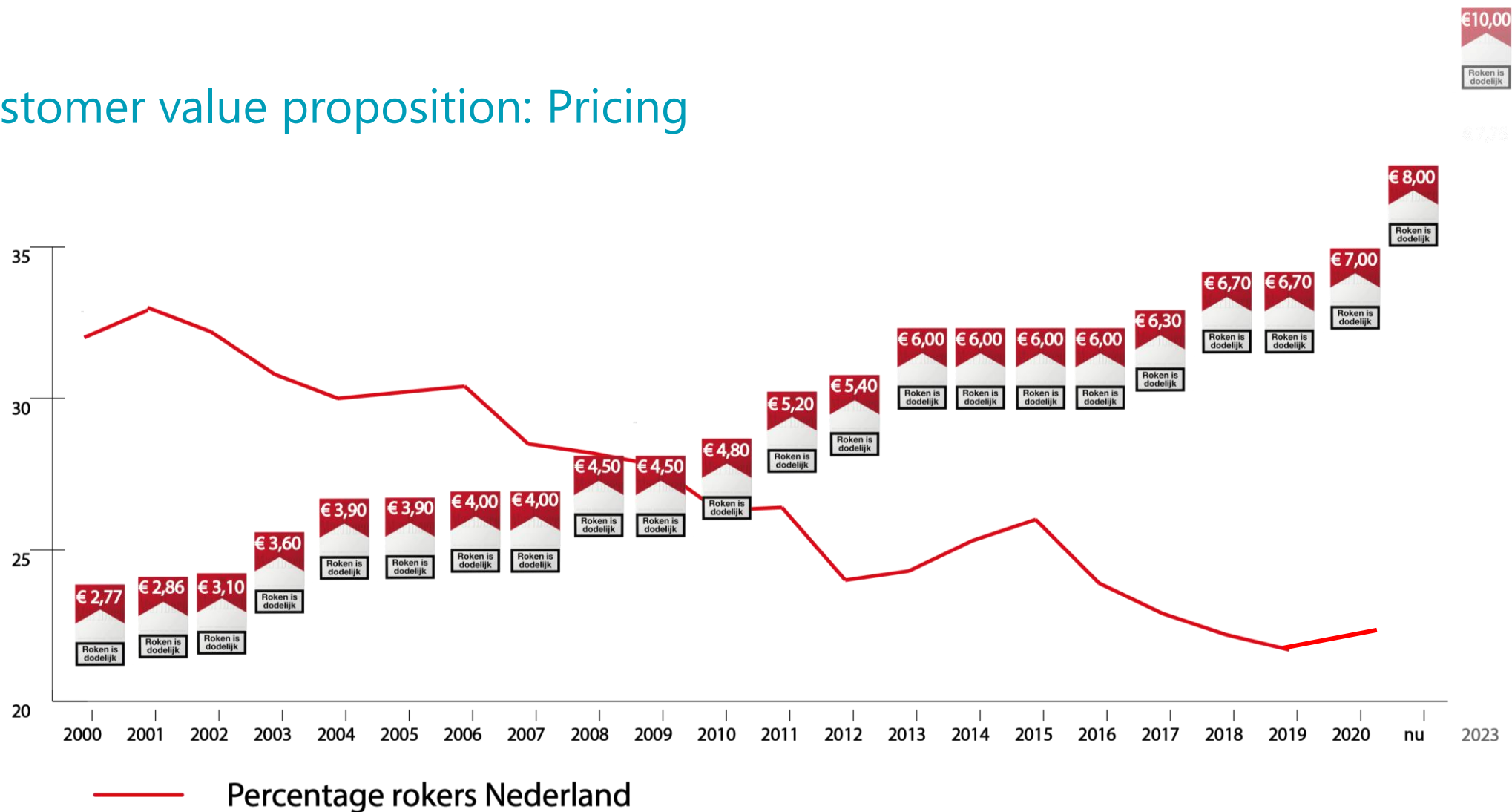
Predictive
Supervised
Propensity

Supervised learning models make true predictions about a customers (future) behaviour by learning from the past regarding **a clearly defined given output**.



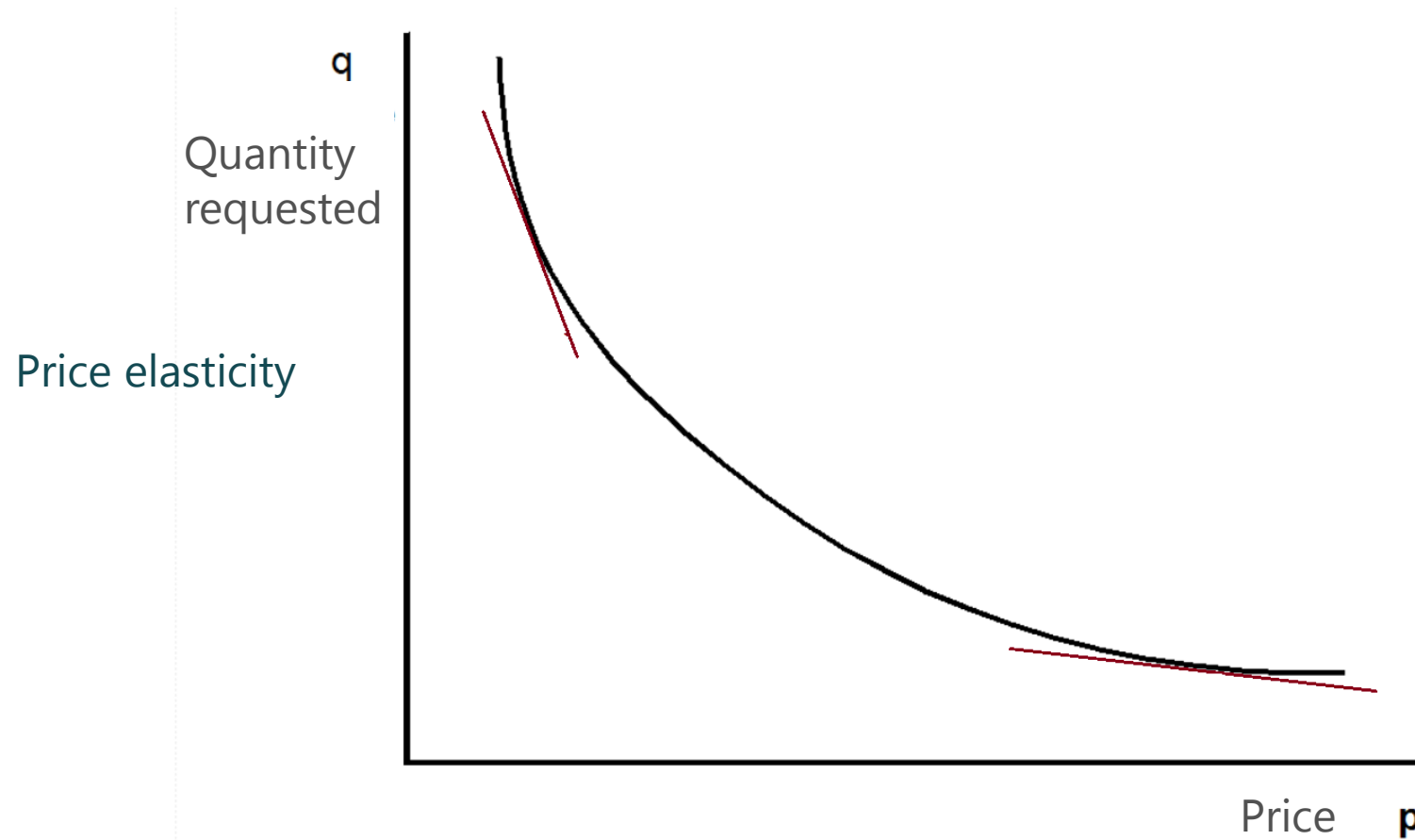
Customer value proposition

Customer value proposition: Pricing



Customer value proposition

Customer value proposition: Pricing

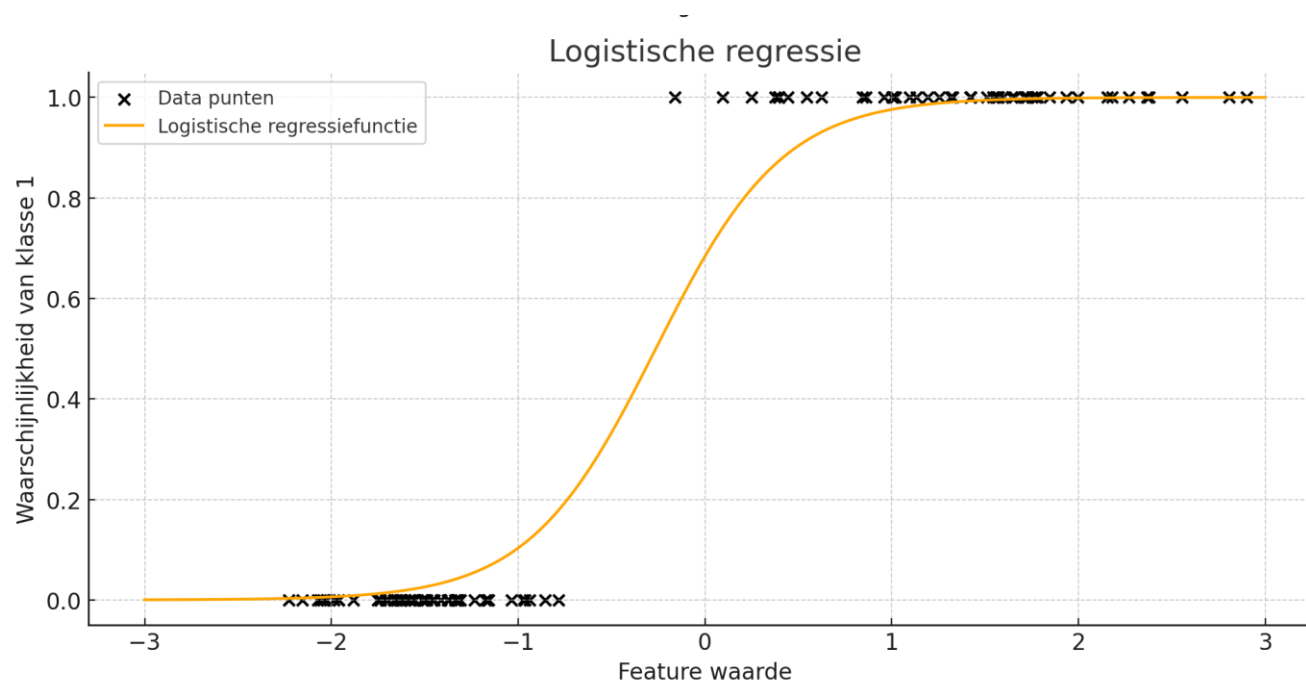


Predictive Analytics - Supervised

A.1 Supervised learning; Propensity models (regression)

Predictive
Supervised
Propensity

Supervised learning models make true predictions about a customers (future) behaviour by learning from the past regarding **a clearly defined given output**.



Predictive Analytics - Supervised

A.2 Supervised learning; Classification

Predictive
Supervised
Classification

1. Beslissingsbomen:

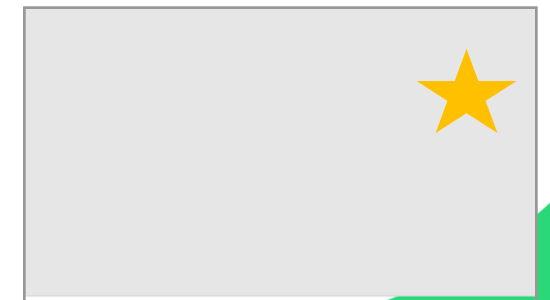
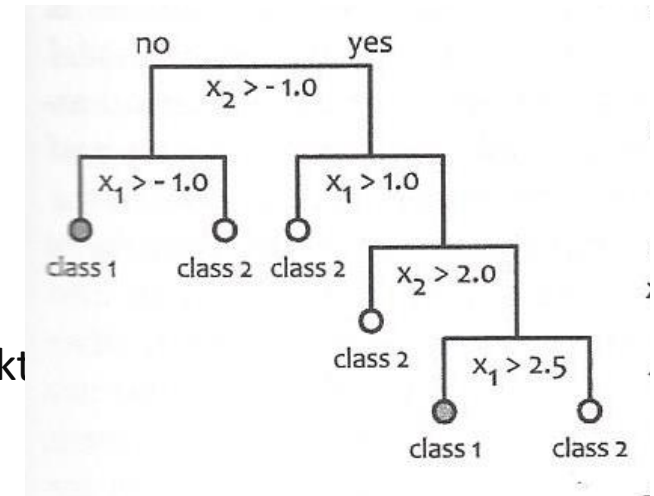
Boomstructuur waarbij elke interne knoop een 'test' op een attribuut vertegenwoordigt, elke tak een uitkomst van de test, en elke bladknoop een klasse label.

2. Random Forest:

Een ensemble-methode die vele beslissingsbomen gebruikt voorspellingen combineert.

3. Gradient Boosting Classifiers:

Deze werken door opeenvolgend zwakkere modellen toe te voegen, om fouten van de voorgaande modellen corrigeren.

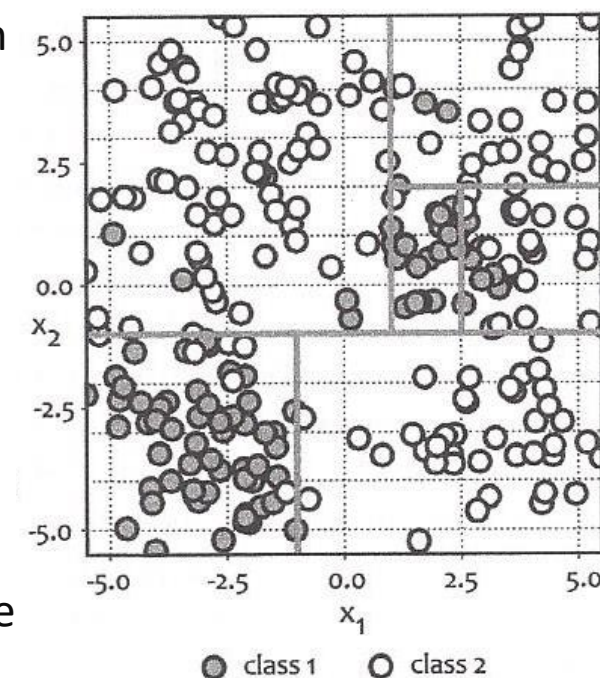


Predictive Analytics - Supervised

A.2 Supervised learning; Classification

Predictive
Supervised
Classification

4. **K-Nearest Neighbors (KNN):** Classificatiemethode die de klasse van een gegeven data punt voorspelt op basis van de klassen van de 'K' dichtstbijzijnde punten.
5. **Support Vector Machines (SVM):** Effectief in high-dimensional spaces, kunnen verschillende soorten kernel- functies gebruiken voor de beslissingsfunctie.
6. **Discriminant Analyse:** Een techniek die wordt gebruikt om de grens tussen de verschillende klassen te vinden door de lineaire combinaties van voorspellende variabelen te vinden die de klassen het beste scheiden.



Predictive Analytics - Supervised

A.2 Supervised learning; Classification



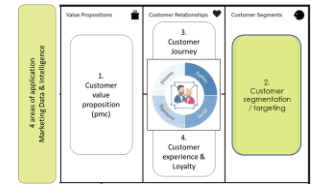
Predictive
Supervised
Classification

7. Naïve Bayes:

Een verzameling van eenvoudige classifiers gebaseerd op het toepassen van de Bayes' stelling met sterke onafhankelijkheidsaannames tussen de kenmerken.

Customer Segmentation / Targeting

Predictive Segmentation



Objective

More differentiation in communication
based on expected next visit

Komend half jaar

Tussen half jaar en jaar

Tussen jaar en 2 jaar

Later

Customer Segmentation / Targeting

Predictive Segmentation

Data available

Expected Visit frequency known
of 10% of guests

Data on 15 socio-demographic variables

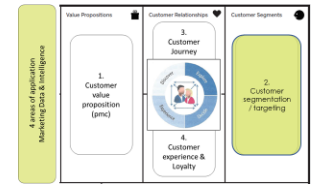
Variables after feature evaluation

Nominal variables like:

JONGVOLW_18_TM_34_KIDS_UNKNOWN
JONGVOLW_18_TM_34_NO_KIDS
VOLW_35_TM_54_KIDS_UNKNOWN
VOLW_35_TM_54_NO_KIDS
SENIOR_55_PLUS
KIDS_TM_11_JAAR
KIDS_12_TM_17_JAAR
GEZIN_MET_KIDS_0_OF_1_JAAR
GEZIN_MET_KIDS_2_TM_4_JAAR
GEZIN_KIDS_JONGER_DAN_12_JAAR
GEZIN_KIDS_12_JAAR_OF_OUDER

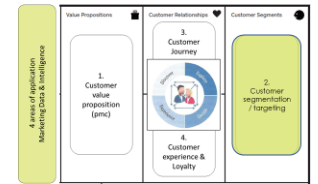
Variables with major missing data:

AGE
AGE_CATEGORY
TRAVEL_DISTANCE (in minutes)
TRAVEL_CATEGORY



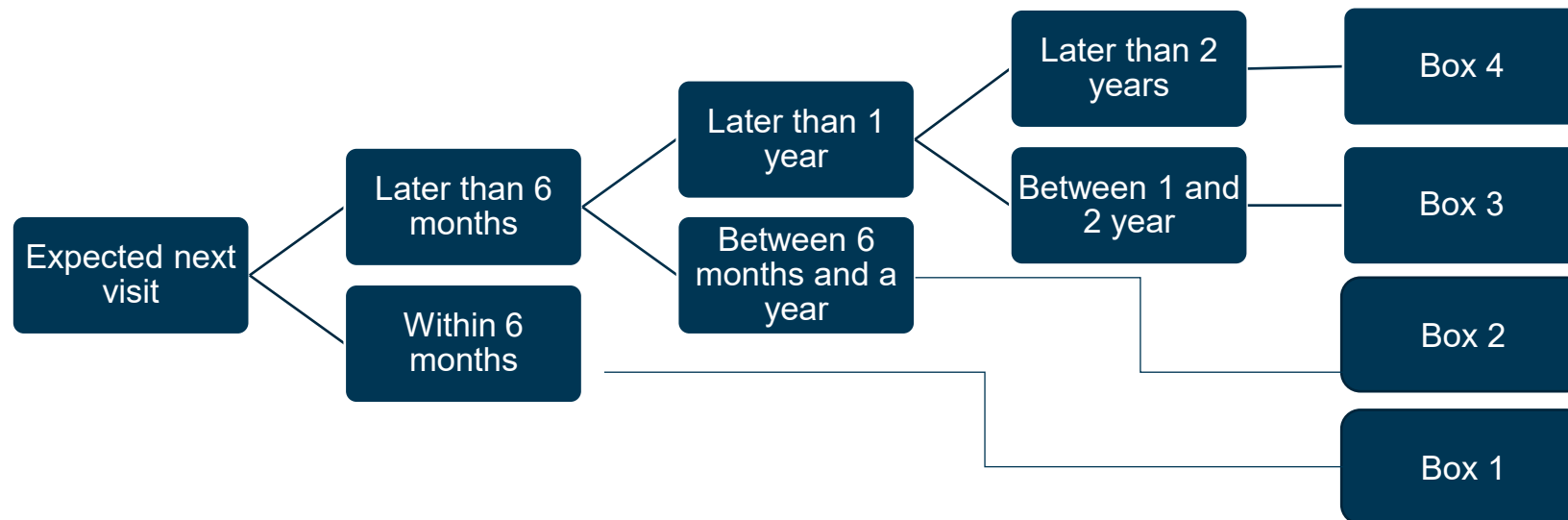
Customer Segmentation / Targeting

Predictive Segmentation



Model

Step-by-step approach

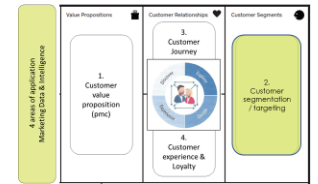
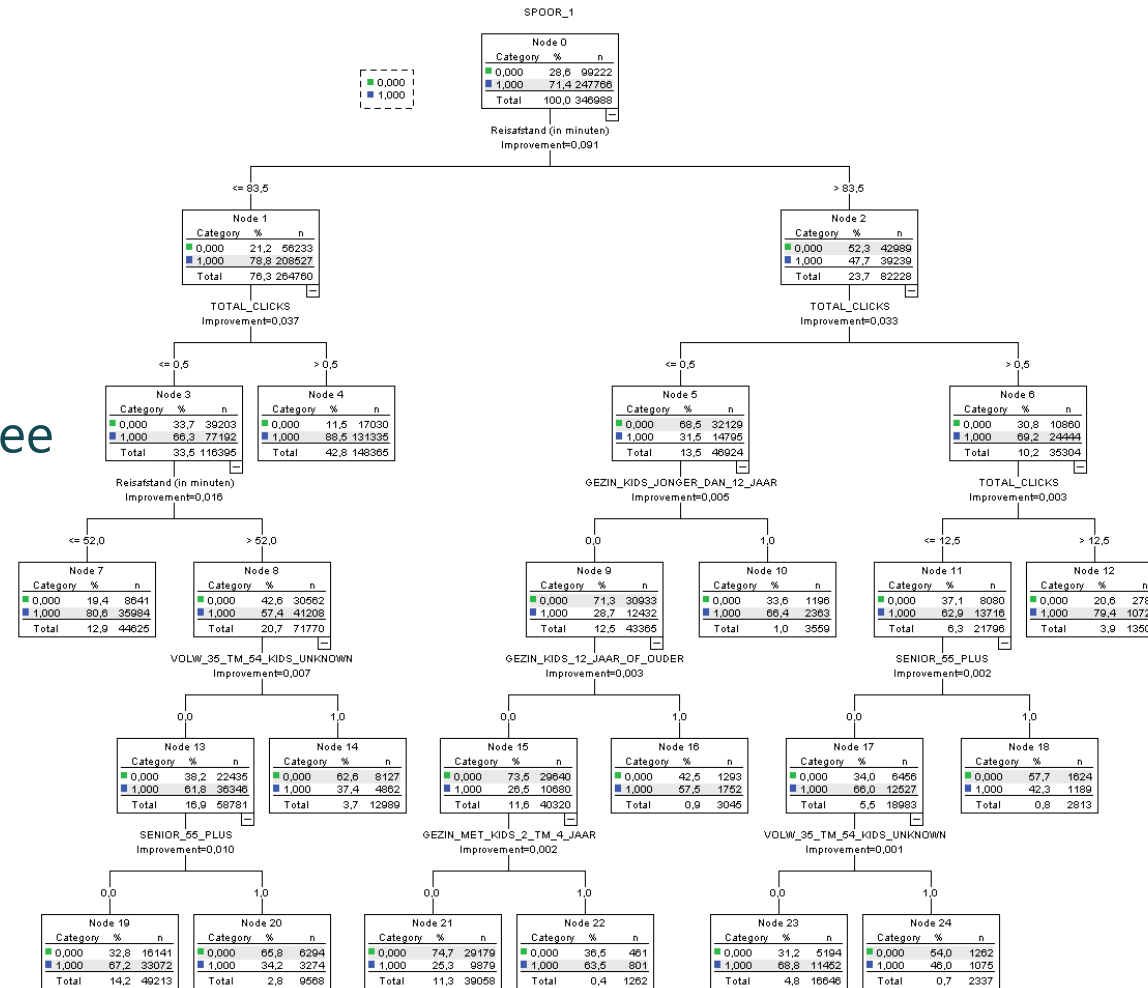


Customer Segmentation / Targeting

Predictive Segmentation

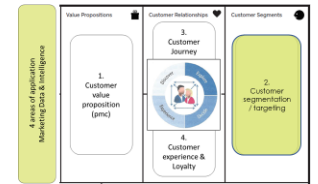
Analytics

Step-by-step current reality tree



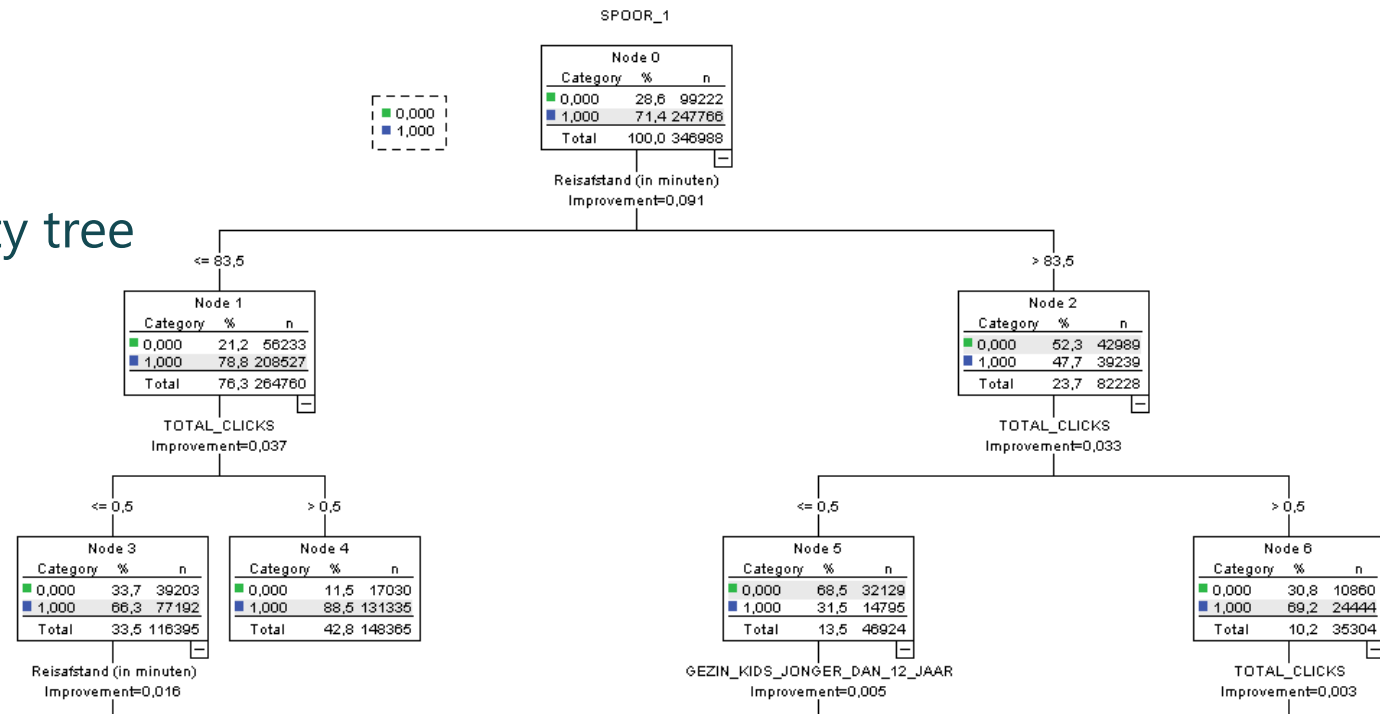
Customer Segmentation / Targeting

Predictive Segmentation



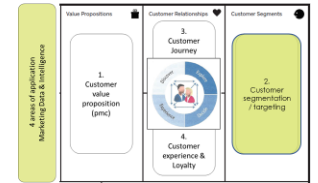
Analytics

Step-by-step current reality tree



Customer Segmentation / Targeting

Predictive Segmentation



Results based on step-by-step Current Reality Tree

Observed	Predicted		
	0	1	Percent Correct
0	13.397	15.198	46,9%
1	5.844	65.560	91,8%
Overall Percentage	19.241	80.759	79,0%

Growing Method: CRT

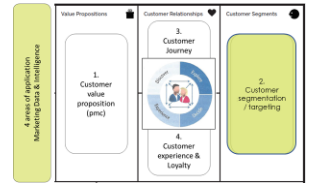
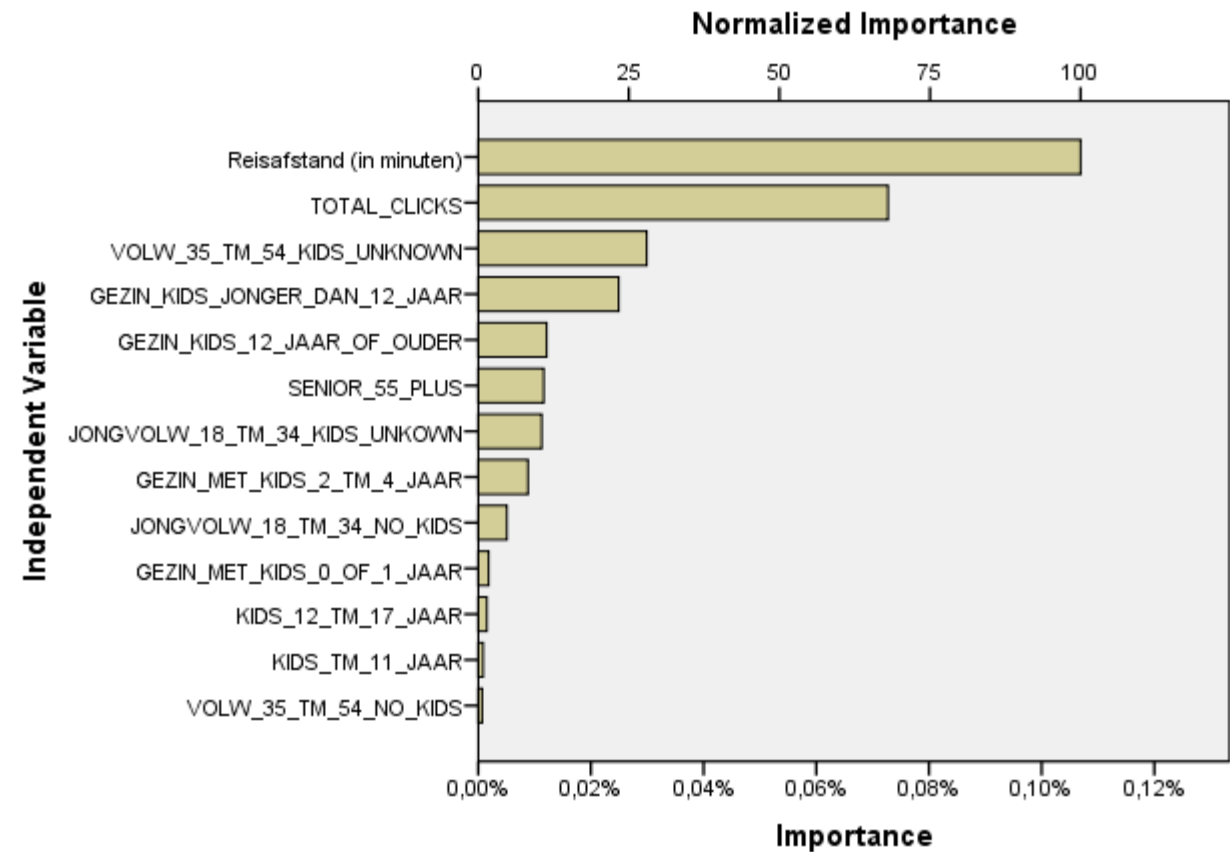
Dependent Variable: SPOOR_1

Customer Segmentation / Targeting

Predictive Segmentation

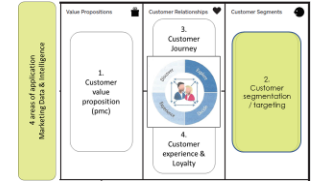
Results

Extent to which the independent variables contributed to the results



Customer Segmentation / Targeting

Predictive Segmentation



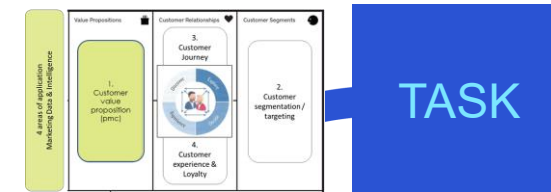
Final Results

83% of visitors in the right segment,
98% in the right segment or adjacent segment

Differentiation communication to
different expected 'next visit' categories,
Based on travel distance, clicks on newsletters &
family phase

Case: Customer value proposition

Dagblad



Om nieuwe abonnees te werven maakt de krant gebruik van 'probeer abonnementen' (PA).

Voor een bepaald bedrag krijgt de proefabonnee de krant 6 weken in de bus. Daarna kan zij/hij een abonnement afsluiten voor een half jaar, een jaar of twee jaar (met nog een introductie korting).

In het afgelopen jaar werd het PA aangeboden voor € 14,50. Maar ook zijn er meerdere acties geweest waarbij andere prijzen gehanteerd werden. Ook in de 2 jaren daarvoor werd het PA aangeboden met verschillende prijzen.

De conversie naar echte abonnees op deze acties zijn redelijk geweest, maar niet echt overweldigend.

Sales wil daarom dit proefabonnement tegen een lager prijs gaan aanbieden, zodat er meer proefabo's komen en -bij gelijkblijvende conversies- veel meer abonnees komen.

Kranten hebben echter veel last van stijgende kosten (redactie, online kosten, papier en bezorging). Daarom wil de financieel directeur de prijs juist verhogen € 14,50 naar € 17,50, om de marge te verbeteren.

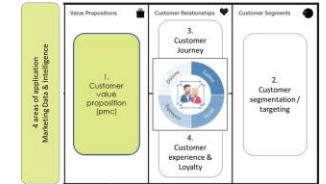
Als data driven marketeer heb je dit allemaal aangehoord. Nu kijken allen in de meeting naar jou.

'Wat stel jij voor wat we moeten doen' vraagt de CEO dan aan jou.

Met welke variabelen hebben we te maken? Welke afhankelijke variabele, welke onafhankelijke variabelen?
Welke analyse categorie? Welk type analyse?

Customer value proposition

Value proposition and acquisition price



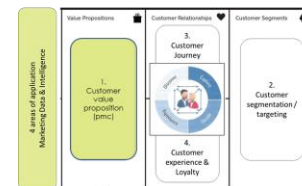
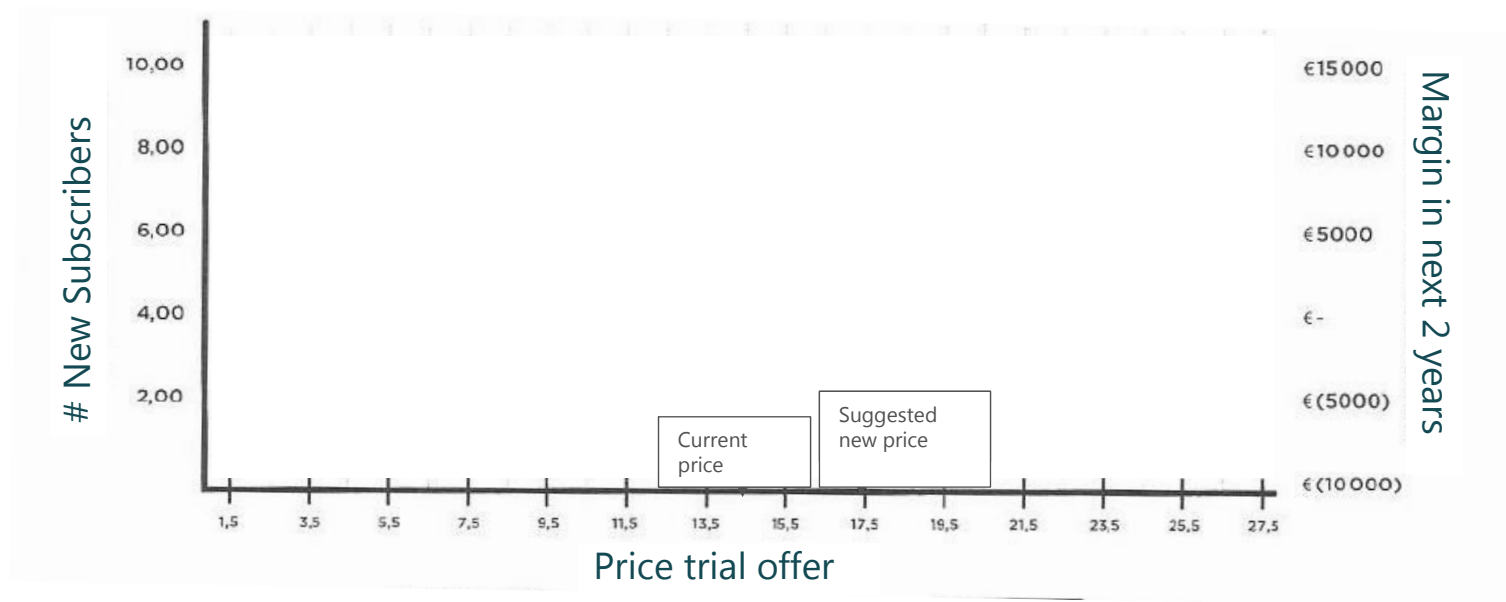
Price discussion



Customer value proposition

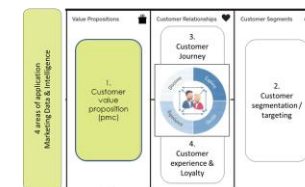
Value proposition and acquisition price

Which KPI's

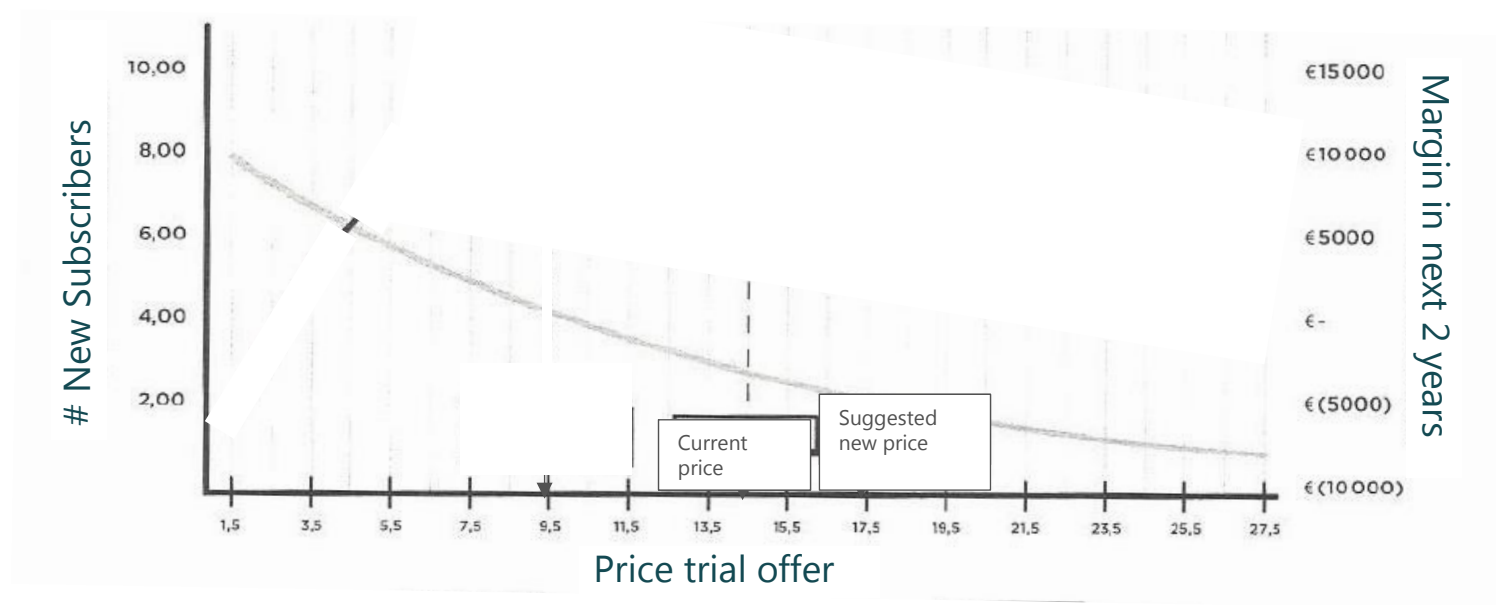


Customer value proposition

Value proposition and acquisition price

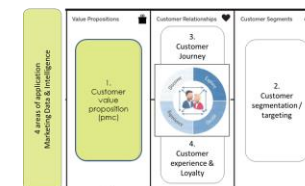


Elasticity demand curve

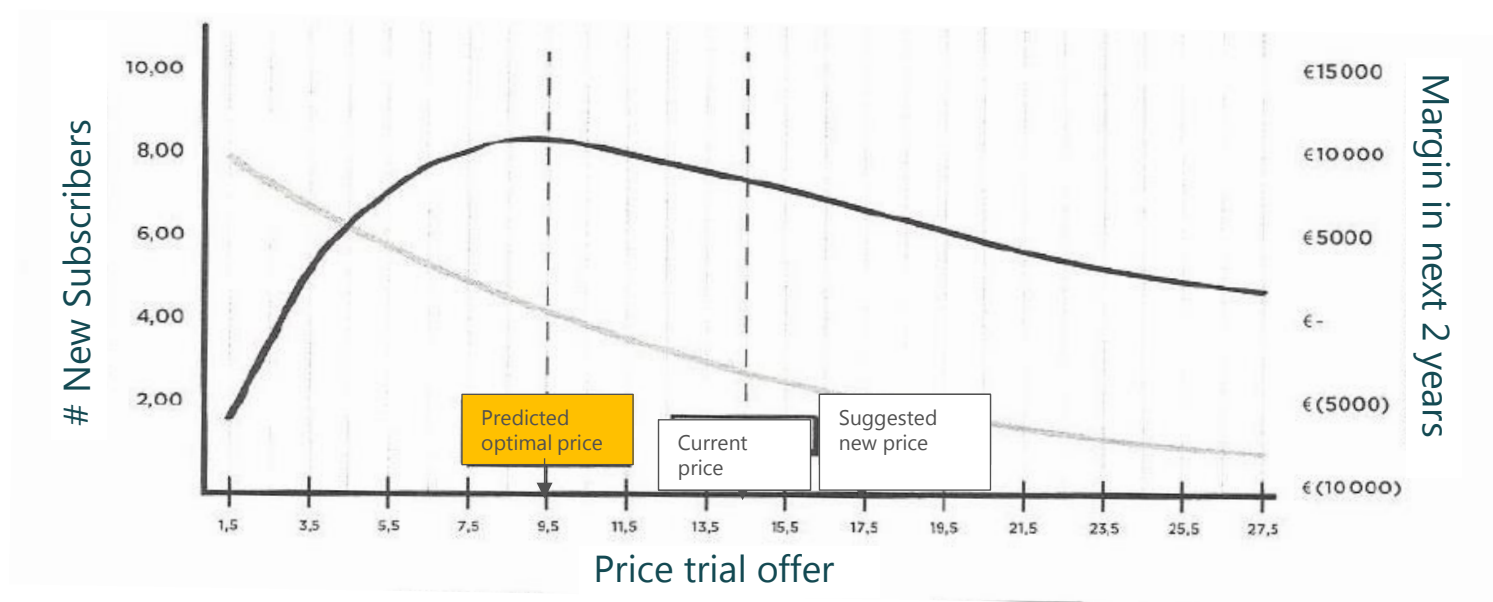


Customer value proposition

Value proposition and acquisition price

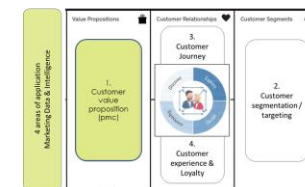


Cumulative margin curve

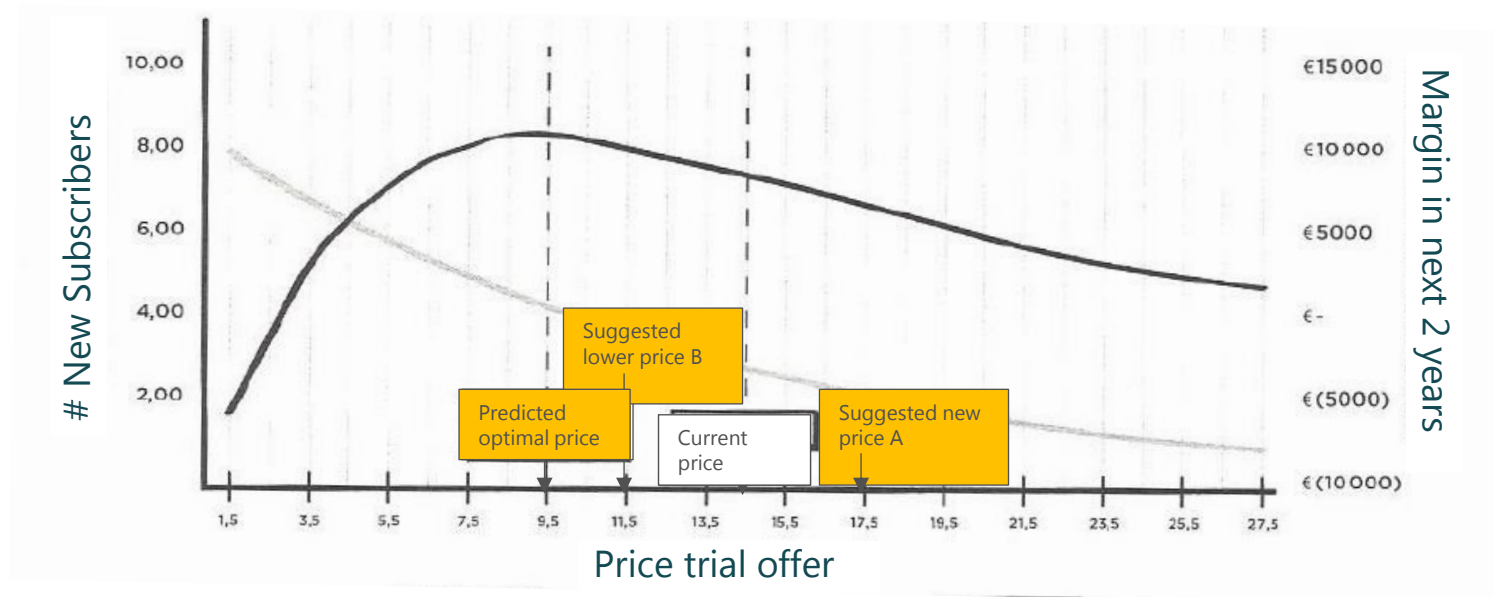


Customer value proposition

Value proposition and acquisition price



Selected price points



Customer value proposition

Value proposition and acquisition price

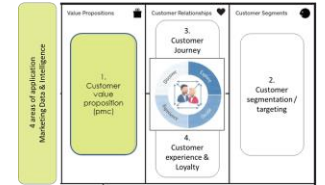
Result of the A/B test

In comparison with control group (regular acquisition price offer of € 14.50)

-  New subscribers
-  Margin over 2 years

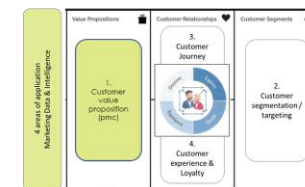
A
Price € 17,50
(regular € 14,50)

B
Price € 11,50
(regular € 14,50)



Customer value proposition

Value proposition and acquisition price



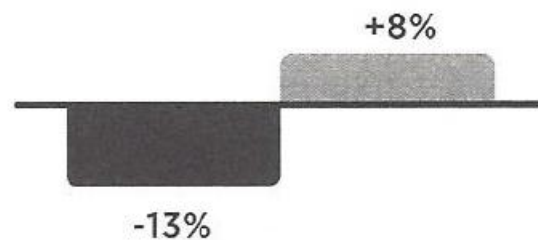
Result of the A/B test

-  New subscribers
-  Margin over 2 years

In comparison with control group (regular acquisition price offer of € 14.50)

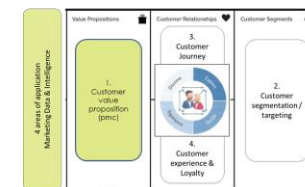
A
Price € 17,50
(regular € 14,50)

B
Price € 11,50
(regular € 14,50)



Customer value proposition

Value proposition and acquisition price



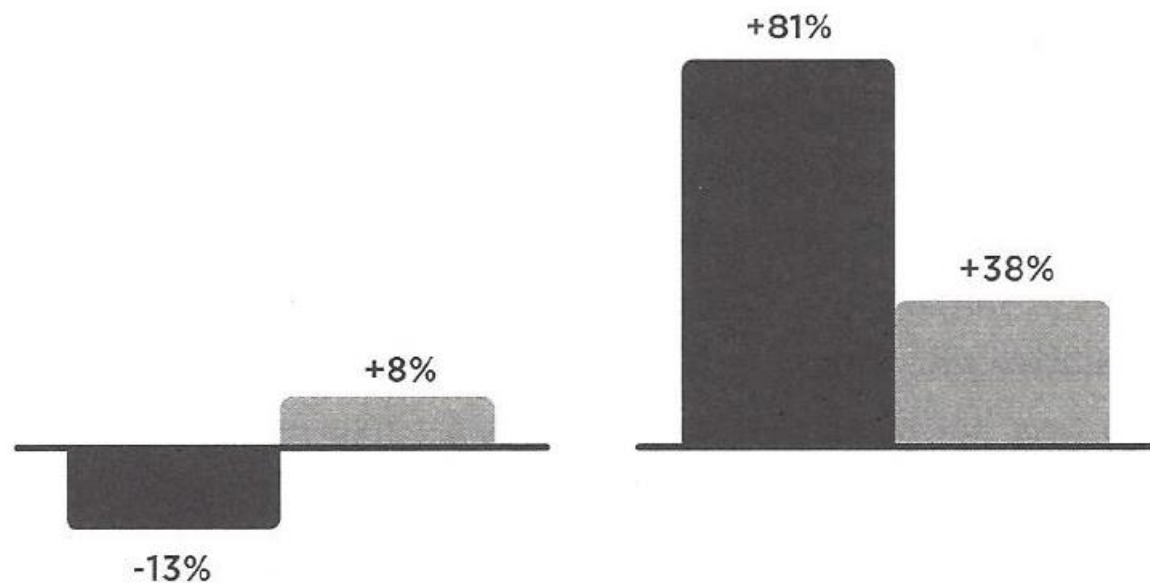
Result of the A/B test

In comparison with control group (regular acquisition price offer of € 14.50)

■ New subscribers
■ Margin over 2 years

A
Price € 17,50
(regular € 14,50)

B
Price € 11,50
(regular € 14,50)





Predictive Analytics ctd

**Predictive
Unsupervised**

- A. Supervised learning
Propensity, classification
- B. Unsupervised learning
Clustering, Principal Component Analysis**
- C. Reinforcement learnings
Collaborative Filtering, recommendation
models based on nearest neighbourhood

Predictive Analytics - Unsupervised

**Predictive
Unsupervised
Clustering**

B Unsupervised learning

Recognising patterns in data without knowing what you are looking for. Does not predict/find anything special

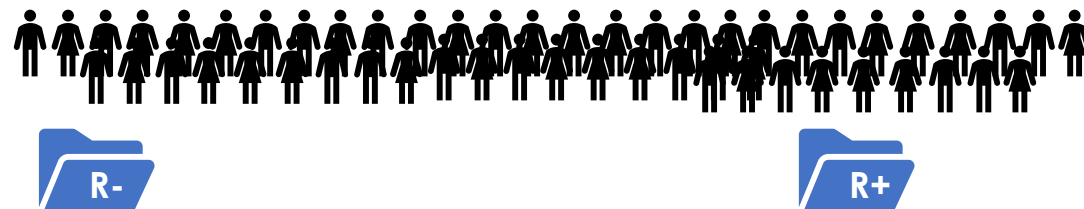
1. Clustering

- **RFM model**
- K Nearest Neighbour

2. Principal Component Analysis

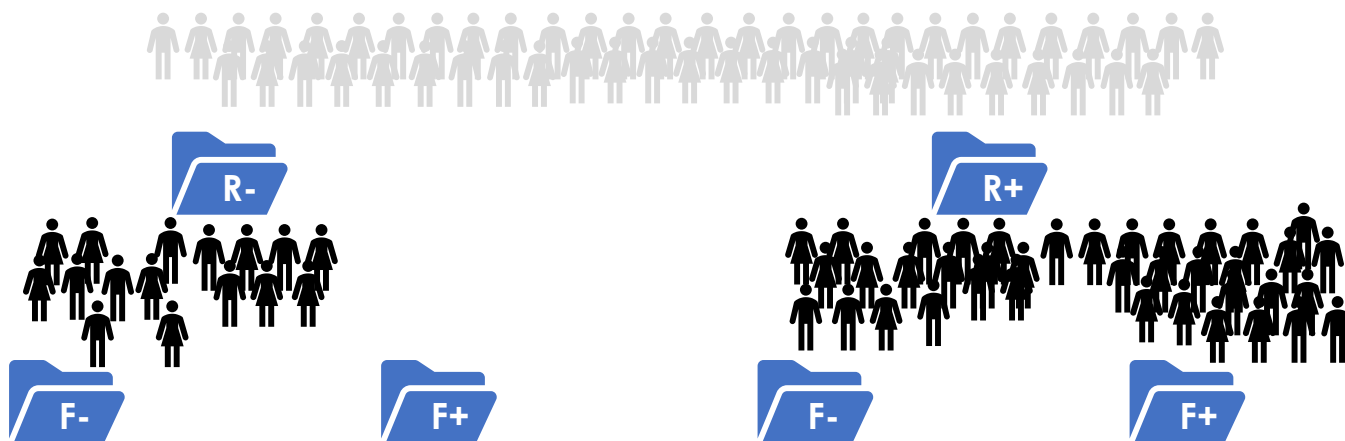
Predictive Analytics - RFM

**Predictive
Unsupervised
Clustering**



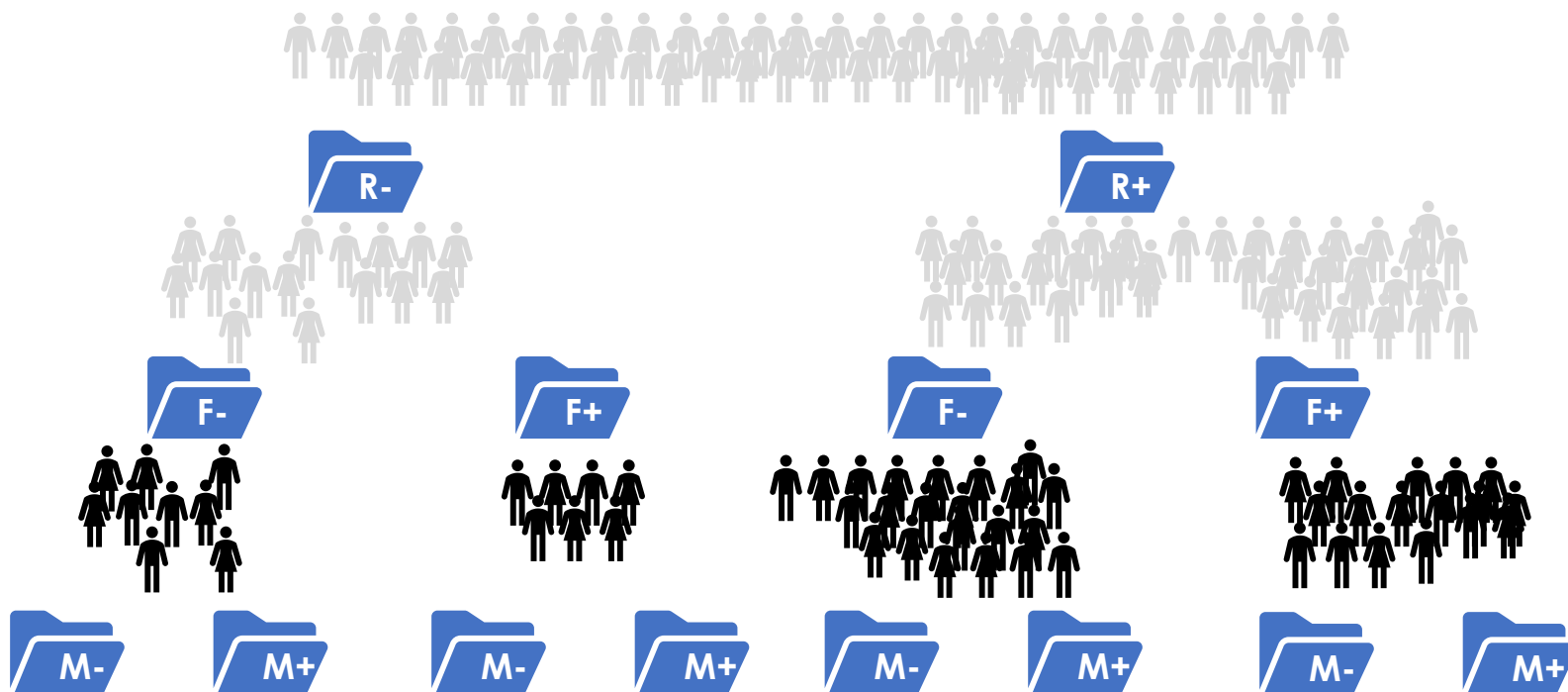
Predictive Analytics - RFM

**Predictive
Unsupervised
Clustering**



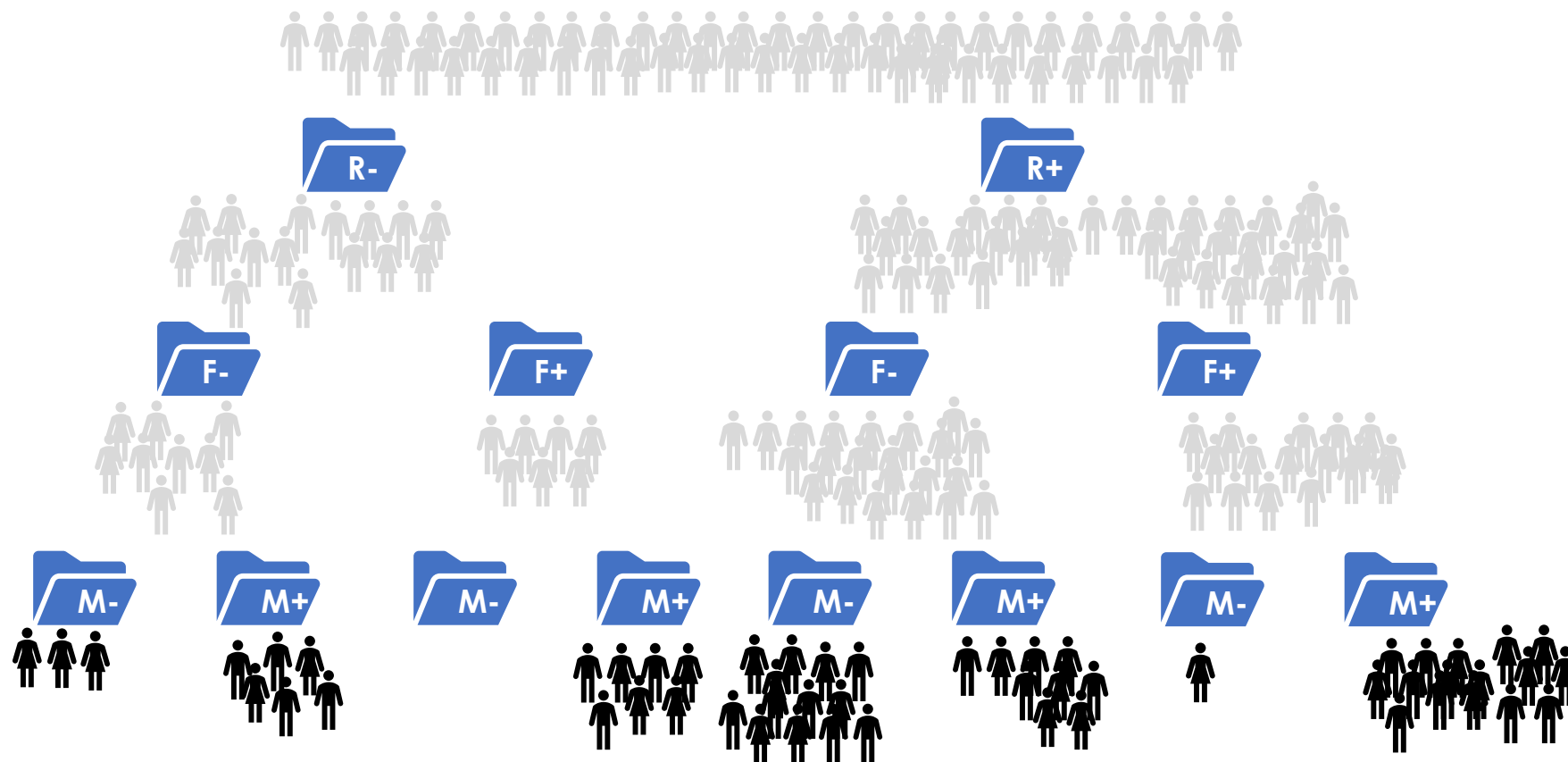
Predictive Analytics - RFM

Predictive
Unsupervised
Clustering



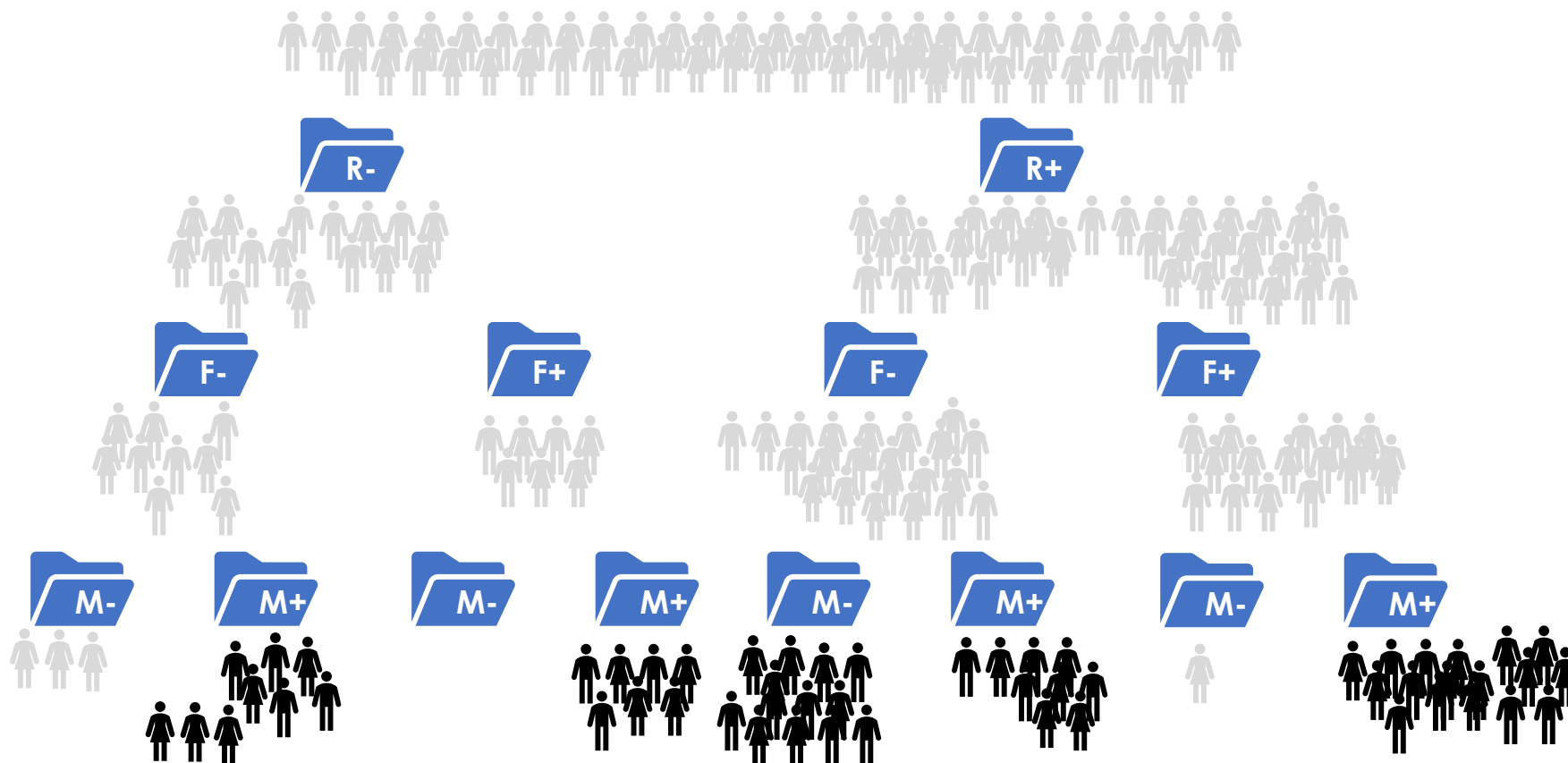
Predictive Analytics - RFM

Predictive
Unsupervised
Clustering

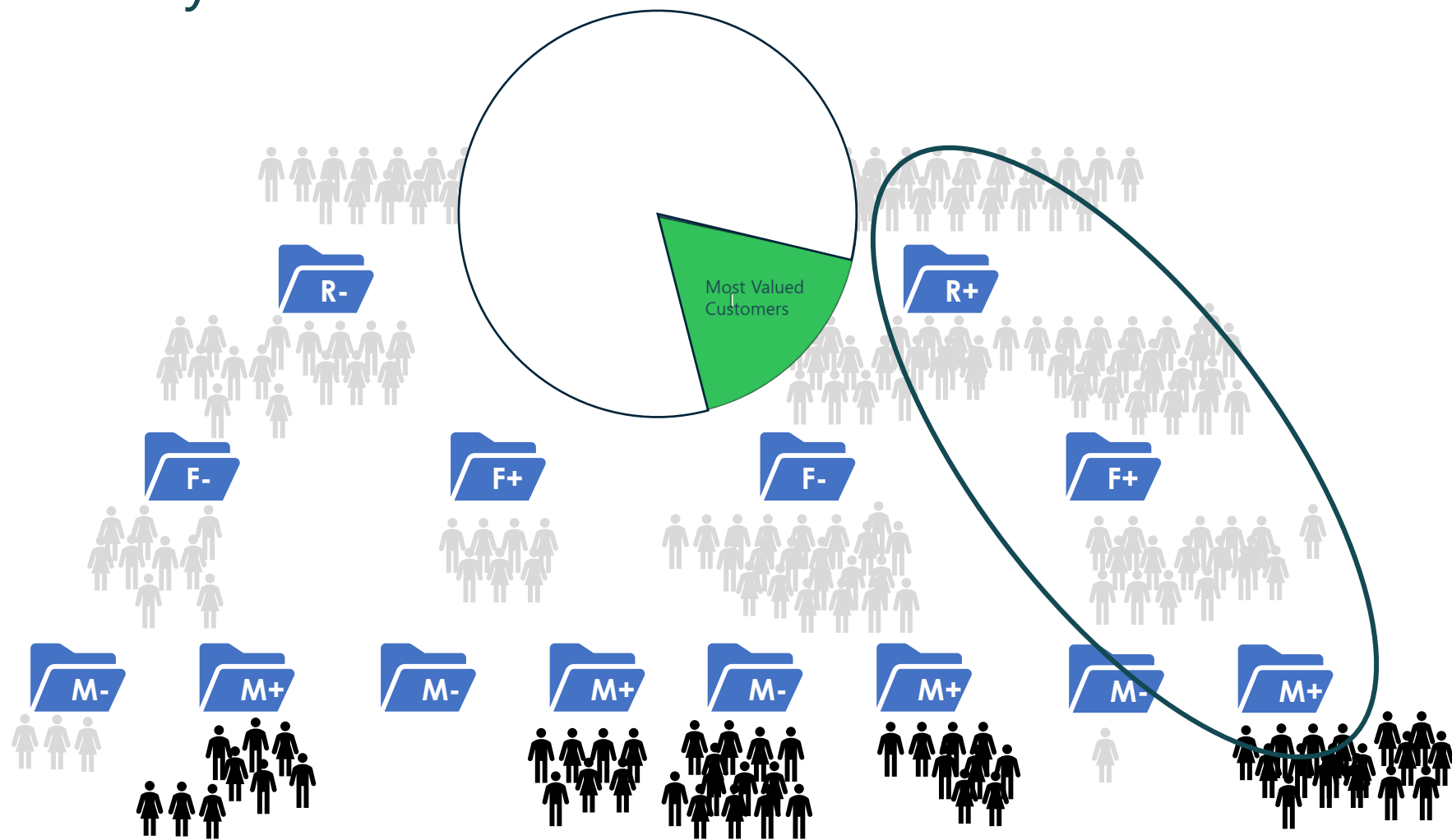


Predictive Analytics - RFM

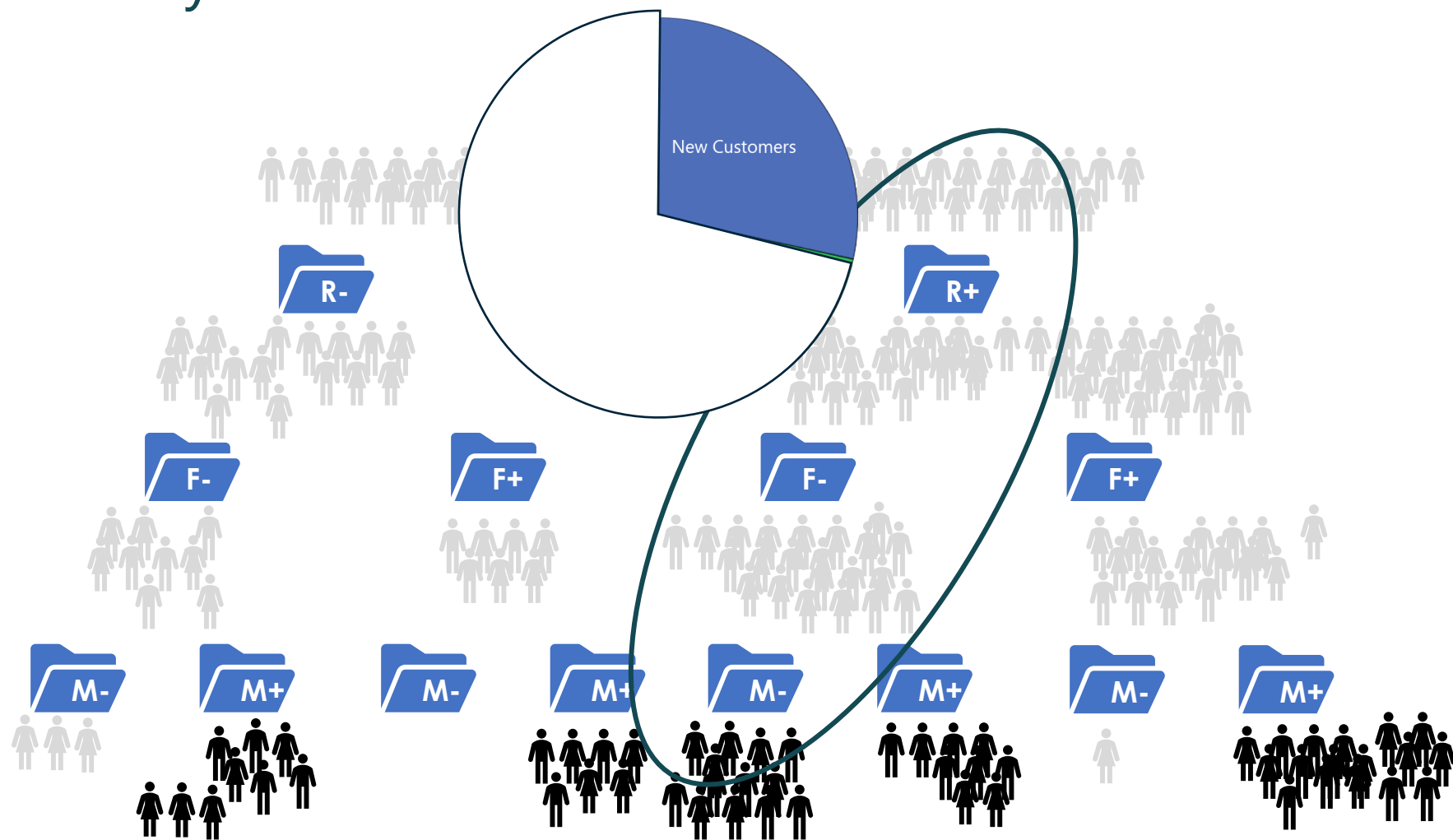
Predictive
Unsupervised
Clustering



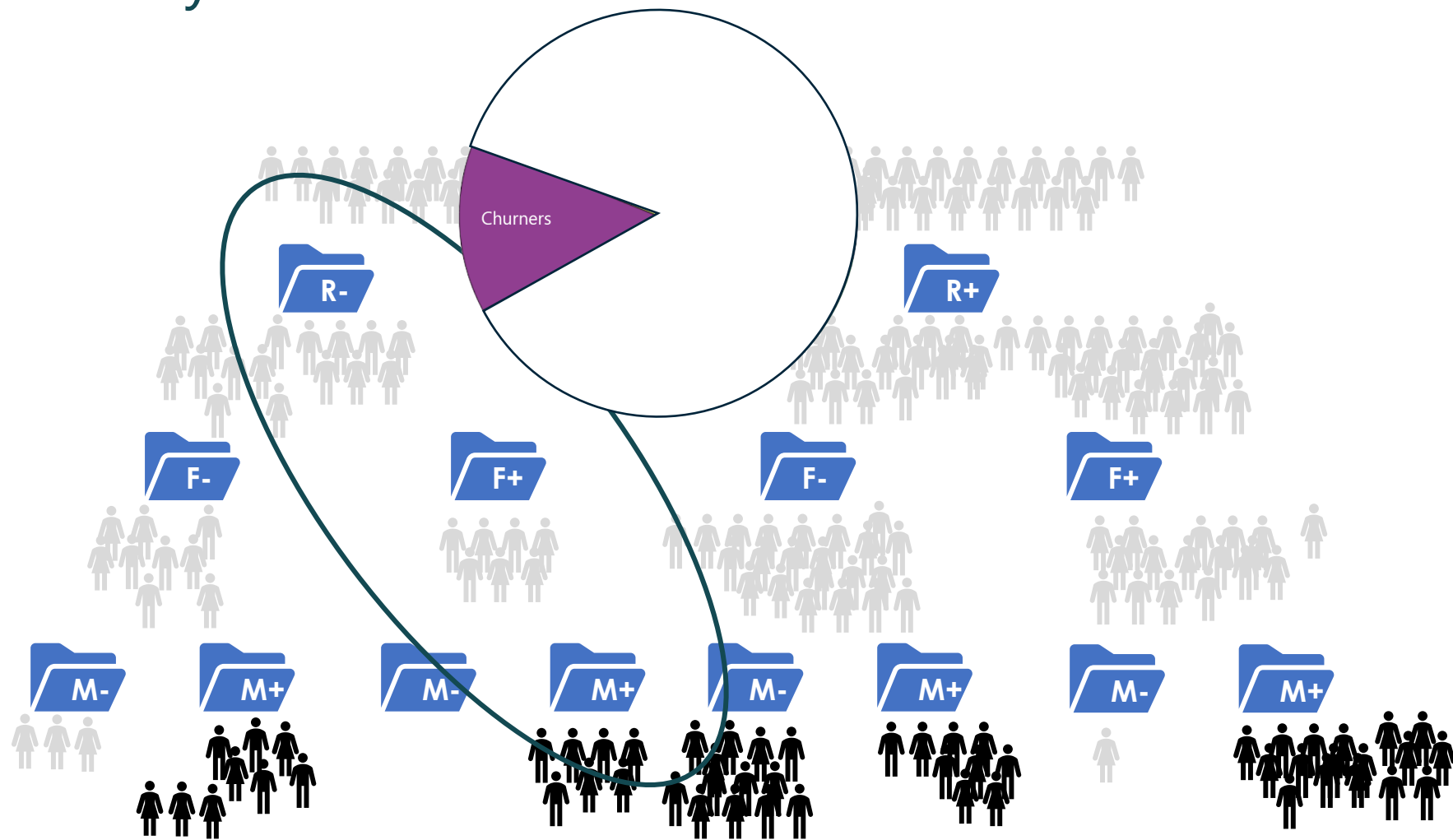
Predictive Analytics - RFM



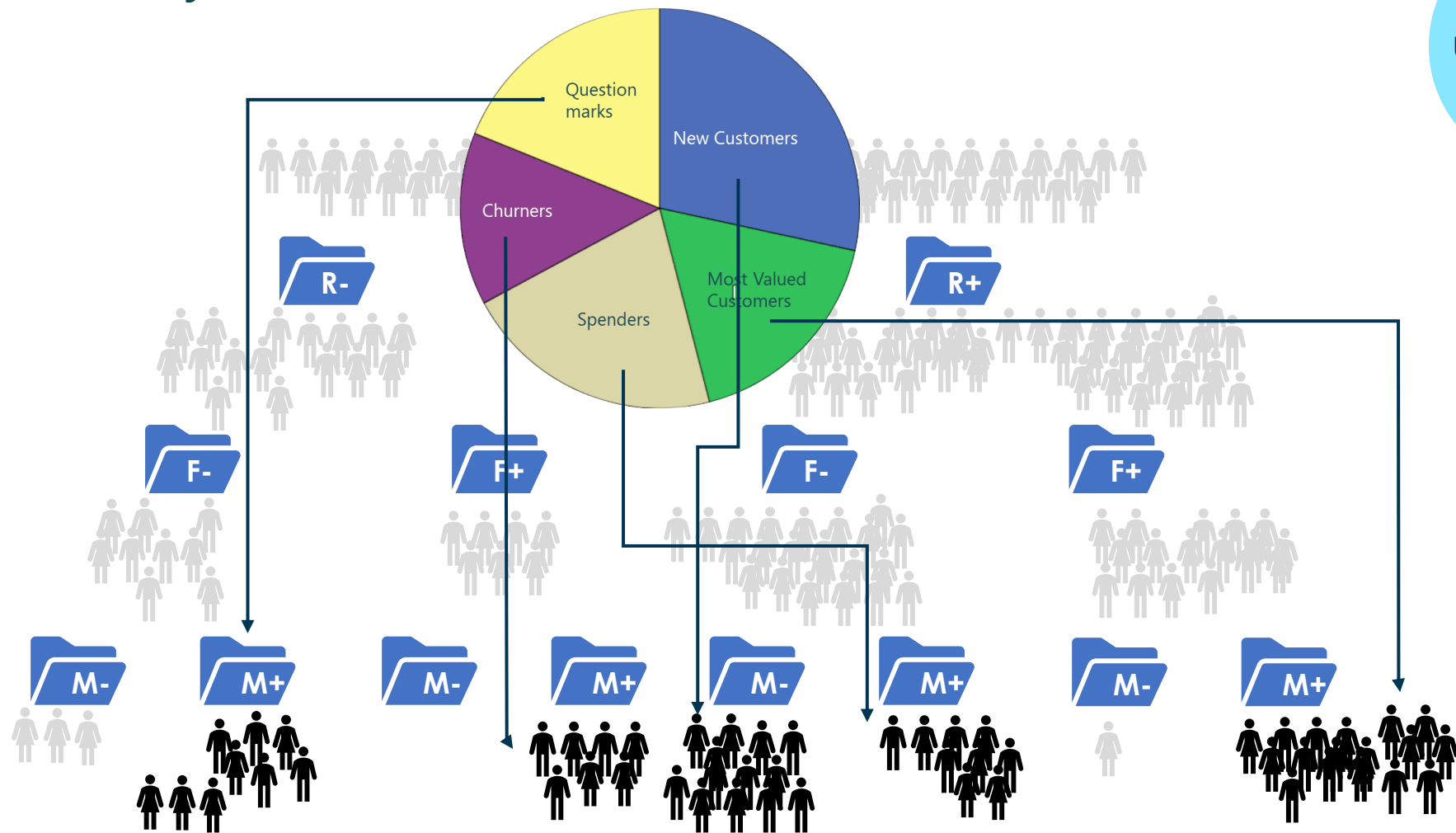
Predictive Analytics - RFM



Predictive Analytics - RFM



Predictive Analytics - RFM



**Predictive
Unsupervised
Clustering**

Predictive Analytics - Unsupervised

Predictive
Unsupervised
Clustering

B Unsupervised learning

Recognising patterns in data without knowing what you are looking for. Does not predict/find anything special

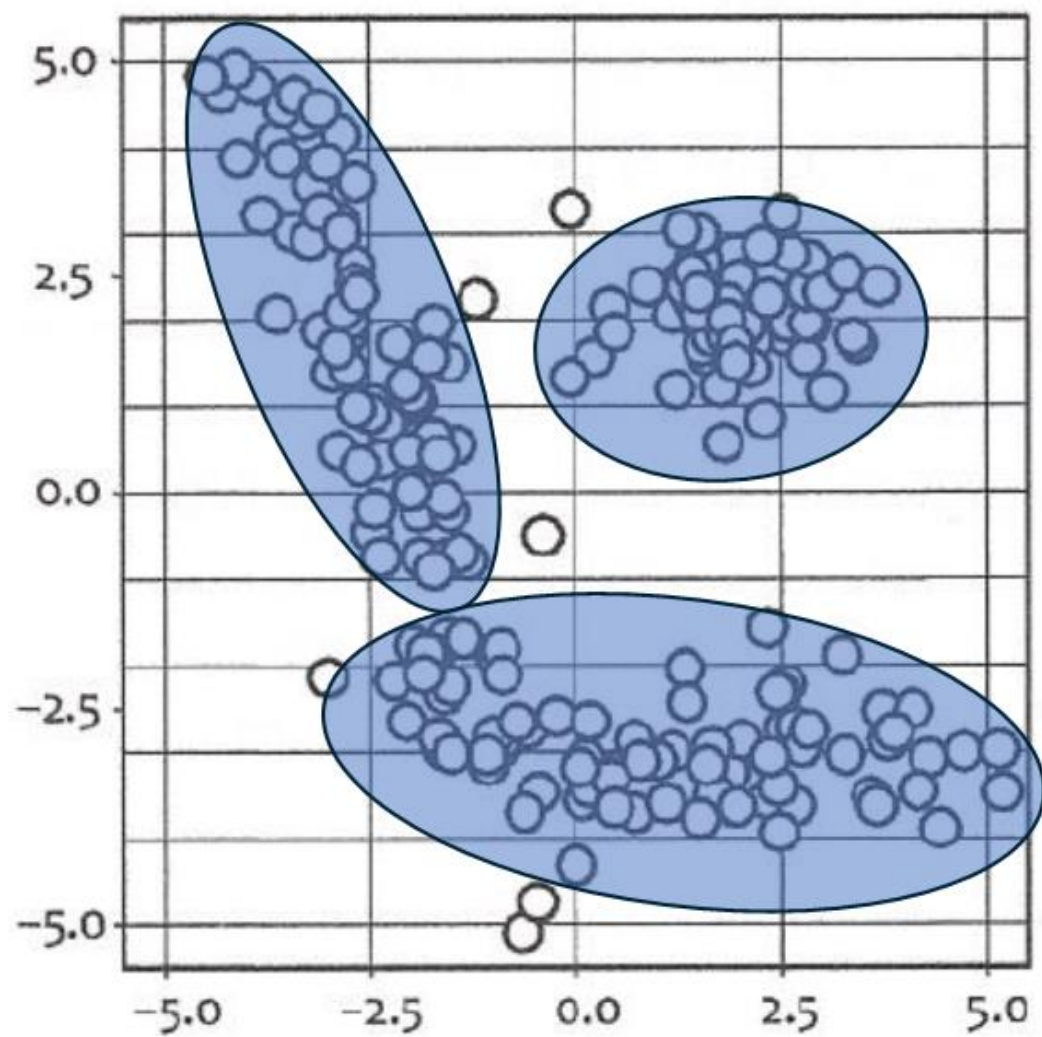
1. Clustering

- RFM model
- **K Nearest Neighbour**

2. Principal Component Analysis

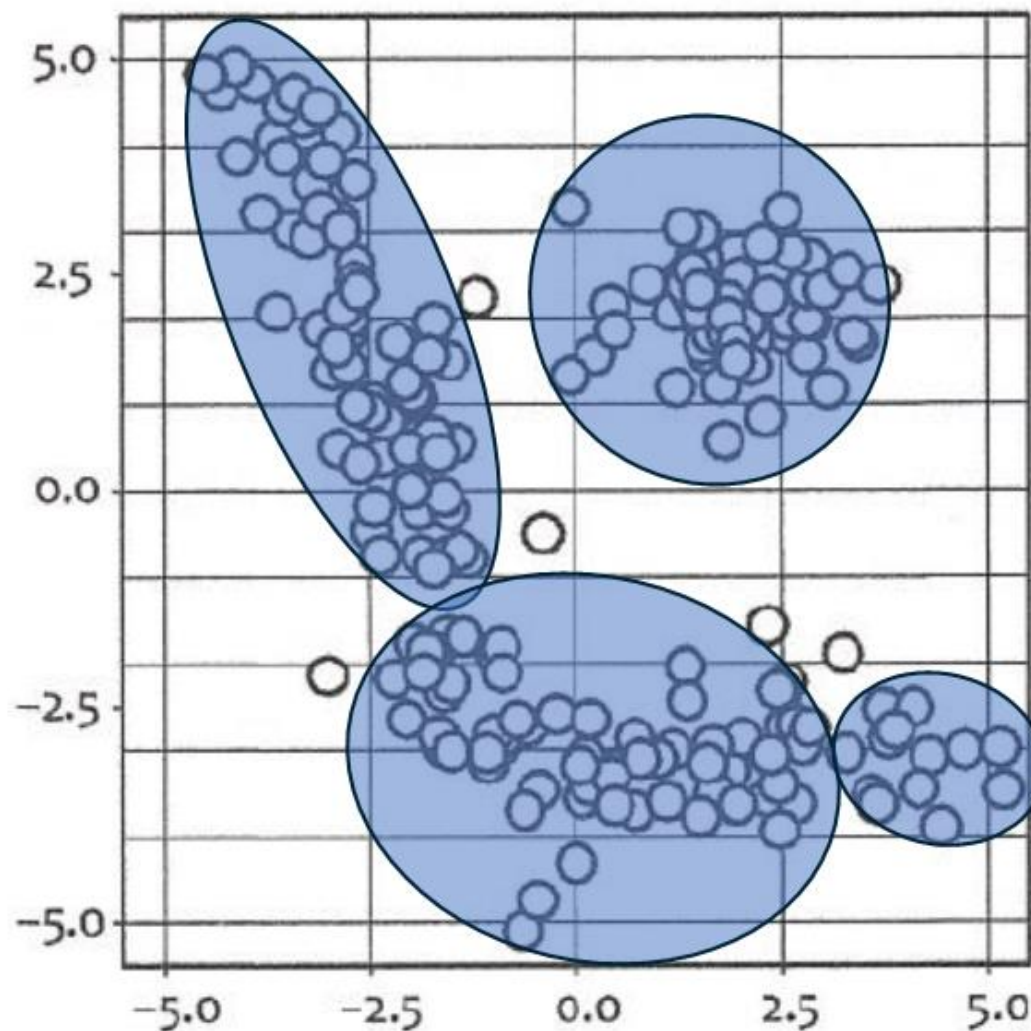
Predictive Analytics - K Nearest Neighbour

**Predictive
Unsupervised
Clustering**



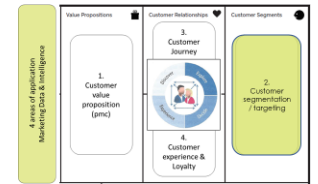
Predictive Analytics - K Nearest Neighbour

**Predictive
Unsupervised
Clustering**



Customer Segmentation / Targeting

Better targeted communication to the different segments



- Theme Park



- Overnight stay



- Theatre & Festivals



Predictive Analytics - Unsupervised

B Unsupervised learning

Predictive
Unsupervised
PCA

Recognising patterns in data without knowing what you are looking for. Does not predict/find anything special

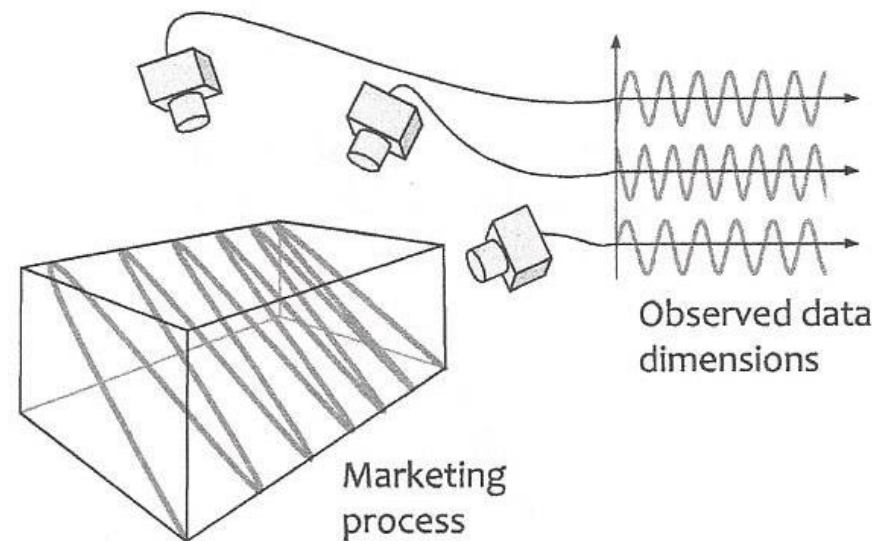
1. Clustering

- RFM model
- K Nearest Neighbour

2. Principal Component Analysis

Decorrelation

Dimensionality Reduction



2. Principal Component Analysis

Predictive
Unsupervised
PCA

Decorrelation

1

Valuation Assignment
'Postmaster DIAI&C'

Variance determined by
variable 'organizational
support' (15.7%)

**Support
of organization**

2. Principal Component Analysis

Predictive
Unsupervised
PCA

Decorrelation

2

Valuation Assignment
'Postmaster DIAI&C'

Available time

Variance determined
by variable 'available
time' (19.4%)

2. Principal Component Analysis

Predictive
Unsupervised
PCA

Decorrelation

3

Variance uniquely
determined by variable
'available time'

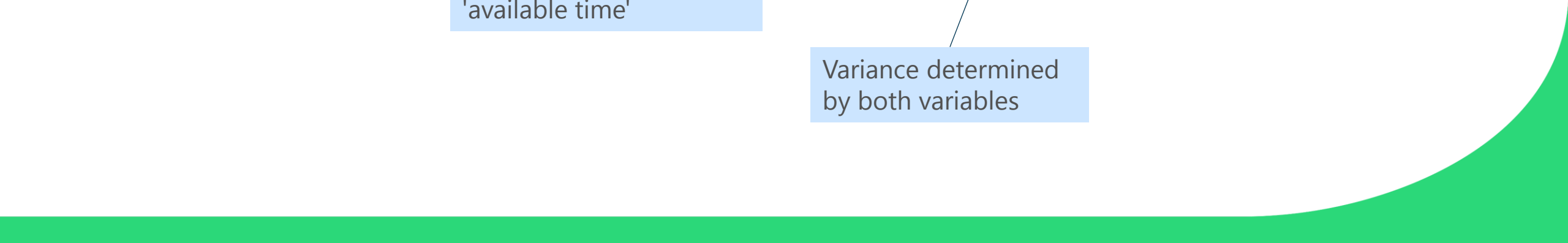
Valuation Assignment
'Postmaster DIAI&C'

Available time

Support
of organization

Variance determined
by both variables

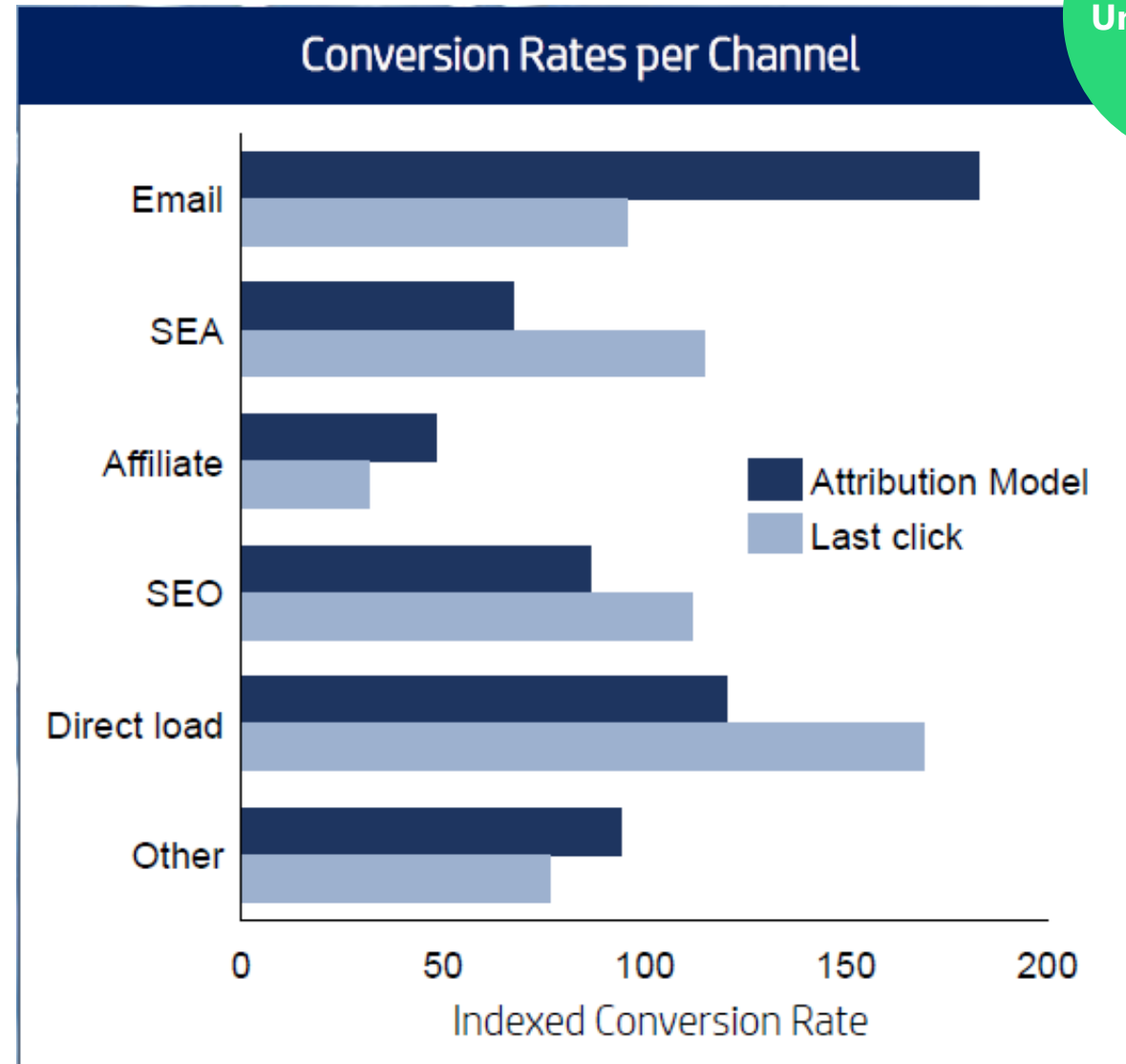
Variance uniquely
determined by variable
'organizational support'



2. Principal Component Analysis

Decorrelation

Channel Attribution to Conversion



Predictive
Unsupervised
PCA

2. Principal Component Analysis

Predictive
Unsupervised
PCA

Dimensionality
reduction

Product Attribute Modelling

Survey elements – specific attributes	Factor loadings		
	Price	Service	Convenience
Low prices	.899		
Attractive offers	.602		
Product quality		.681	
Customer friendly personnel		.747	
Good supply of fresh products		.732	
Large assortment		.790	
Long opening hours		.573	
Store attractiveness		.857	
Fast checkout		.713	
Good supply of additional services		.624	
Child friendliness of the store		.644	
Tidy store		.852	
Spacious store		.768	
Knowledgeable personnel		.769	
Much attention for new products		.768	
The store is nearby			.749
Sufficient supply of other stores close to the focal store			.671

Customer Value & Segmentation

Actual Customer Value

Method -

Customer Value:

- Customer Pyramid based on Customer Value

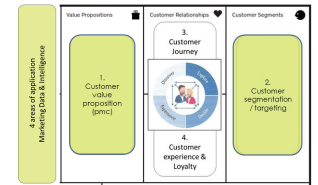
25% Top Customers

25% Large Customers

25% Mid size Customers

25% Small Customers

% of Sales



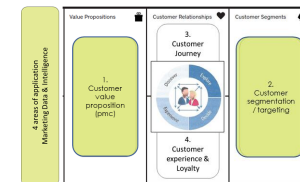
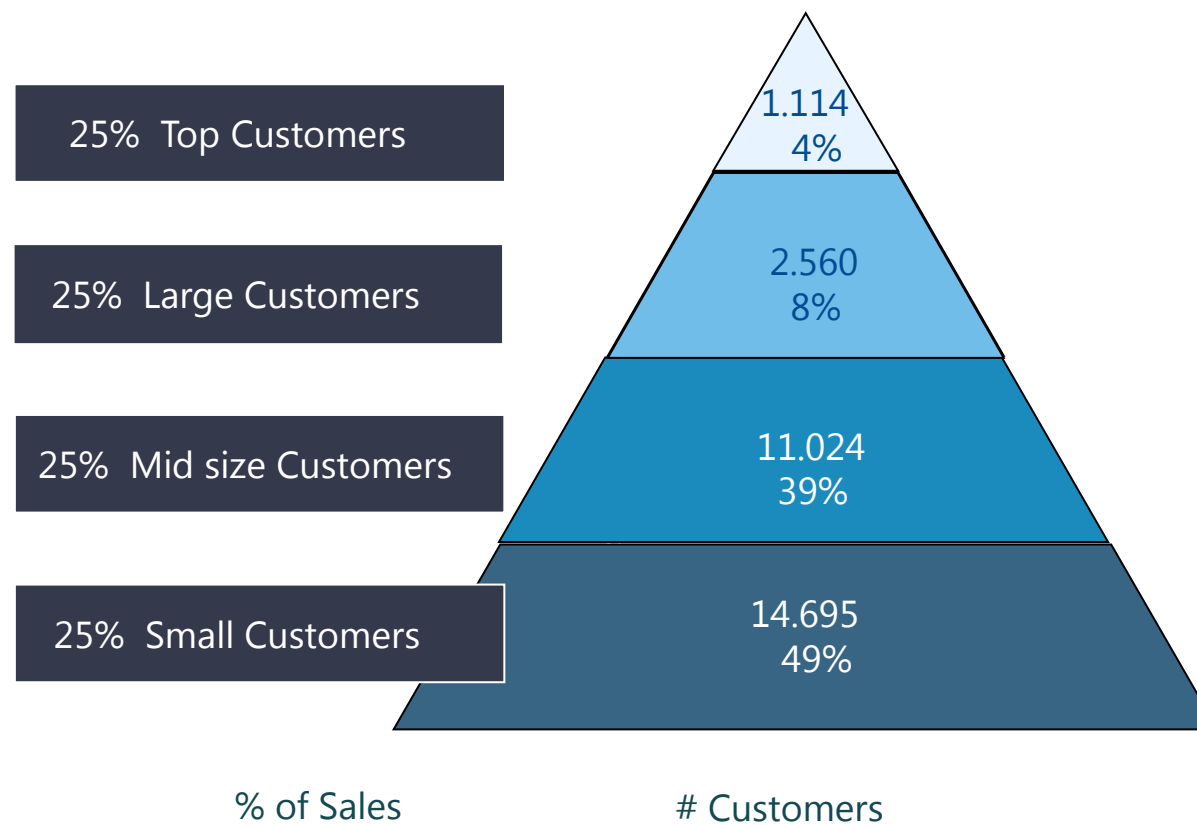
Customer Value & Segmentation

Actual Customer Value

Method -

Customer Value:

- Customer Pyramid based on Customer Value



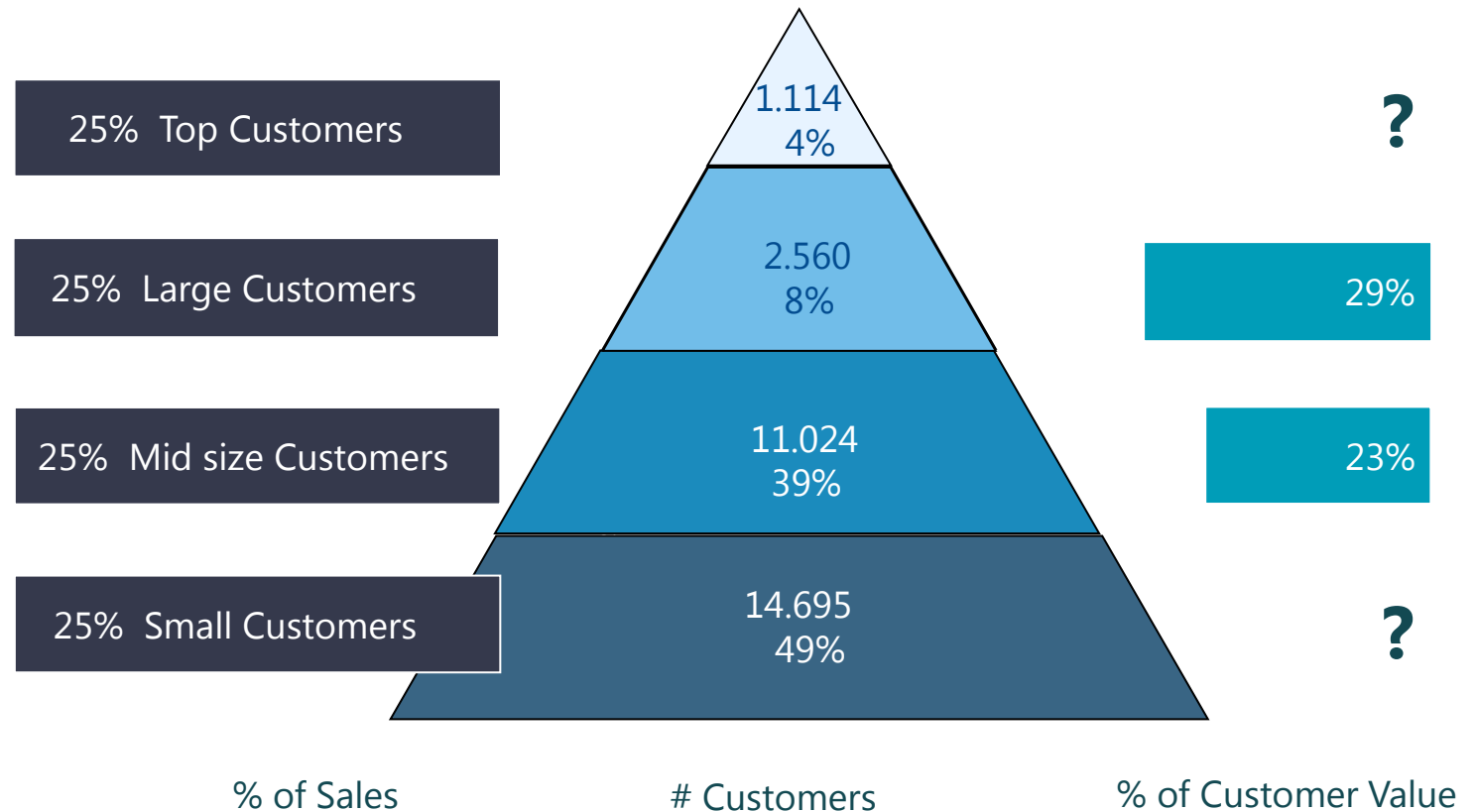
Customer Value & Segmentation

Actual Customer Value

Method -

Customer Value:

- Customer Pyramid based on Customer Value



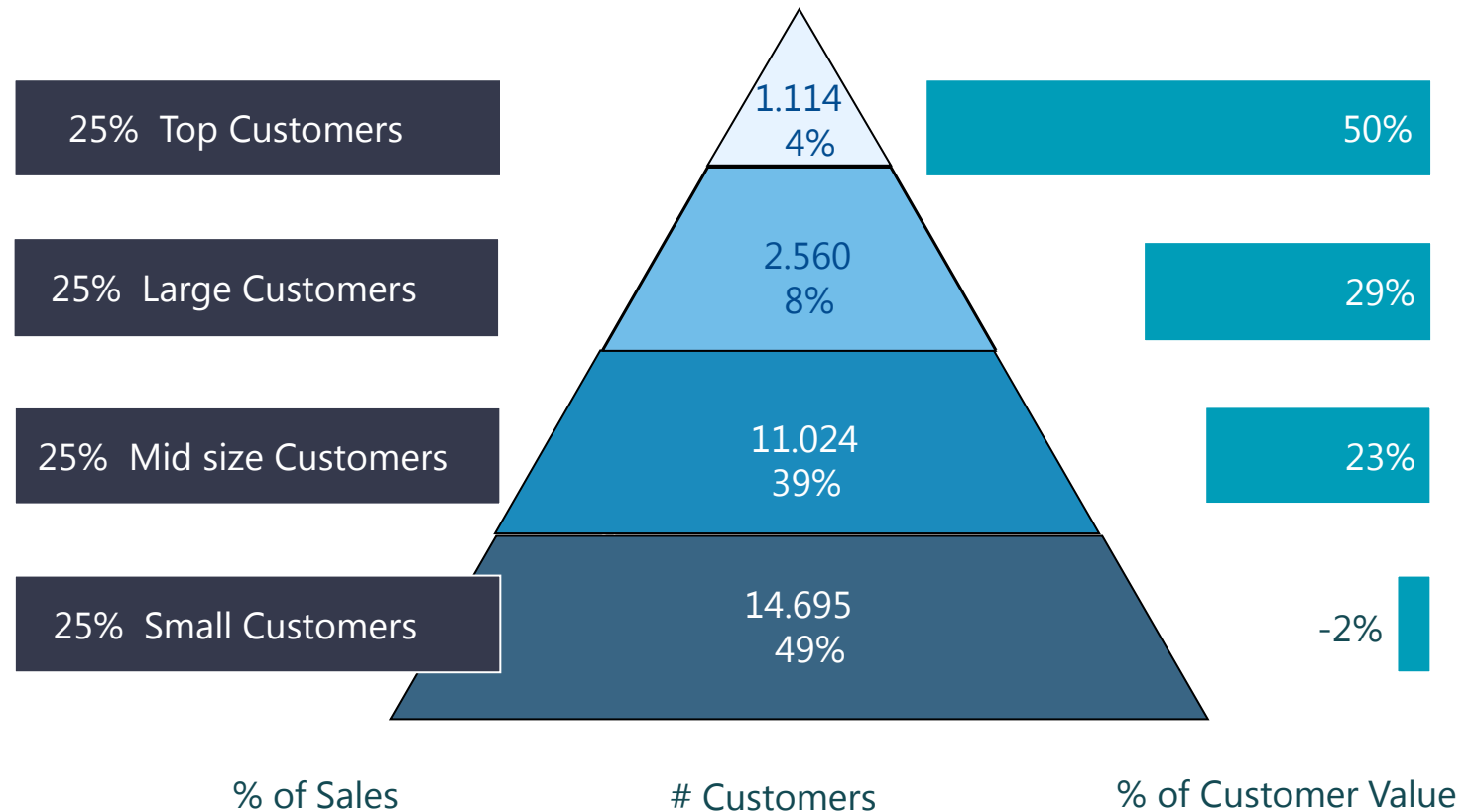
Customer Value & Segmentation

Actual Customer Value

Method -

Customer Value:

- Customer Pyramid based on Customer Value



Customer Value & Segmentation

Customer Value Proposition



TASK

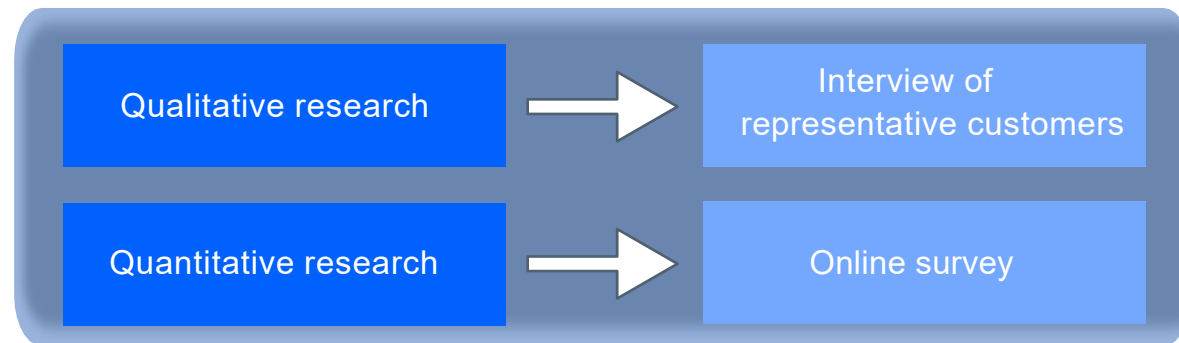
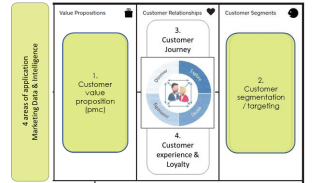
+

Bedenk een aantal 'features' die van belang zouden kunnen zijn bij een reisorganisatie voor business reizen

Customer Value & Segmentation

Method;
determining '**product features**' and assigning value to the features

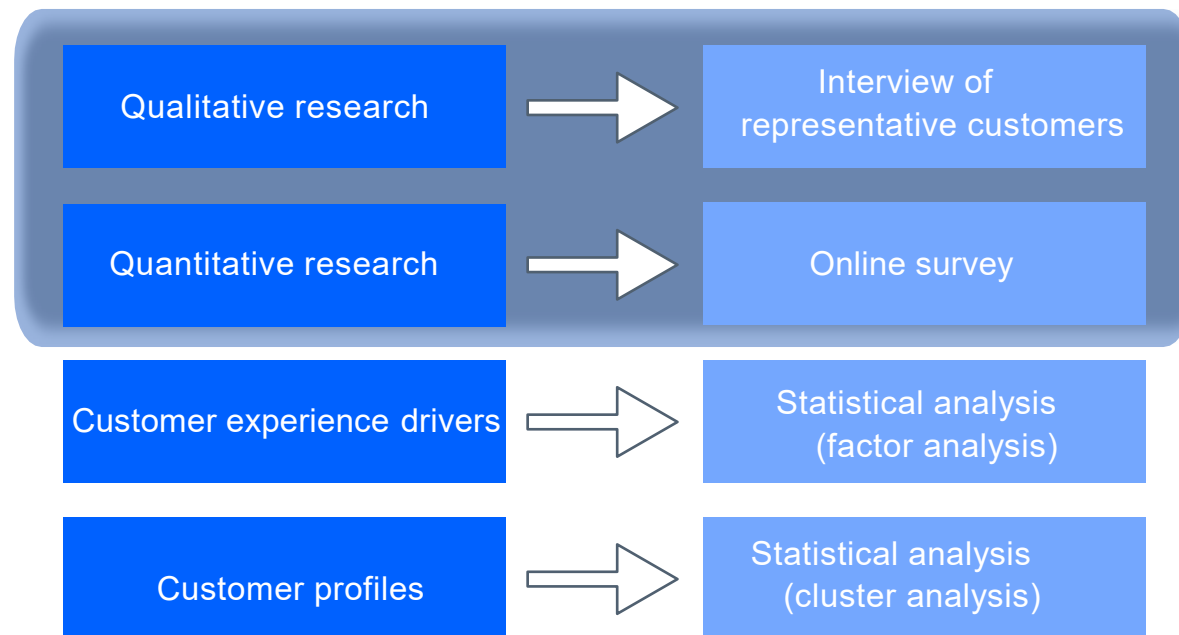
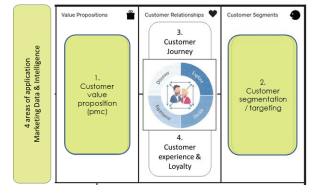
82



Customer Value & Segmentation

Method;
determining '**product features**' and assigning value to the features

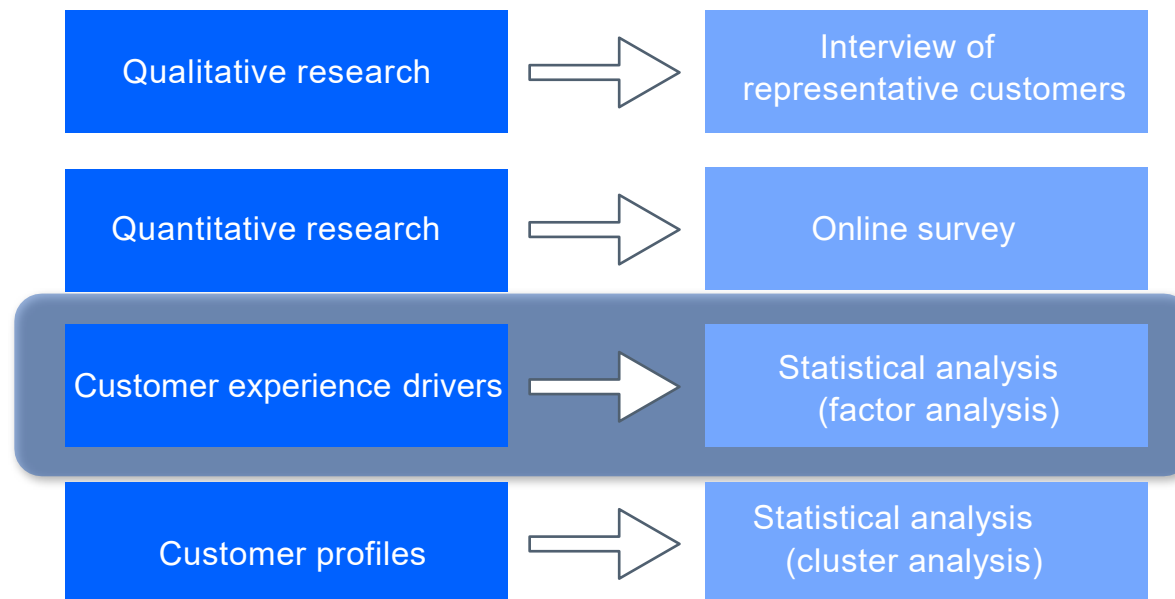
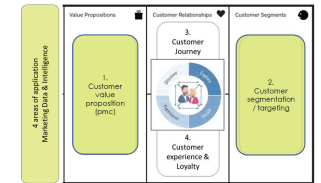
82



Customer Value & Segmentation

Methode; Principle Component Analyses (Factor analyse)

82



Customer Value & Segmentation

Principal Component Analyses

4 Principal Components (Factors):

PC 1 mainly explained by awareness, image, reputation and ethics (**Brand**)
Factor with the greatest explained variance.

PC 2 mainly determined by **customer focus** factors.

PC 3 is the **price(quality)** factors

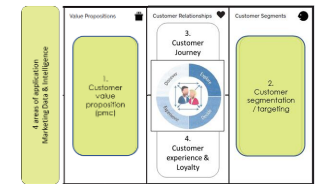
Factor with the smallest explained variance (*).

PC 4 is a **service support** factor

(*) This does not mean that price and quality are not important, but it does mean that all respondents gave relatively the same response!

+

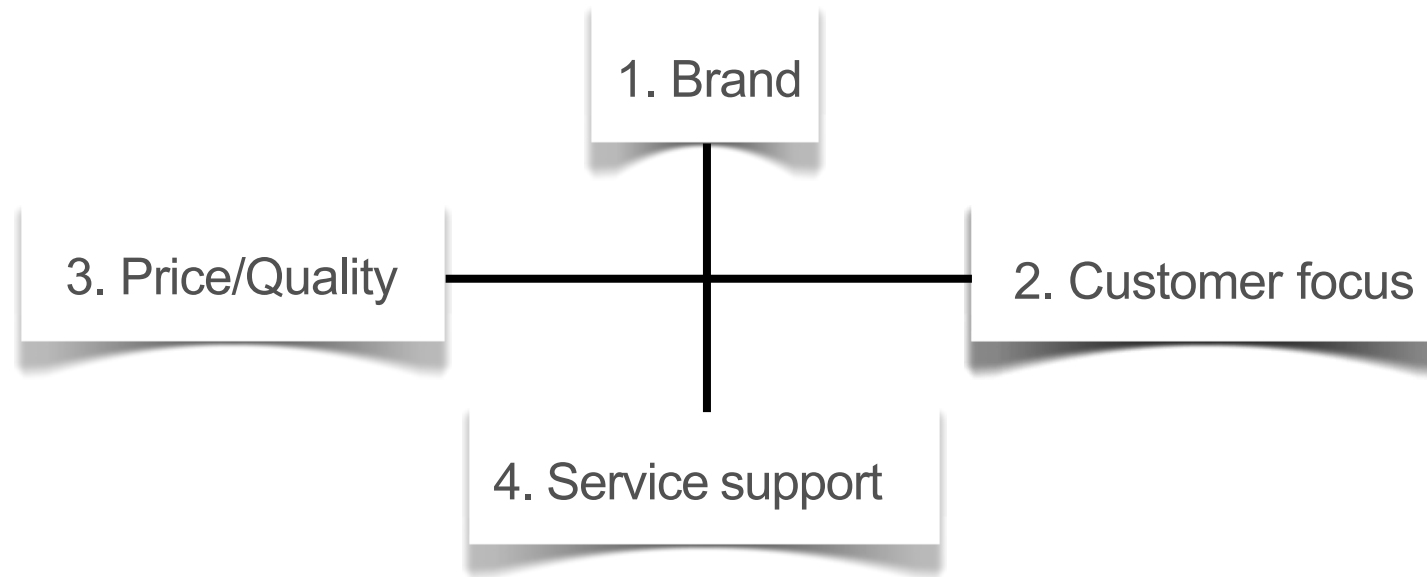
84



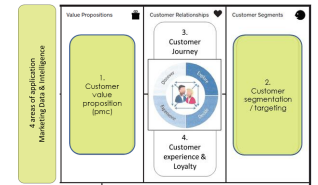
Variable	Factor Loadings (Varimax normalized) (eigenities 2 Extraction: Principal components (Marked loadings are >,600000)			
	Factor 1	Factor 2	Factor 3	Factor 4
P	0,113117	0,017985	0,839708	0,038106
Q	0,021652	0,374679	0,313546	0,365602
P_Q	0,044158	0,162272	0,834757	0,086941
S_OPL	0,179238	0,226012	-0,064182	0,647743
S_SUP	0,028601	0,263092	0,100543	0,594210
S_SELF	0,136992	-0,027162	0,072335	0,759561
AWARENESS	0,792584	0,059953	0,052453	0,045477
IMAGE	0,829292	0,063139	0,062369	0,042512
REPUT	0,669228	0,105731	0,206790	0,101979
INFO	0,419962	0,282386	0,164984	0,341564
ETHICS	0,628973	0,279323	-0,014504	0,161042
CUSTFOCUS	0,074193	0,778894	0,076019	0,102718
EVENTS	0,527046	0,315932	-0,150128	0,134946
CUSTKNOW	0,138309	0,702494	0,144978	0,173246
CUSTVAL	0,207262	0,706569	0,074557	0,117173
RELATION	0,304764	0,676608	-0,035334	0,171694
Expl.Var	2,840931	2,621625	1,652677	1,751124

Customer Value & Segmentation

Principal Component Analyses: 4 customer value dimensions



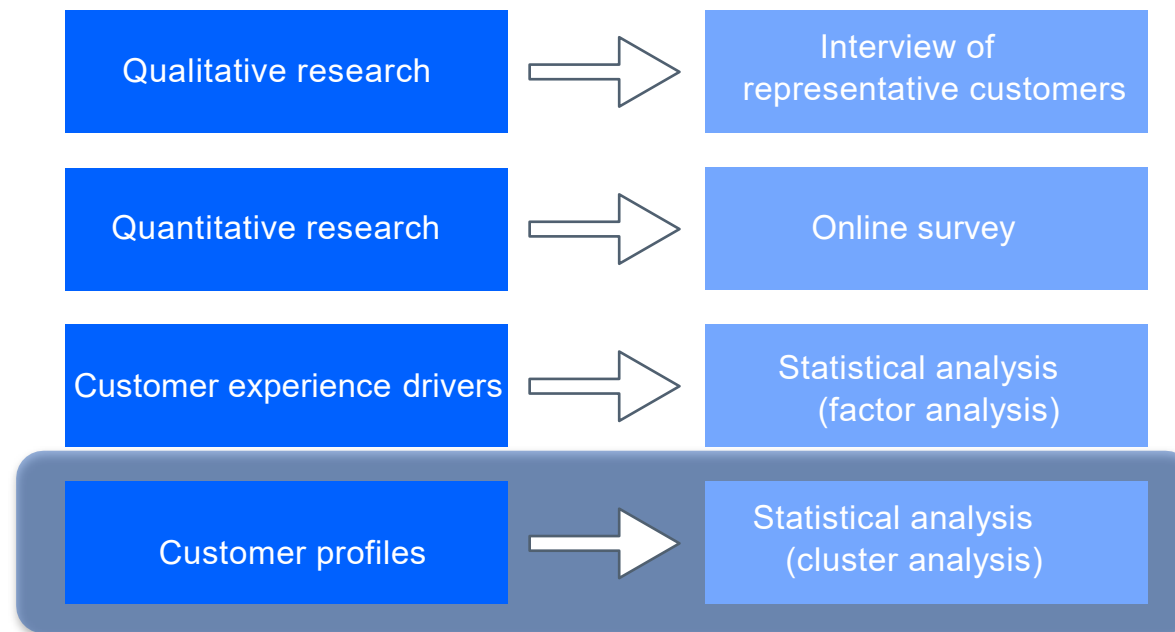
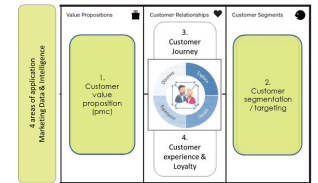
85



Customer Value & Segmentation

Method: determine which PCs are important to which customers

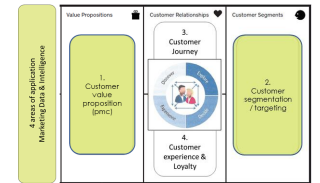
86



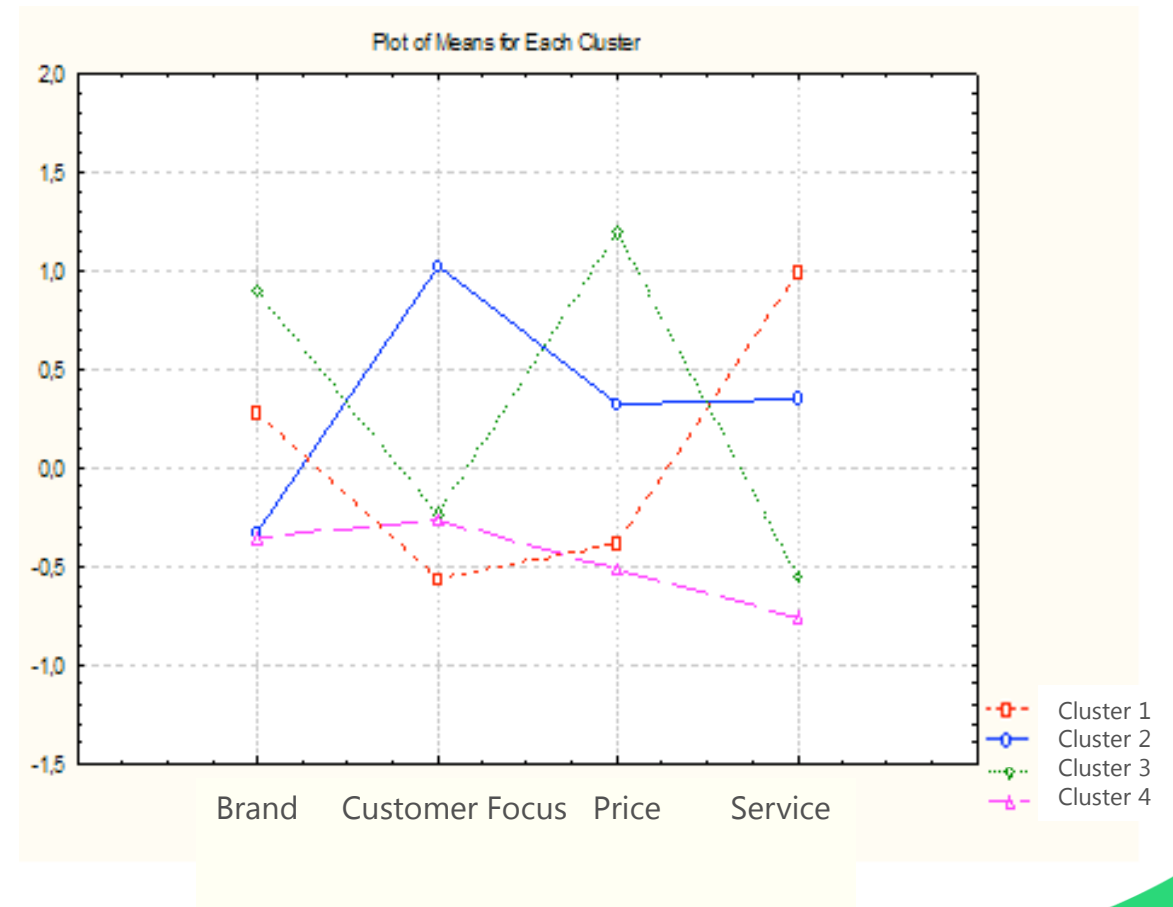
Customer Value & Segmentation

87

Clusteranalyse



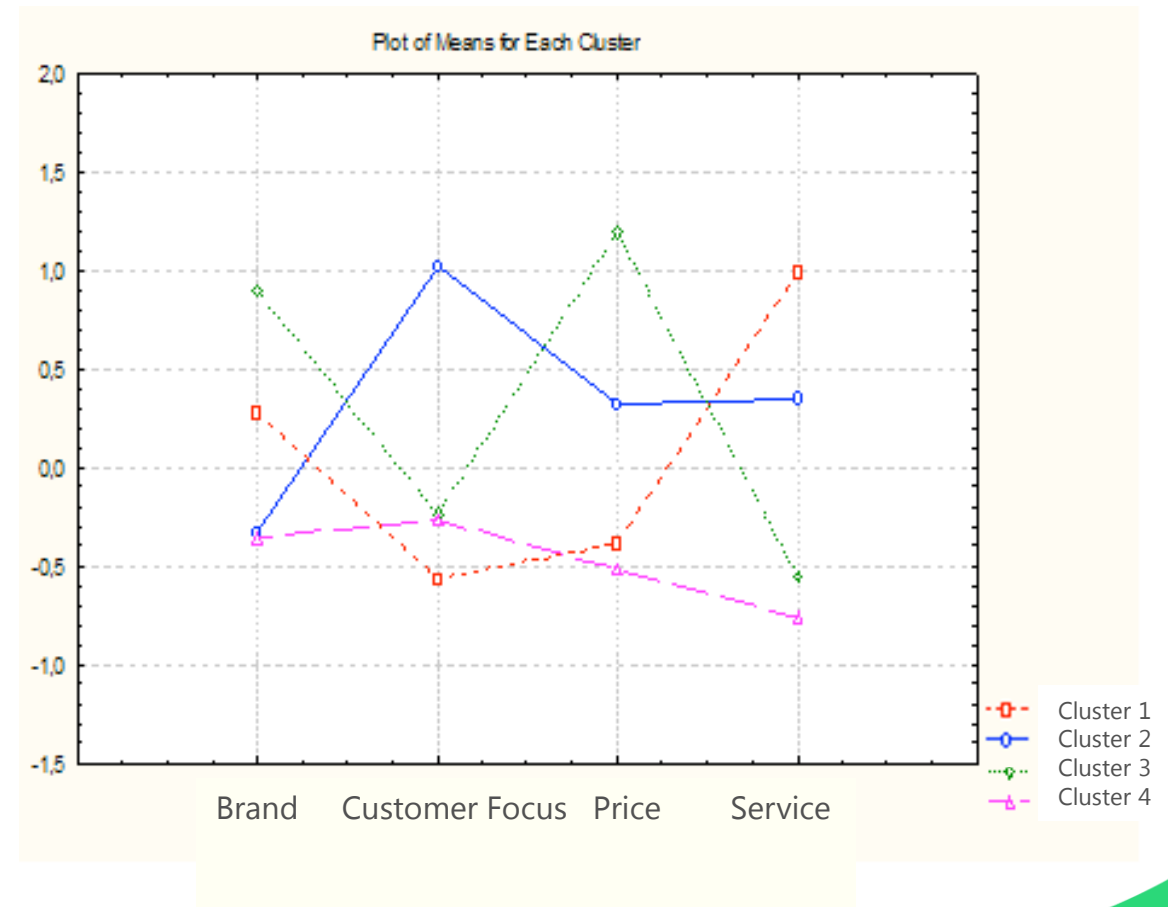
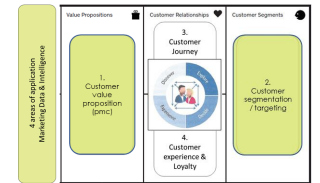
- **Each respondent** ultimately has a score for each of the 4 customer value dimensions (PCs) found.
- Based on these scores, we want to investigate whether we can identify **groups of customers** with very similar characteristics (clusters)



Customer Value & Segmentation

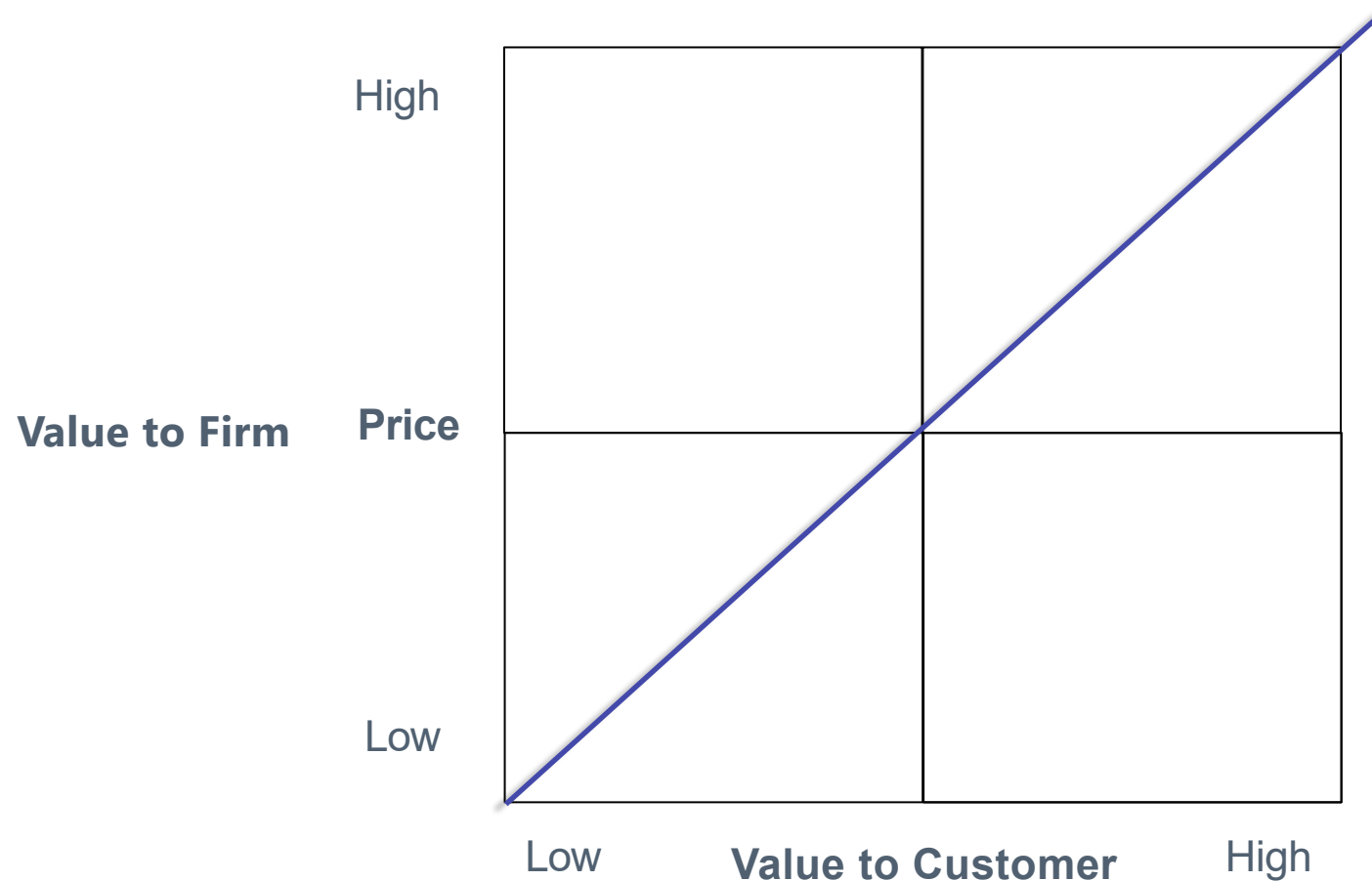
87

Clusteranalyse



Customer Value & Segmentation

Cluster en Customer Value



Cluster 1
Service
20%

Cluster 2
Solution focus
48%

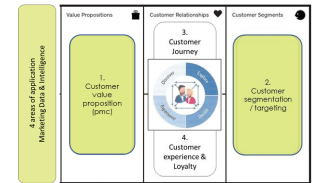
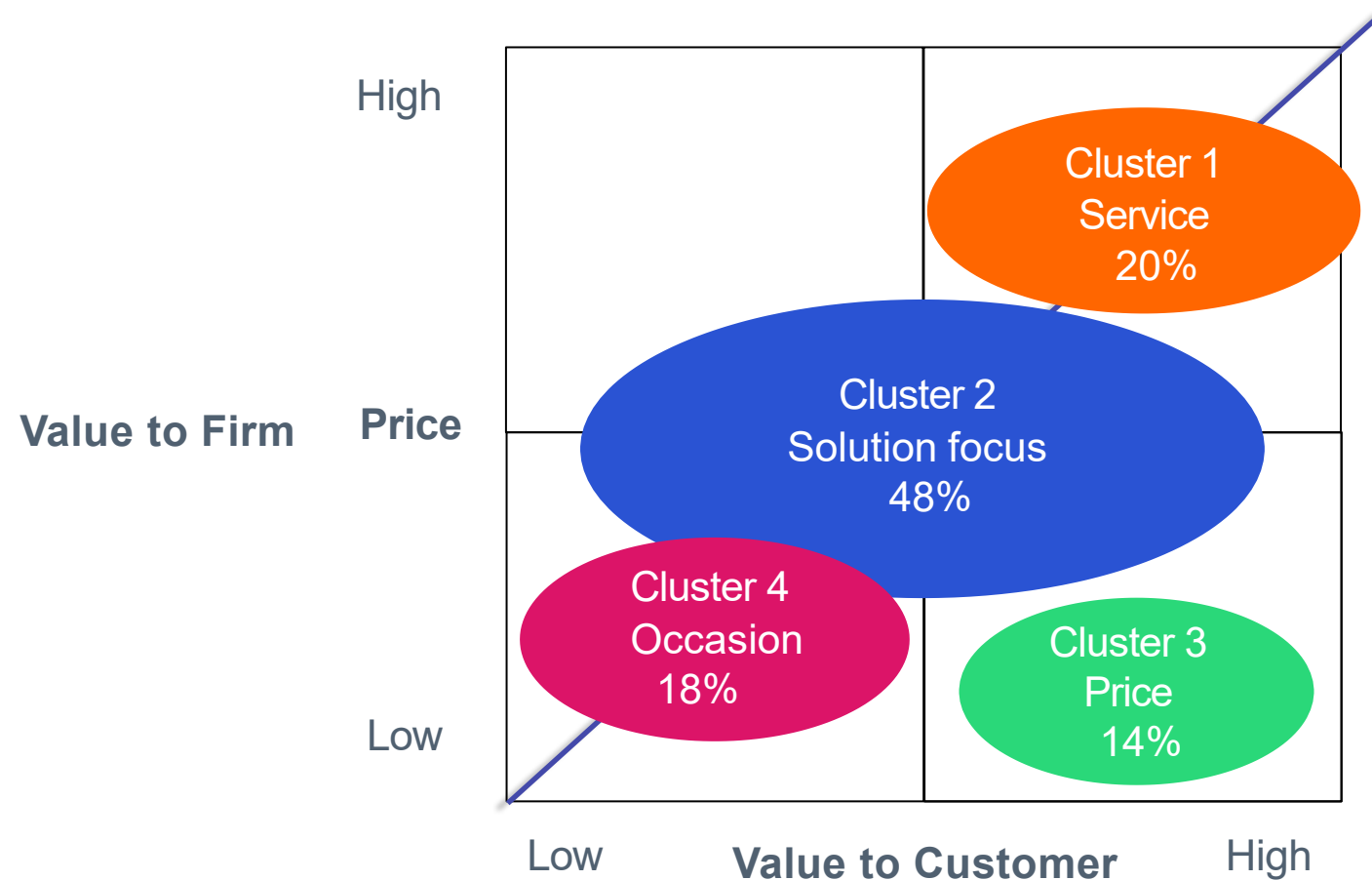
Cluster 3
Price
14%

Cluster 4
Occasion
18%

TASK

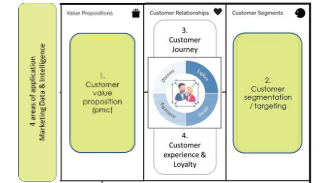
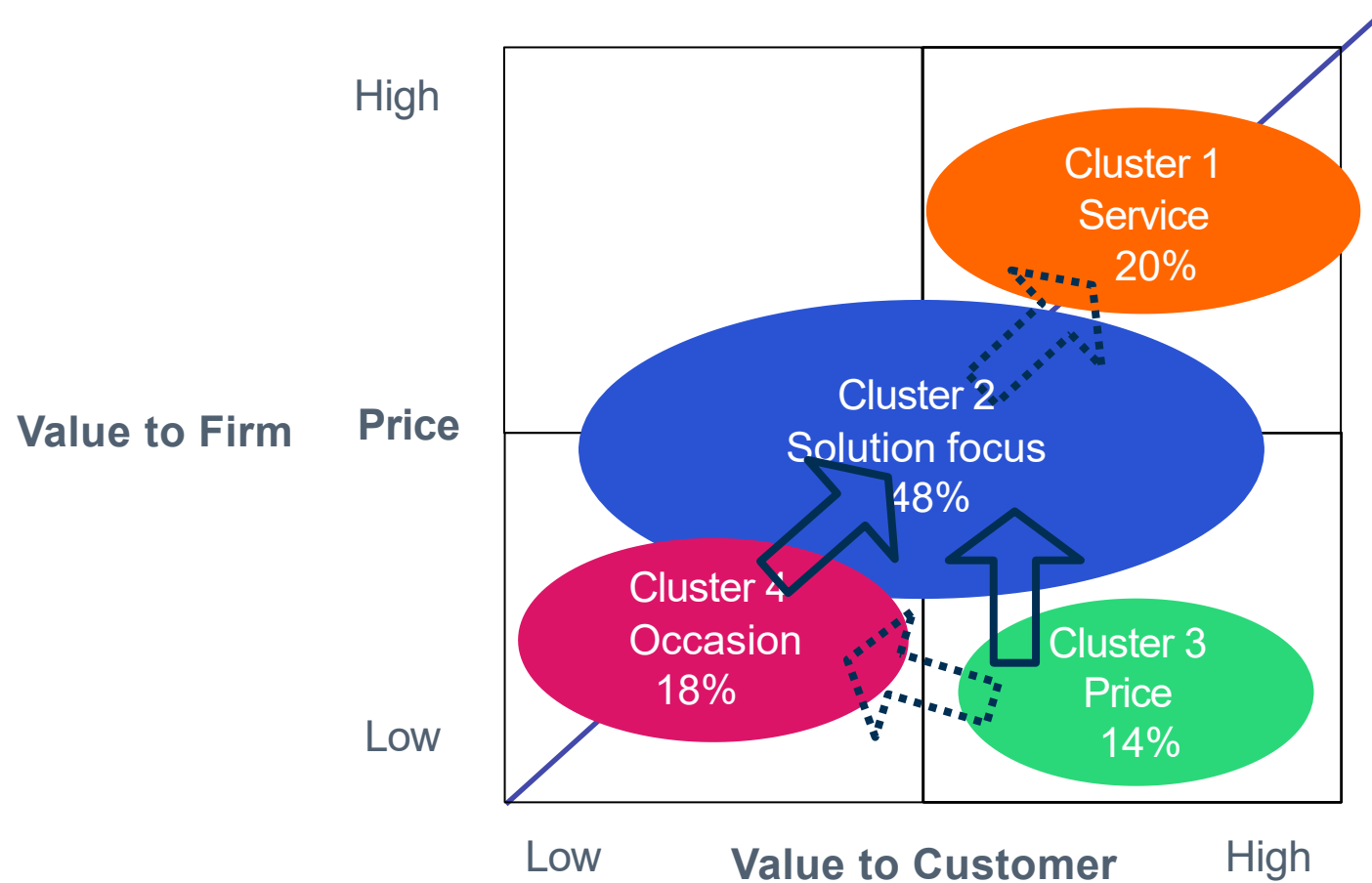
Customer Value & Segmentation

Cluster en Customer Value



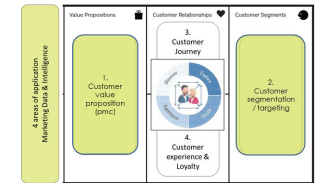
Customer Value & Segmentation

Cluster en Customer Value



Customer Value & Segmentation

Actual Customer Value & Proposition



4 Customer Segment / Service Clusters

- For each cluster 3 go-to-market strategies:
 - Retention strategy
 - Growth strategy
 - Acquisition / Win Back strategy
- Prediction of results including prospects
- Monitoring results of activities

Service concept Value segment	Solution Oriented	Price Oriented	Service Oriented
A. Top Customers	4%		
B. Large Customers	7%	1%	
C. Mid size Customers	17%	18%	4%
D. Small Customers	12%	24%	13%



Predictive Analytics ctd

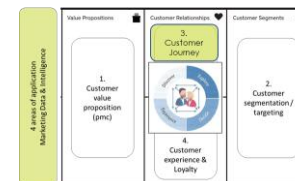
Predictive Reinforcement

- A. Supervised learning
Propensity, classification
- B. Unsupervised learning
Clustering, Principal Component Analysis
- C. Reinforcement learnings**
Collaborative Filtering, recommendation models based on nearest neighbourhood

Customer Journey - Decide

2. C Reinforcement Learning & Collaborative Filtering

What did others read, looked to, bought,.....



Gerelateerde producten voor dit item

Pagina 1 van 35

Gesponsord



JETech Case met Screen Protector Compatibel met Samsung Galaxy Watch 6 40mm, Overall...
★★★★☆ 407
€ 11,99 (€ 6,00/stuk)



Diruite 4-Pack Schermbechermer voor Samsung Galaxy Watch 6/5/4 40 mm Screen...
★★★★☆ 6.324
€ 9,88 (€ 2,47/stuk)



JETech Screen Protector voor Samsung Galaxy Watch 6 44mm, Gehard Glas Film Schermbe...
★★★★☆ 3.408
€ 8,99 (€ 3,00/stuk)



Diruite 4-Pack Schermbechermer voor Samsung Galaxy Watch 6/5/4 44mm Screen...
★★★★☆ 3.431
€ 9,88



Diruite 4-Stuks Screen Protector voor Samsung Galaxy Watch 6 Classic 47mm Schermbe...
★★★★☆ 1.420
€ 9,99



[2 Pack] 20mm Siliconen Bandjes voor Samsung Galaxy Watch 4, Galaxy Watch5/5 Pro Ban...
€ 9,99



Siliconen Vervangende Band Compatibel met Samsung Galaxy Watch 5/4 44mm/40mm...
★★★★☆ 6
€ 6,99

Klanten die dit item hebben bekeken, bekeken ook

Pagina 1 van 3



Smartwatch Samsung Galaxy Watch 6 44mm Silver
★★★★☆ 76
€241,20
Je hebt het al in huis op **Morgen, 2 apr.**
Gratis bezorging door Amazon



Samsung Watch 6 Bluetooth 44MM Graphit
★★★★☆ 159
€269,61
Ontvang het **11 - 15 apr.**
€ 9,99 verzendingskosten
Nog slechts 3 op voorraad.



Samsung Galaxy Watch6 44 mm Digital Touchscreen Silver
€249,99
Je hebt het al in huis op **woensdag, 3 apr.**
Gratis bezorging door Amazon



Smartwatch Samsung Galaxy Watch 6 Classic 43mm Black
★★★★☆ 55
€274,21
Je hebt het al in huis op **woensdag, 3 apr.**
Gratis bezorging door Amazon
Nog slechts 1 op voorraad.



SAMSUNG Smartwatch, standaard
★★★★☆ 23
€304,90
Je hebt het al in huis op **woensdag, 3 apr.**
Gratis bezorging door Amazon



Samsung Galaxy Watch6 Classic 47 mm Numérique Écran tactile 4G Argent
★★★★☆ 11
€333,20
Ontvang het **10 - 13 apr.**
€ 13,38 verzendingskosten
Wordt gewoonlijk verzond...

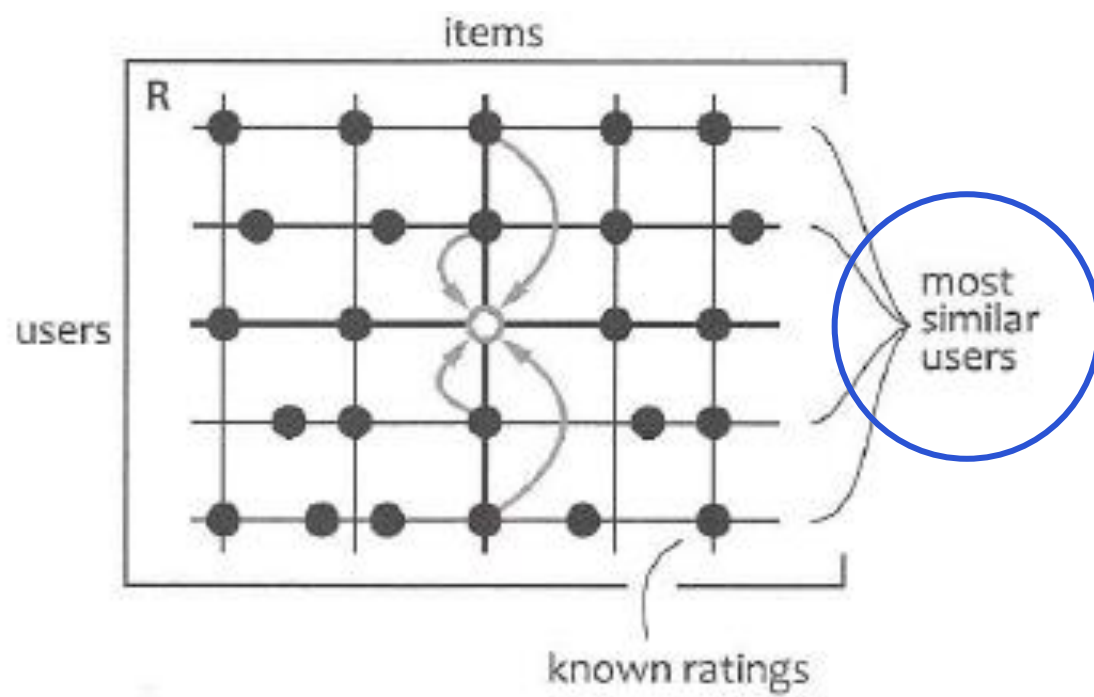


Samsung Galaxy Watch 6 Bluetooth WLAN GPS 40MM Graphit
★★★★☆ 79
€248,96
Je hebt het al in huis op **woensdag, 3 apr.**
Gratis bezorging door Amazon
Nog slechts 2 op voorraad.

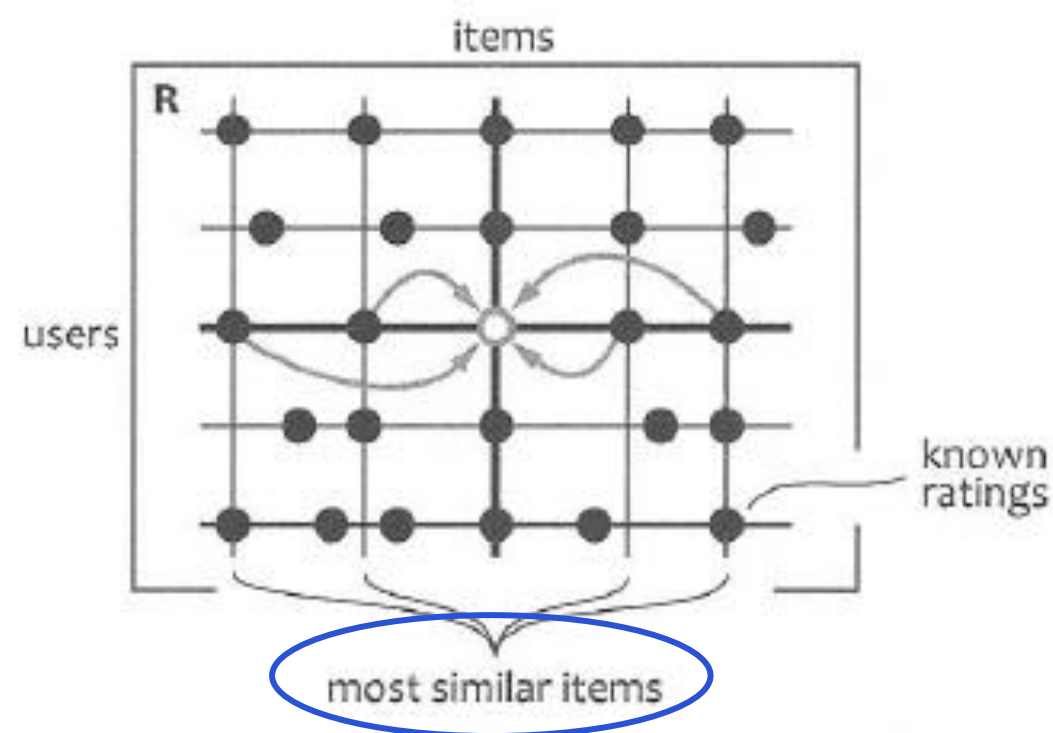
Customer Journey - Decide

2. C Reinforcement Learning & Collaborative Filtering

Predictive
Reinforcement



User based



Item based

Customer Journey - Decide

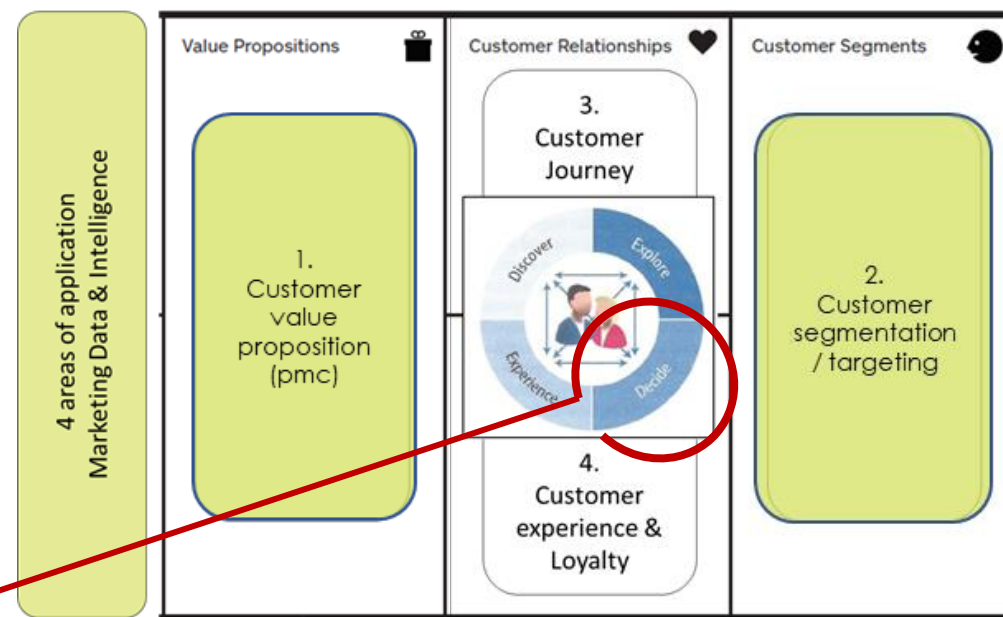
2. C Reinforcement Learning & Collaborative Filtering

Predictive
Reinforcement

CF models can recommend:

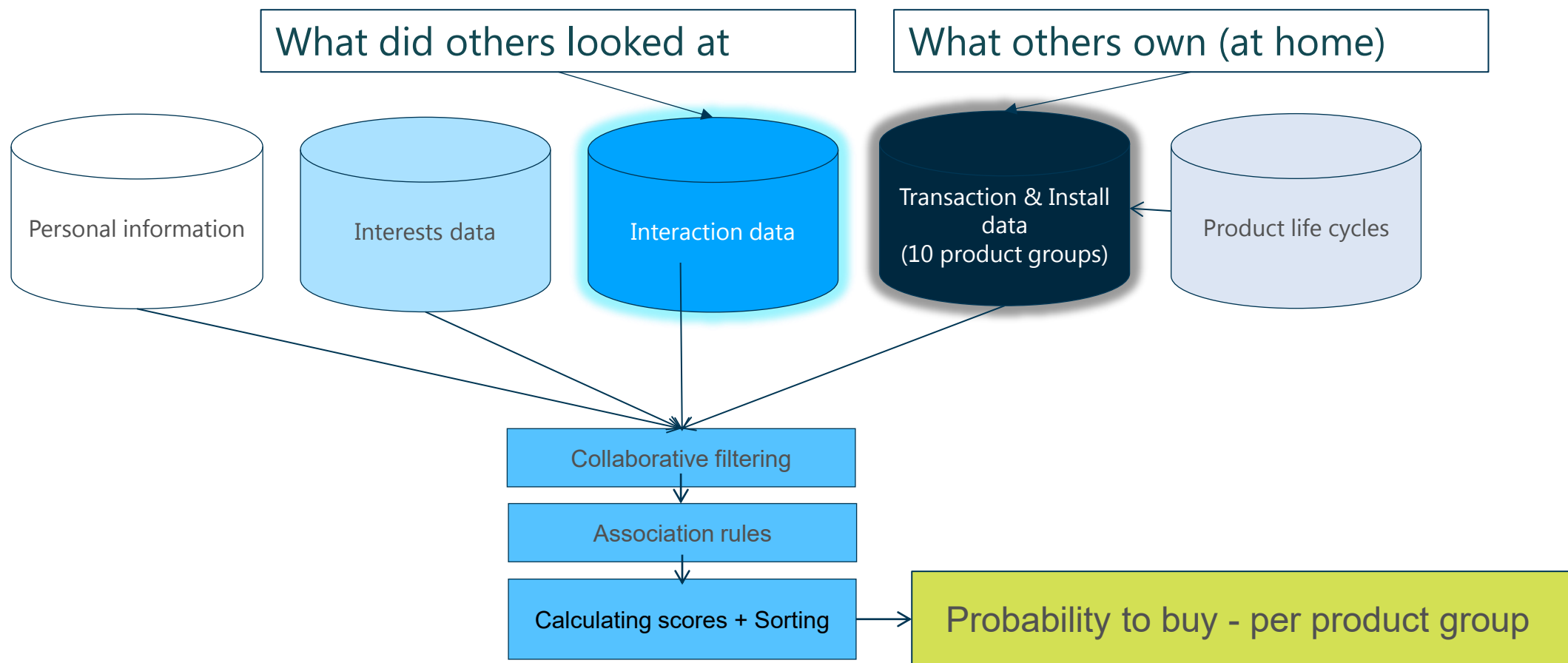
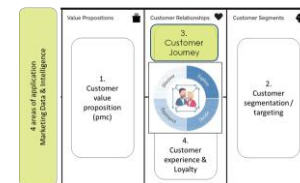
- products bought,
- products looked at,
- services,
- content

Use recommenders in the 'decide phase' of the buying process.



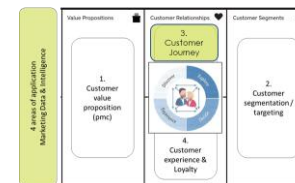
Case Customer Journey - Decide

'Best offer' based on 2 integrated recommenders



Case Customer Journey - Decide

Next Best Offer



Based on: 1. Probability to buy score

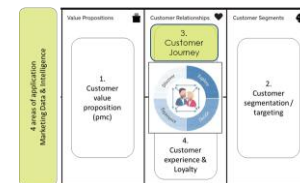
Customer_ID	Highest probability to buy	Probability highest score
101	Digital TV	0,84
102	Cooking	0,95
103	Smartphone	0,65
104	Wearable	0,45
105	Cool/Freeze	0,65

Case Customer Journey - Decide

Next Best Offer

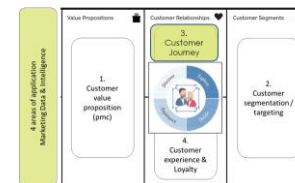
Based on: 1. Probability to buy score
2. Customer Value

Customer_ID	Highest probability to buy	Probability highest score	Customer Value
101	Digital TV	0,84	100
102	Cooking	0,95	60
103	Smartphone	0,65	30
104	Wearable	0,45	60
105	Cool/Freeze	0,65	85



Case Customer Journey - Decide

Next Best Offer



Based on:

1. Probability to buy score
2. Customer Value
3. Potential Customer Value



Next Best Offer
for every customer

Customer_ID	Highest probability to buy	Probability highest score	Customer Value	Potential Customer Value
101	Digital TV	0,84	100	240
102	Cooking	0,95	60	70
103	Smartphone	0,65	30	320
104	Wearable	0,45	60	240
105	Cool/Freeze	0,65	85	140

Case Customer Journey - Decide

Next Best Offer

Based on:

1. Probability to buy score
2. Customer Value
3. Potential Customer Value



Next Best Offer
for every customer

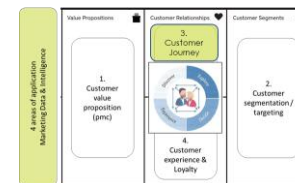


Next Best Action
for every customer

Customer_ID	Highest probability to buy	Probability highest score	Customer Value	Potential Customer Value	Proposed action
101	Digital TV	0,84	100	240	
102	Cooking	0,95	60	70	Bring to attention
103	Smartphone	0,65	30	320	Bring to attention & offer
104	Wearable	0,45	60	240	No direct action yet
105	Cool/Freeze	0,65	85	140	

Case Customer Journey - Decide

Next Best Offer



Based on:

1. Probability to buy score
2. Customer Value
3. Potential Customer Value



Next Best Offer
for every customer

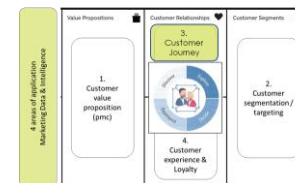


Next Best Action
for every customer

Customer_ID	Highest probability to buy	Probability highest score	Customer Value	Potential Customer Value	Proposed action
101	Digital TV	0,84	100	240	Bring to attention & offer
102	Cooking	0,95	60	70	Bring to attention
103	Smartphone	0,65	30	320	Bring to attention & offer
104	Wearable	0,45	60	240	No direct action yet
105	Cool/Freeze	0,65	85	140	No direct action yet

Case Customer Journey - Decide

Results Best offer & Potential Customer Value



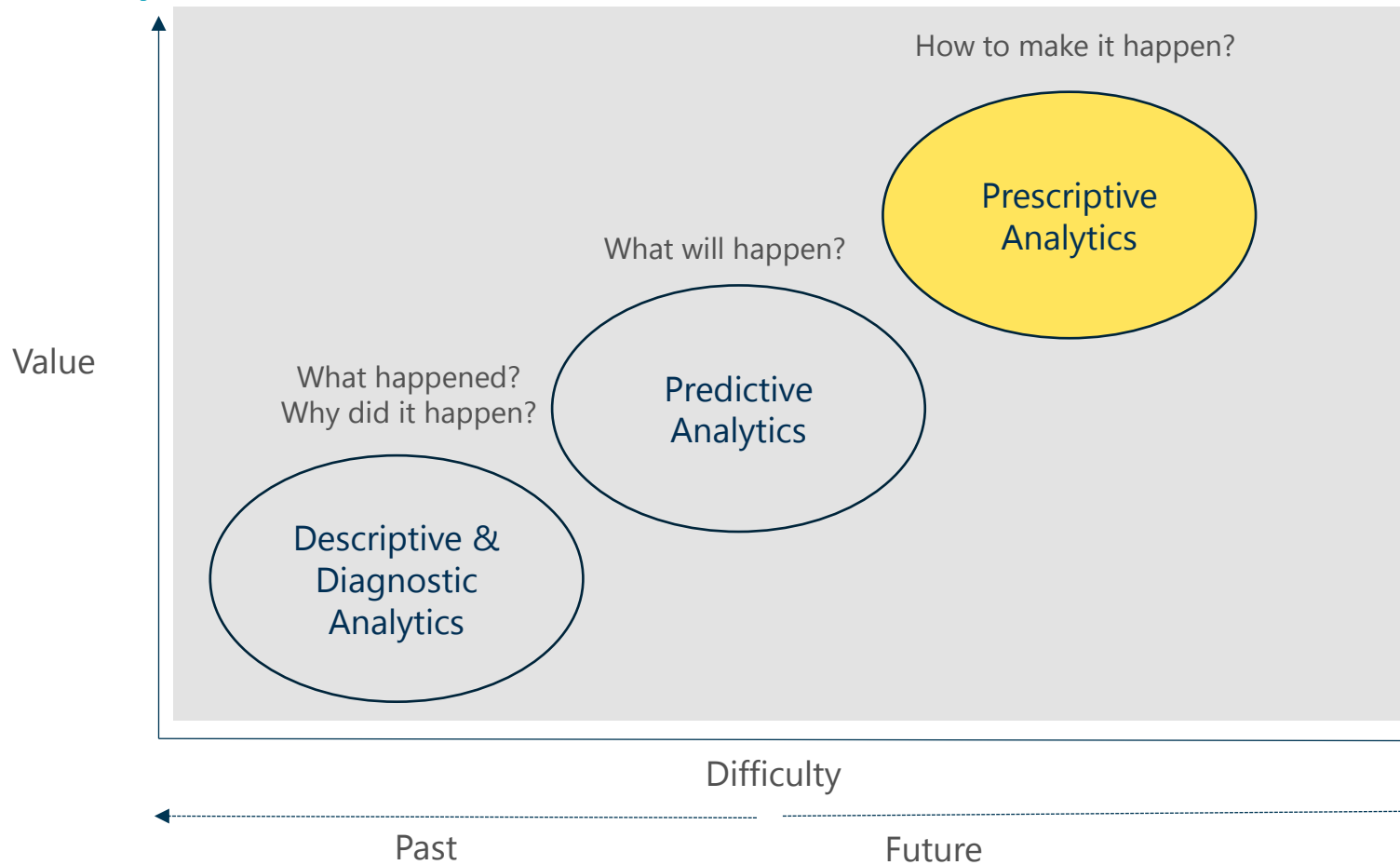
MAIL CAMPAIGN - TABLET						
	GENERIC MAIL NOT TARGETED		TARGETED MAIL RECOMMENDER GROUP		DIFFERENCE RECO GROUP vs TOTAL	
		% of total		% of total	numbers	percentages
TOTAL	242.348	100,0%	34.592	100,0%	14,3%	100,0%
UNIQUE VIEWS	52.645	21,7%	11.292	32,6%	21,4%	150,3%
UNIQUE CLICKS	3.764	1,6%	3.024	8,7%	80,3%	562,9%
CONVERSION	466	0,2%	722	2,1%	154,9%	1085,5%

Results:

1. Conversion rate almost 11x higher compared to generic approach
2. Absolutely 55% more sales
3. More than 200,000 customers NOT bothered with most likely irrelevant message

Development Data Driven Models

Levels Data Analytics





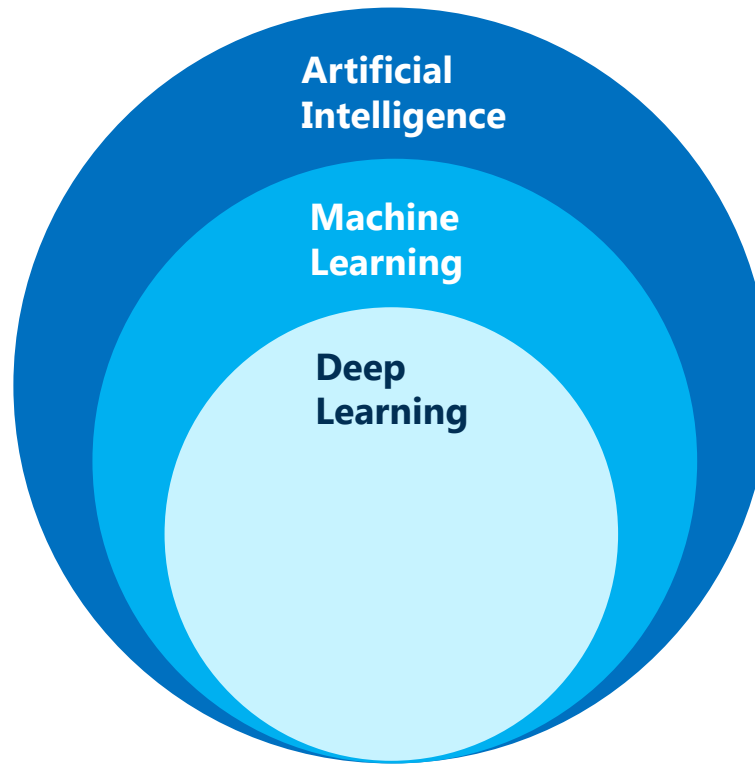
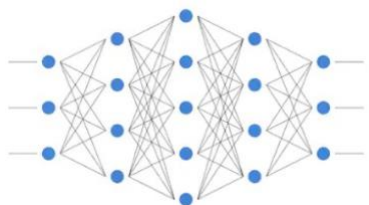
Prescriptive Analytics

AI, ML & DL

AI is an umbrella term for all automated repetitive decisions and actions made by machines, that mimic human decision-making and creation processes.

ML is the practice of using algorithms to parse data, learn from it through feedback, and then make a determination or prediction about something in the world.

DL is a subset of machine learning that uses neural networks





AI impact on marketing: 7 areas (*)

Inhoud creëren (72%) (**)

- Geautomatiseerd schrijven
- Visuele en videocontent

Gegevensanalyse (67%)

- Sentimentanalyse
- A/B testen

Adverteren (65%)

- Programmatische reclame

(*) Igor Nowé, Marketing, Data Strategy, 2025, chap 5

(**) NielsenIQ, CMO outlook 2025, est savings 5% to 15%, \$ 500 mld savings)

AI impact on marketing: 7 areas (*)

Personalisatie (**)

Targeting

- Customer Segmentation
- Gepersonaliseerde content

Voorspellende analyses

- Vraagvoorspelling
- Klantgedrag

Automatisering

- Chatbots en virtuele assistenten
- Geautomatiseerde campagnes

Klantbeleving

- Verbeterde klantinteracties
- Loyaliteitsprogramma's

(*) Igor Nowé, Marketing, Data Strategy, 2025, chap 5

(**) Jean-Pierre Thomassen, AI personalisatie strategie, 2025

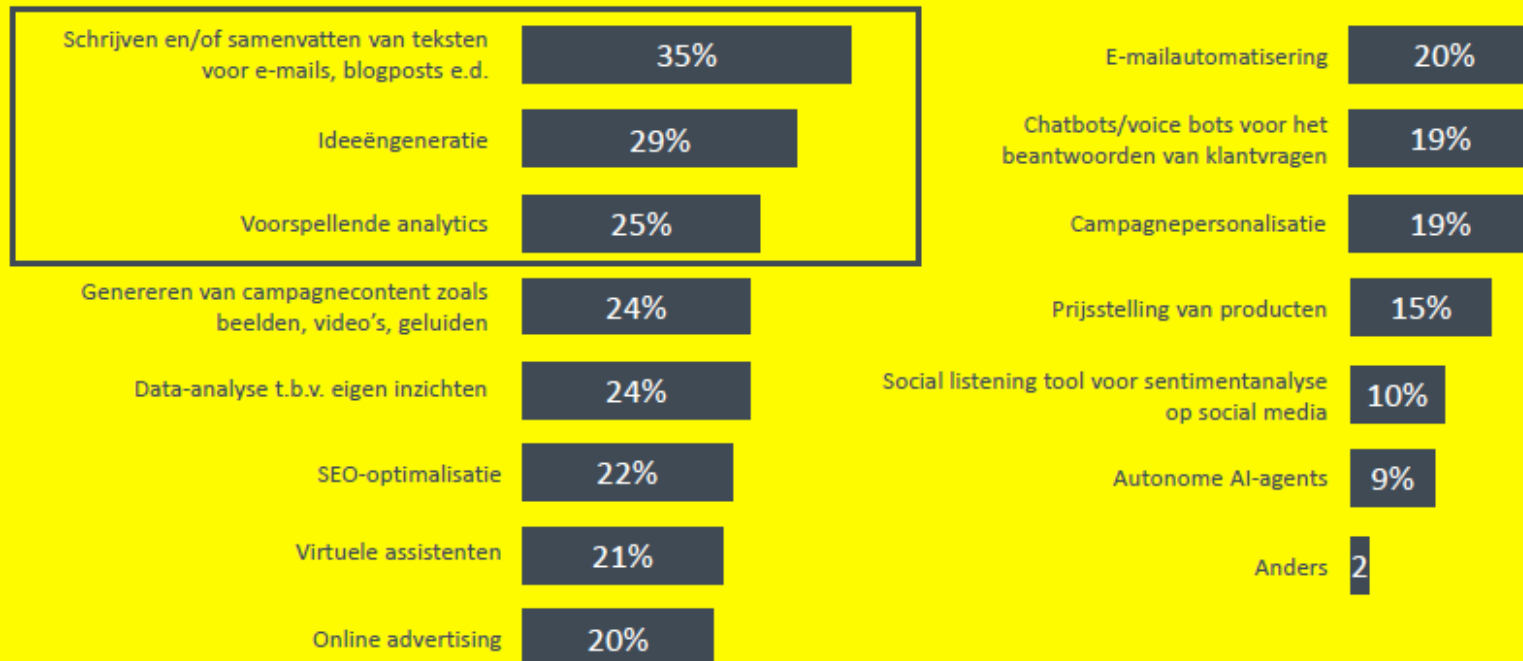


AI in marketing 2026

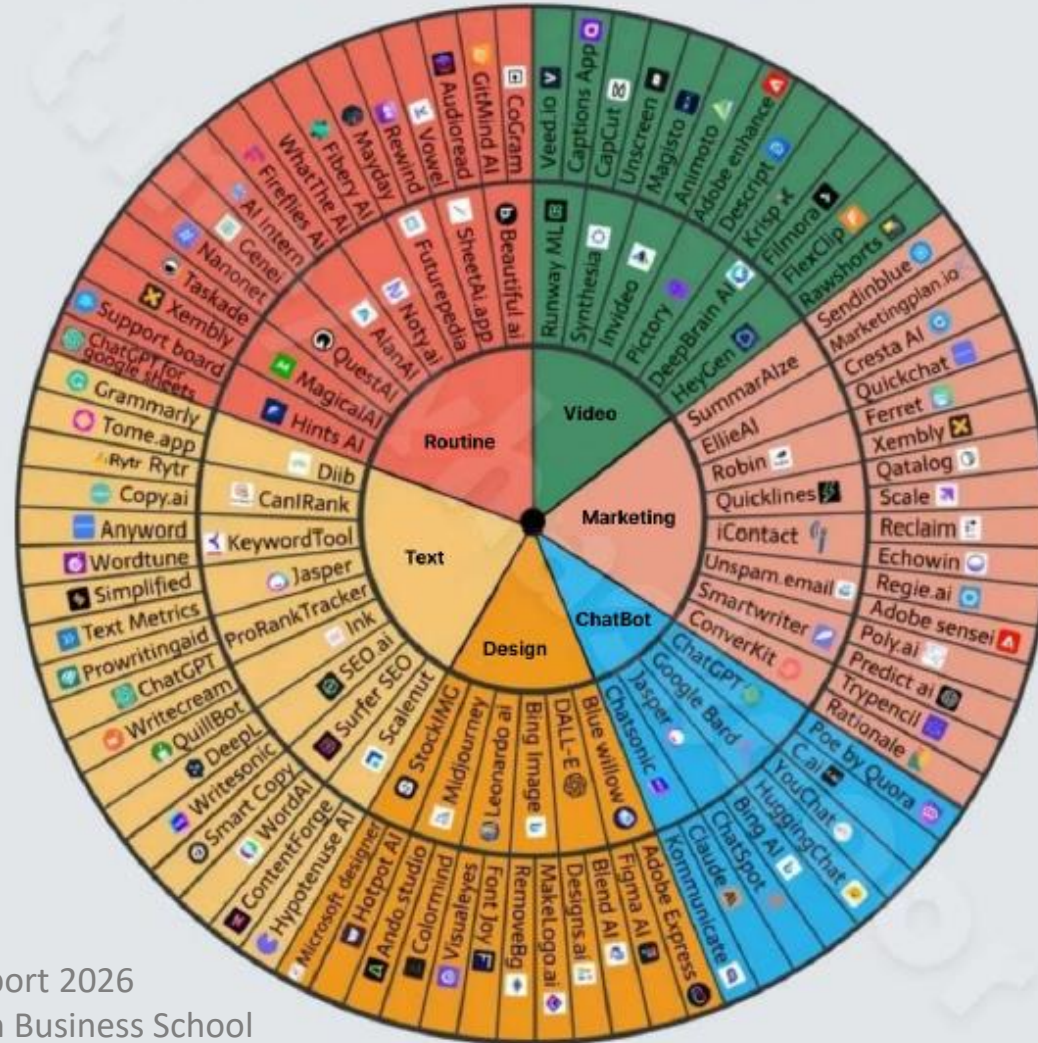
Voor welke marketingtaken willen organisaties AI in 2026 meer inzetten?

DDMA

TOP 3

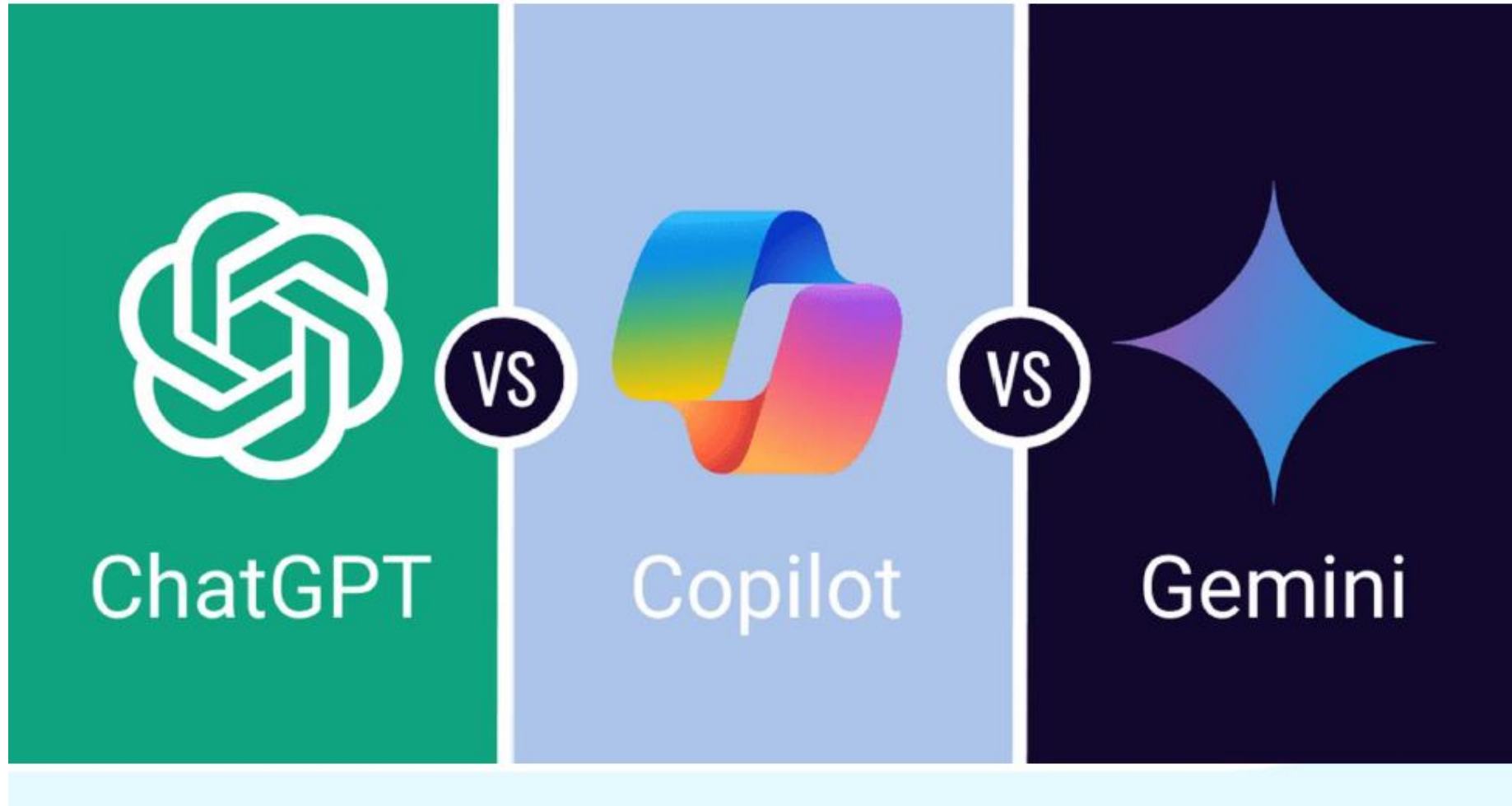


AI tools today



Source: AI & DM trendrapport 2026
Bart Meerdink -Beeckestijn Business School

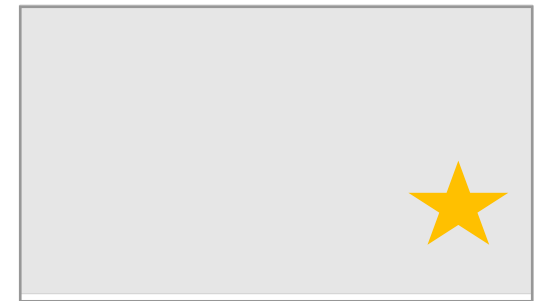
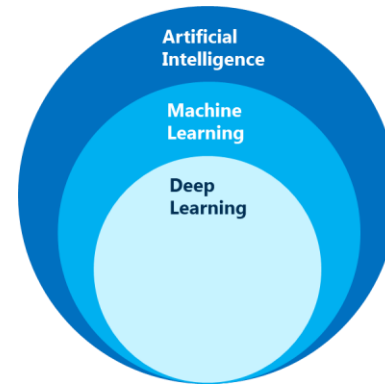
Central AI adoption



Source: AI & DM trendrapport 2026 - Bart Meerdink -Beeckestijn Business School

Prescriptive Analytics

- Modelling of the dependency between decisions and future outcomes for optimal decision making
- In programmatic applications, where automated decision making is key, the main focus is on prescriptive analytics, which in turn, uses predictive models as building blocks



Loyalty - Churn Reduction case

Peugeot churn reduction campaign

Based on prescriptive modeling

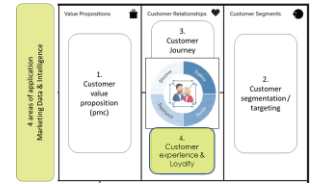


CASE PEUGEOT



Goal:

Increase repurchases Peugeot cars of existing Peugeot Owners
(new and/or top occasions)



CASE PEUGEOT

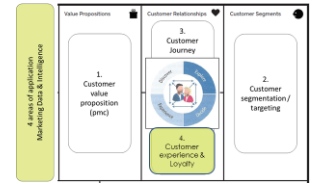


Goal:

Increase repurchases Peugeot cars of existing Peugeot Owners
(new and/or top occasions)

Questioning:

Reduce the churn with current Peugeot owners
(new or top occasion)
from 62% with at least 10 percent points (branch average)
in 1 year



CASE PEUGEOT

Questioning:

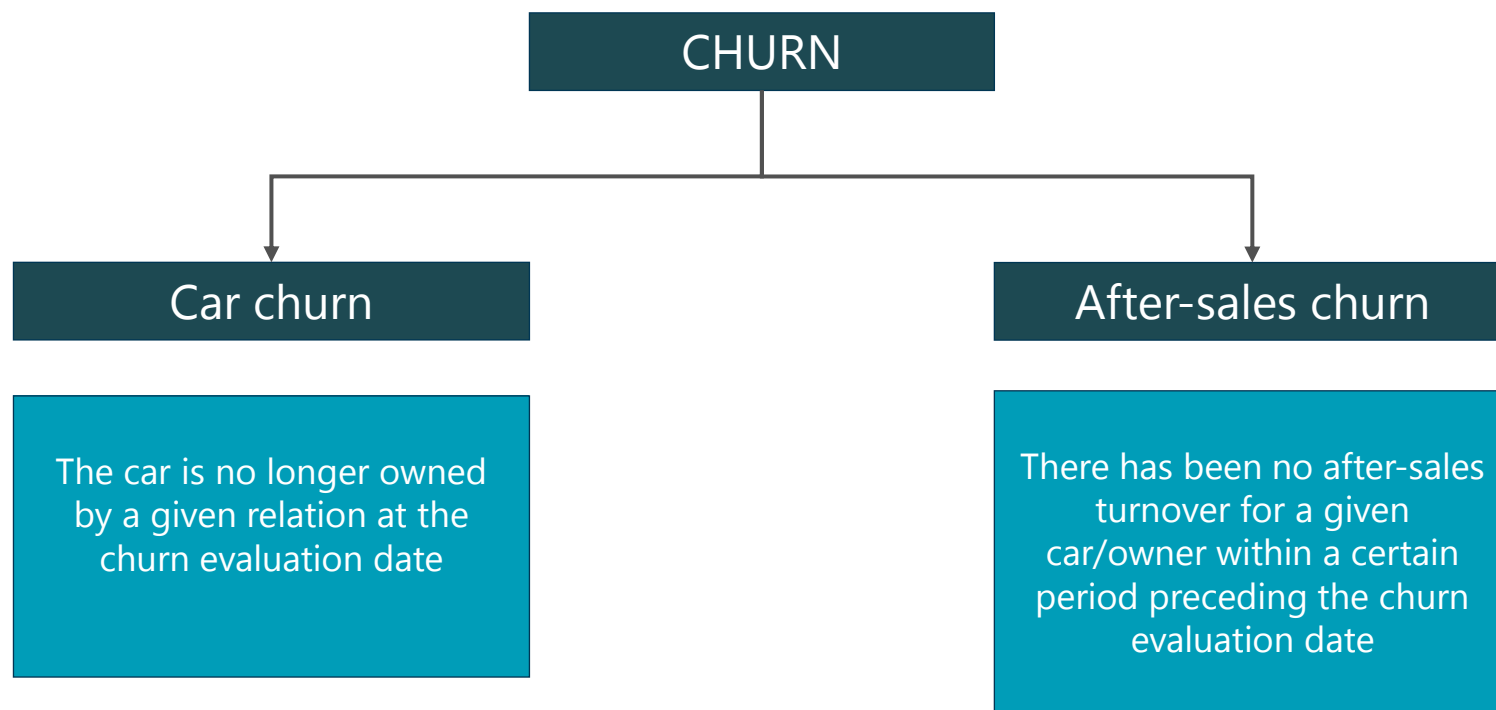
Reduce the churn with current Peugeot owners (new or top occasion) from 62% with at least 10 percent points (branch average) in 1 year

Strategy:

- Predict Peugeot drivers with **highest churn probability** using predictive analytics and
- Predict the **most likely reason** for the specific Peugeot driver.
- **Communicate** with the targeted Peugeot drivers the 'most likely offer' based on the churn reason and other information that is available and relevant.

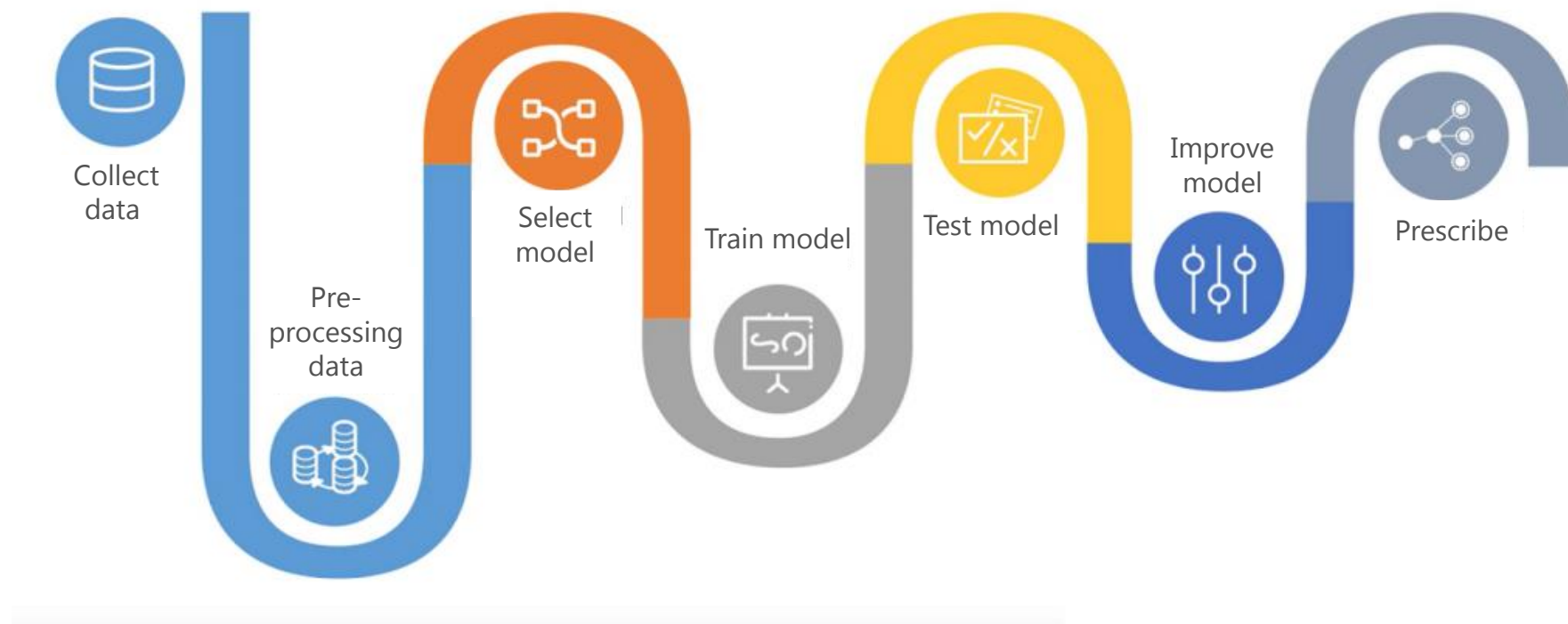


CASE PEUGEOT



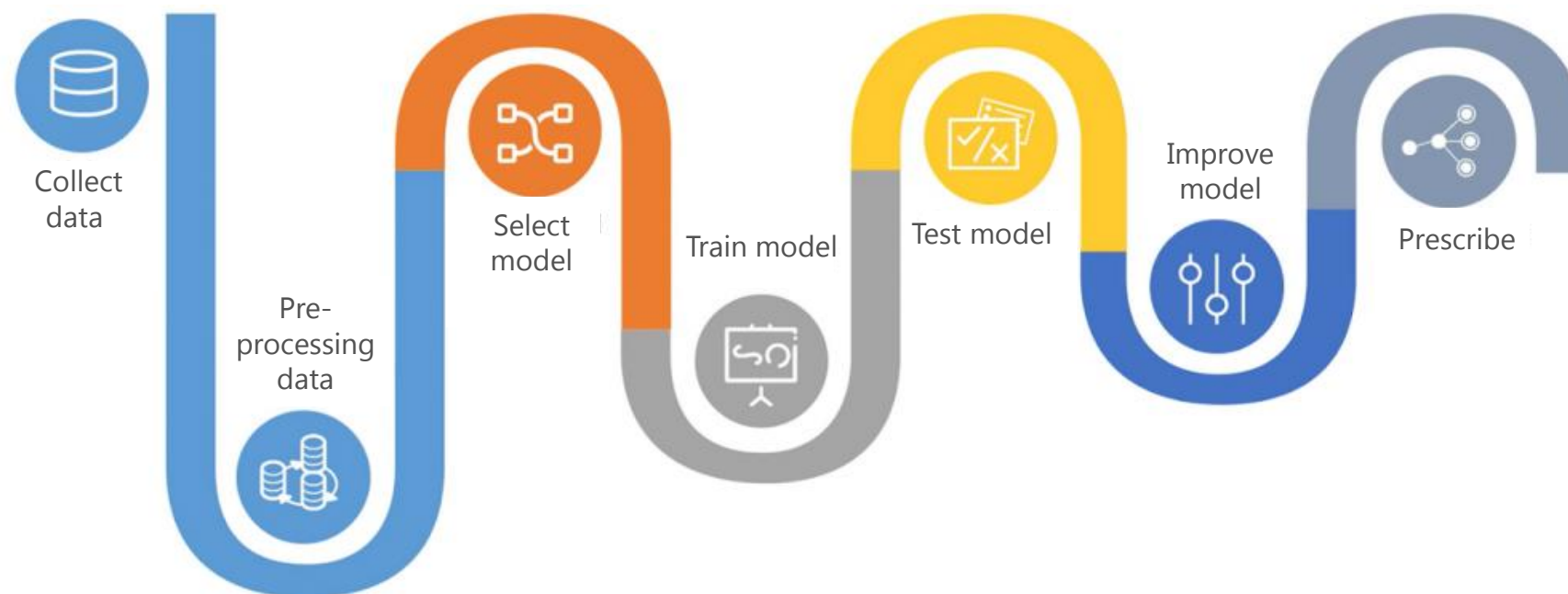
Loyalty - Churn Reduction case

Car churn model process overview - 7 steps machine learning



Loyalty - Churn Reduction case

Car churn model process overview



BRAND DATA

DEALER DATA

Loyalty - Churn Reduction case

Independent variables

- "Directly" available data:
 - Customer type: P (individual), Z (business), or other (trade/undefined)
 - Fuel type: G, H, A, B, C, X, D, E, other
 - Car type: F, A,V, M, C, D, P, other
 - Engine power
 - Cylinder volume
 - Number of cylinders

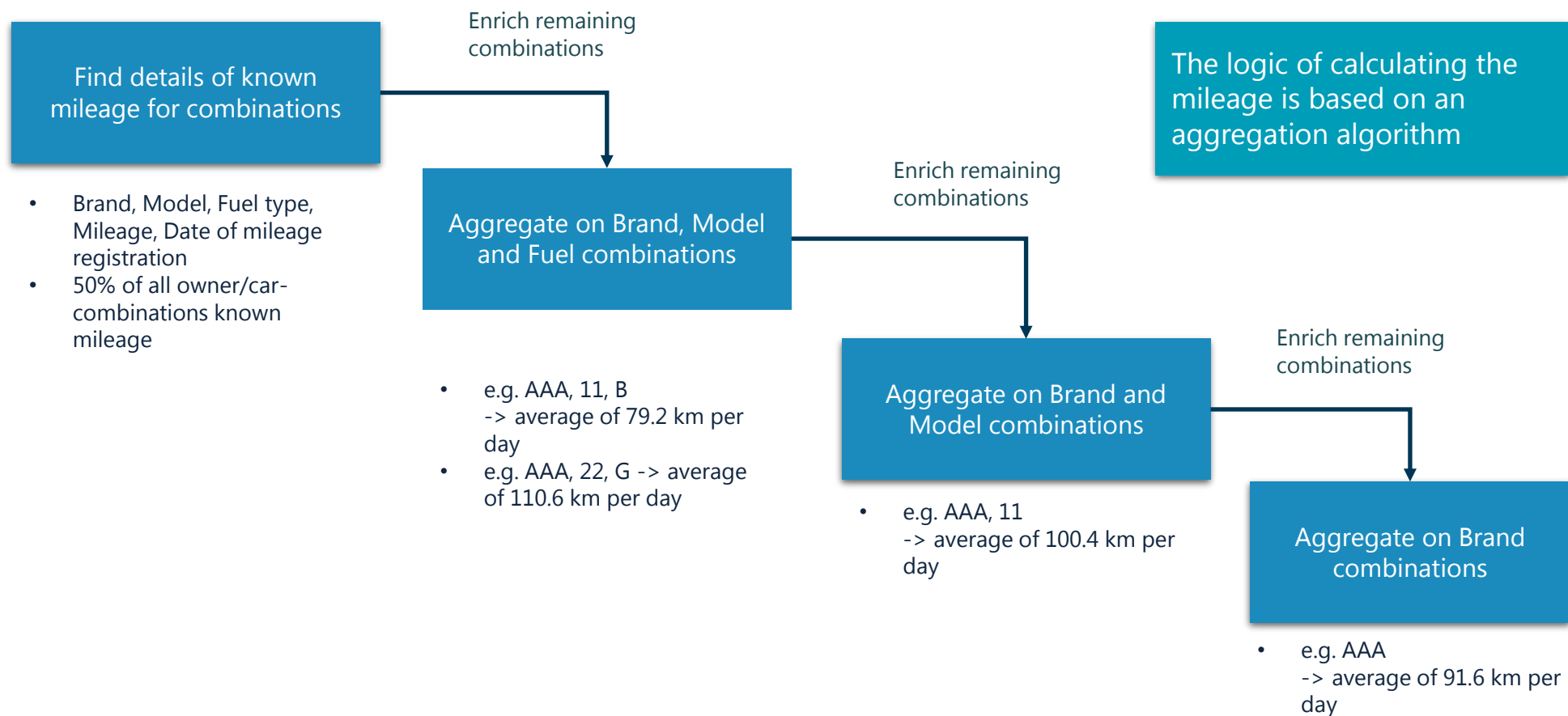
Loyalty - Churn Reduction case

Independent variables

- Data available from CV-model:
 - Construction year
 - Number of dealer activities in a given period (deduped)
 - Total income from dealer activities in a given period (deduped)
 - Recency of the last dealer activity in a given period
- Calculated data based on source data:
 - Duration of the car ownership
 - Average mileage per day
 - Estimated mileage run during ownership of the car
 - Estimated total mileage at the end of the ownership of a car

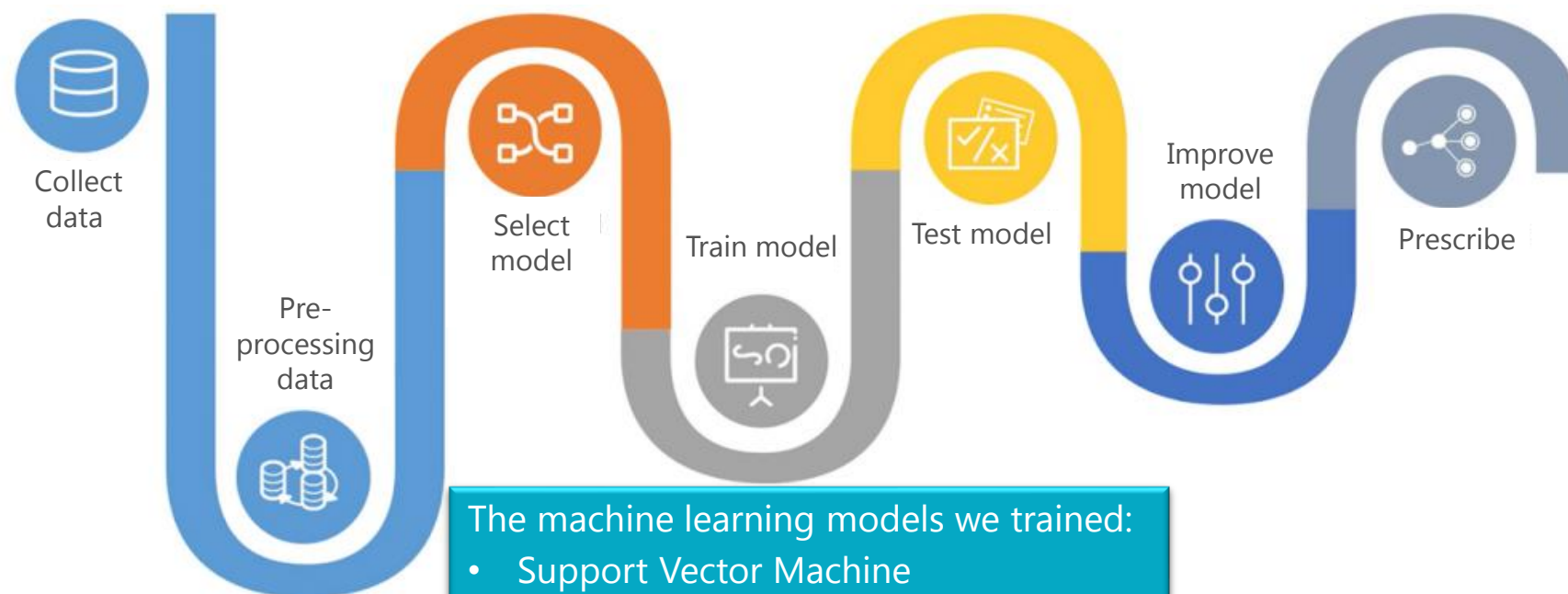
Loyalty - Churn Reduction case

Treatment of missing data (in big data environment)



Loyalty - Churn Reduction case

Car churn model process overview



The machine learning models we trained:

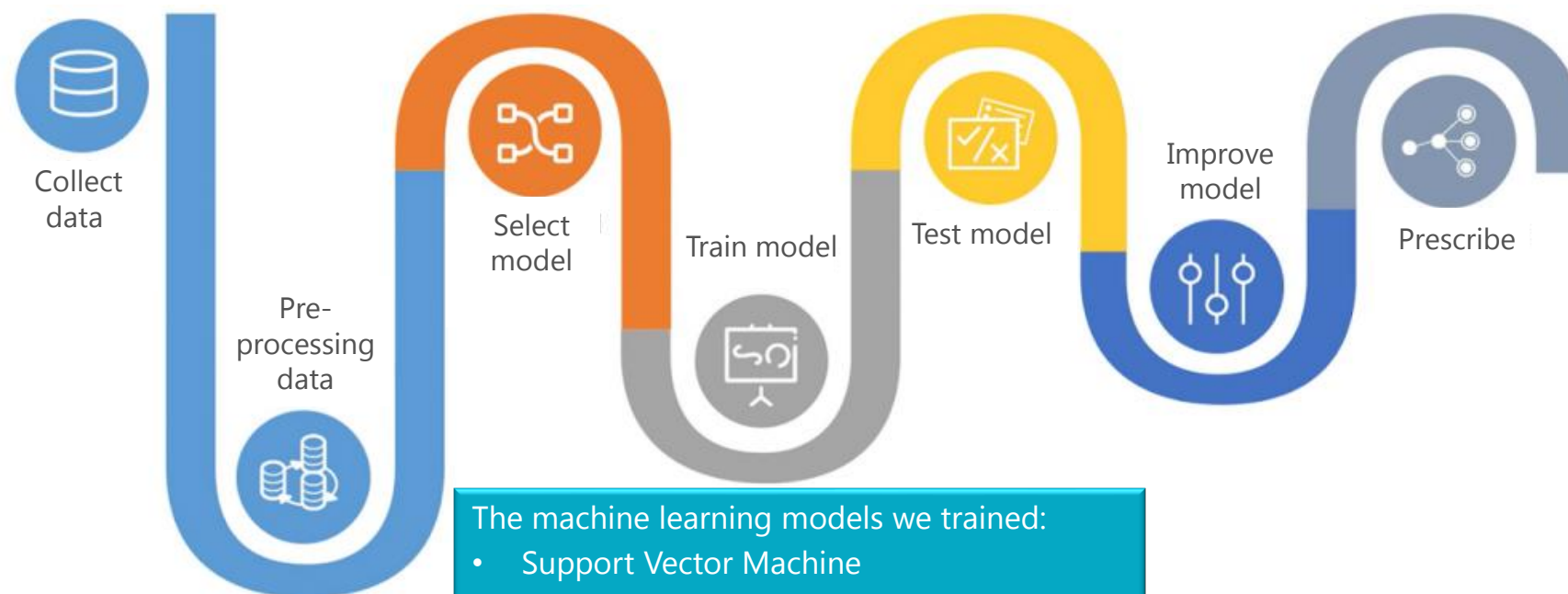
- Support Vector Machine
- Random Forest
- Gradient Boosted Tree

BRAND DATA

DEALER DATA

Loyalty - Churn Reduction case

Car churn model process overview



The machine learning models we trained:

- Support Vector Machine
- Random Forest
- Gradient Boosted Tree



Loyalty - Churn Reduction case

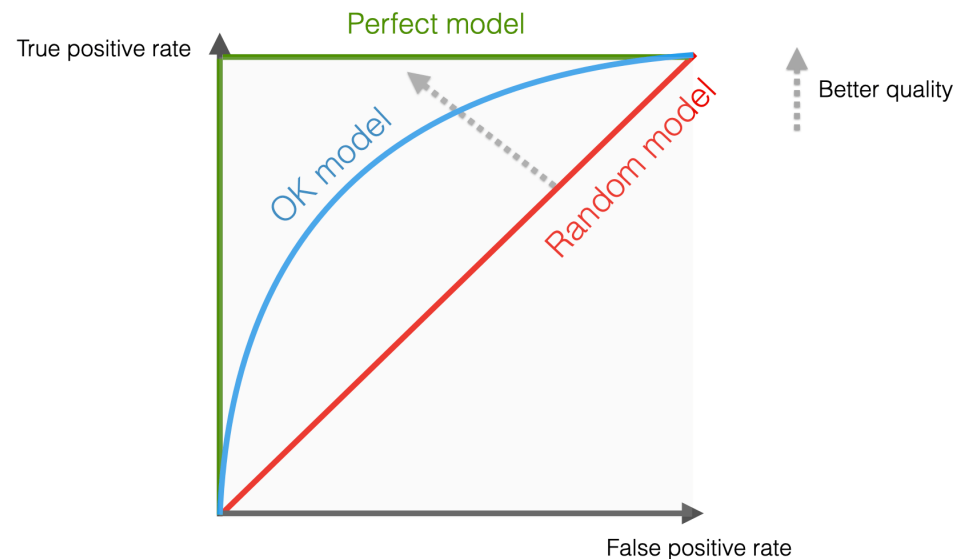
Building the model & test model

The machine learning models we trained:

- Support Vector Machine
- Random Forest
- Gradient Boosted Tree



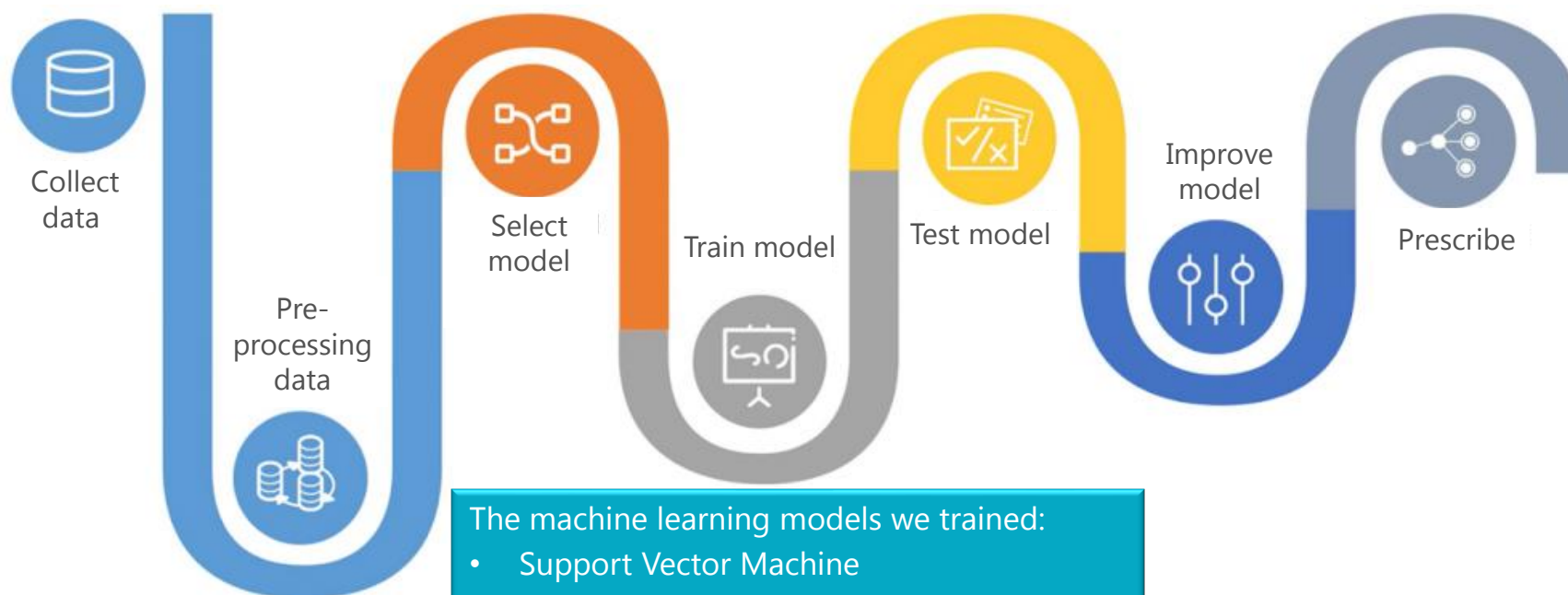
SVM Best Result!
Area under ROC:
Car churn = 86.1%



		Predicted		
		Spam	Not	
Actual	Spam	600 (TP)	300 (FN)	$\rightarrow \text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}$
	Not	100 (FP)	9000 (TN)	$\rightarrow \text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}$

Loyalty - Churn Reduction case

Car churn model process overview



The machine learning models we trained:

- Support Vector Machine
- Random Forest
- Gradient Boosted Tree

BRAND DATA

DEALER DATA

Train Data

Test Data

- Car churn
 - Evaluation period 3 years
 - Churn prediction period 3 months

Loyalty - Churn Reduction case

Model output

Churn score

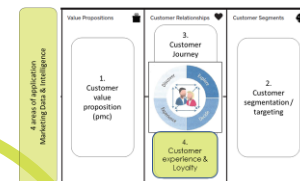
For each customer:

1. Churn probability score
2. Most important reason causing churn



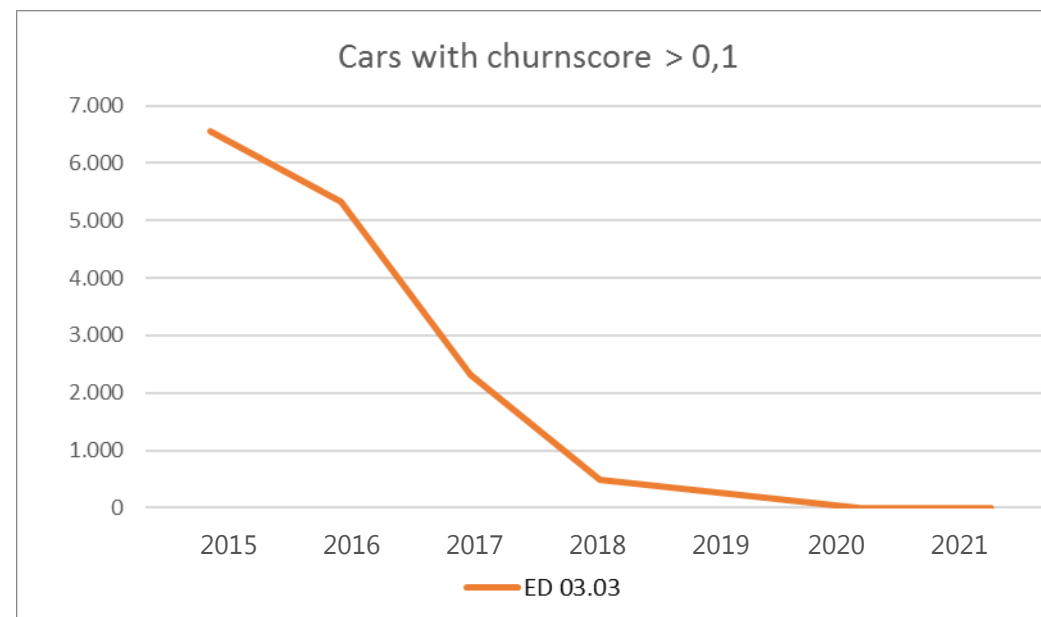
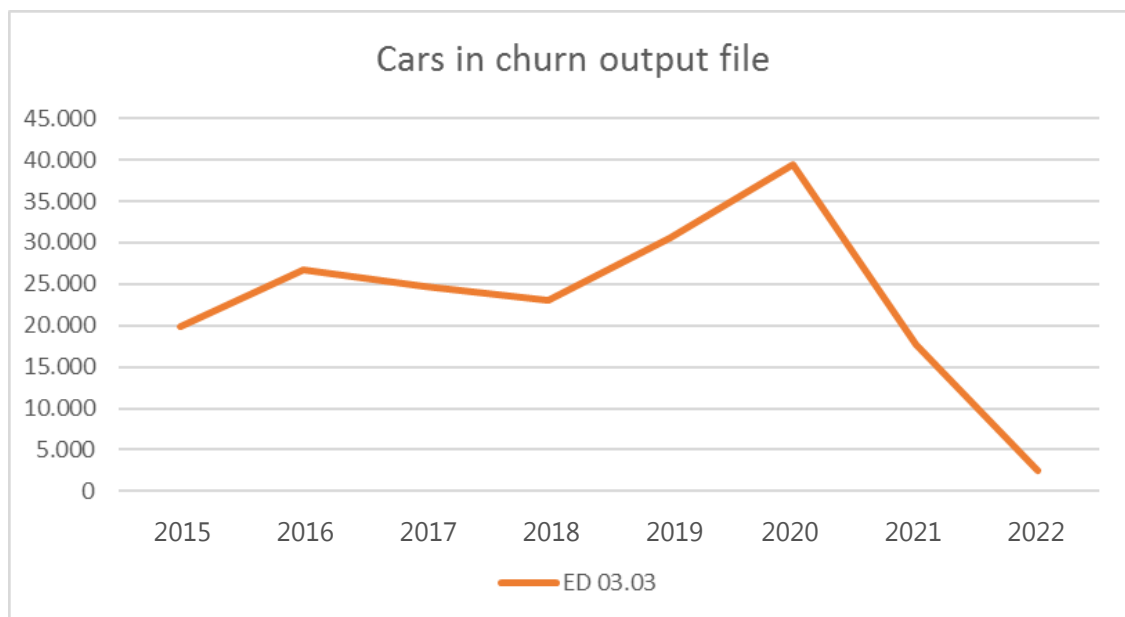
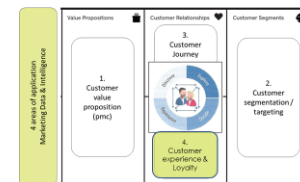
Personal offer
to targeted customer

	IdRelatie	Id_Car	Evaluatie_Dat...	Churn_Datum	Churn_Score
1	3732744	1437253	2017/02/07	2017/05/07	4,087702
2	3450679	1598959	2017/02/07	2017/05/07	3,826379
3	3653504	1988588	2017/02/07	2017/05/07	3,751895
4	3402204	1710653	2017/02/07	2017/05/07	3,224687
5	3821877	1935102	2017/02/07	2017/05/07	3,156151
6	1936755	2054171	2017/02/07	2017/05/07	3,119258
7	3709466	1477608	2017/02/07	2017/05/07	3,108842
8	1268006	1470141	2017/02/07	2017/05/07	3,079751
9	3621038	1477704	2017/02/07	2017/05/07	3,066849
10	1625954	2084841	2017/02/07	2017/05/07	3,025954
11	3717462	1409832	2017/02/07	2017/05/07	2,951475
12	3817399	1863421	2017/02/07	2017/05/07	2,950090
13	1308133	2094668	2017/02/07	2017/05/07	2,935278
14	3807744	1408492	2017/02/07	2017/05/07	2,883967
15	3823125	1780083	2017/02/07	2017/05/07	2,816415
16	3831759	1469581	2017/02/07	2017/05/07	2,794492
17	3673732	2036958	2017/02/07	2017/05/07	2,789654
18	3557595	1540159	2017/02/07	2017/05/07	2,781380



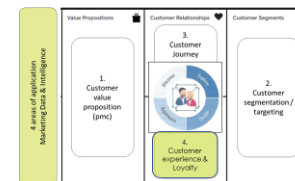
Loyalty - Churn Reduction case

Model output: Scores all cars & churn



Loyalty - Churn Reduction case

Model output: Churn drivers



Feature information (available for all cars)

Indication most important features

		Customer_type_Z	Fuel_type_B	Ownership_duration	Construction_year	Engine_power	Cylinder_volume	Avg_km_per_day	Estimated_total_km	Recency_last_dealer_visit	Total
N	Valid	7236	20105	13294	21222	112	1414	2962	5117	9793	
	Missing	17339	4470	11281	3353	24463	23161	21613	19458	14782	
Mean		,14900	,14400	,46872	1,01014	,20	,17664	,37780	,15530	,10859	
Sum		1078,164	2895,120	6231,227	21437,180	23	249,762	1119,046	794,680	1063,400	34891,13
		3,1%	8,3%	17,9%	61,4%	0,1%	0,7%	3,2%	2,3%	3,0%	100,0%

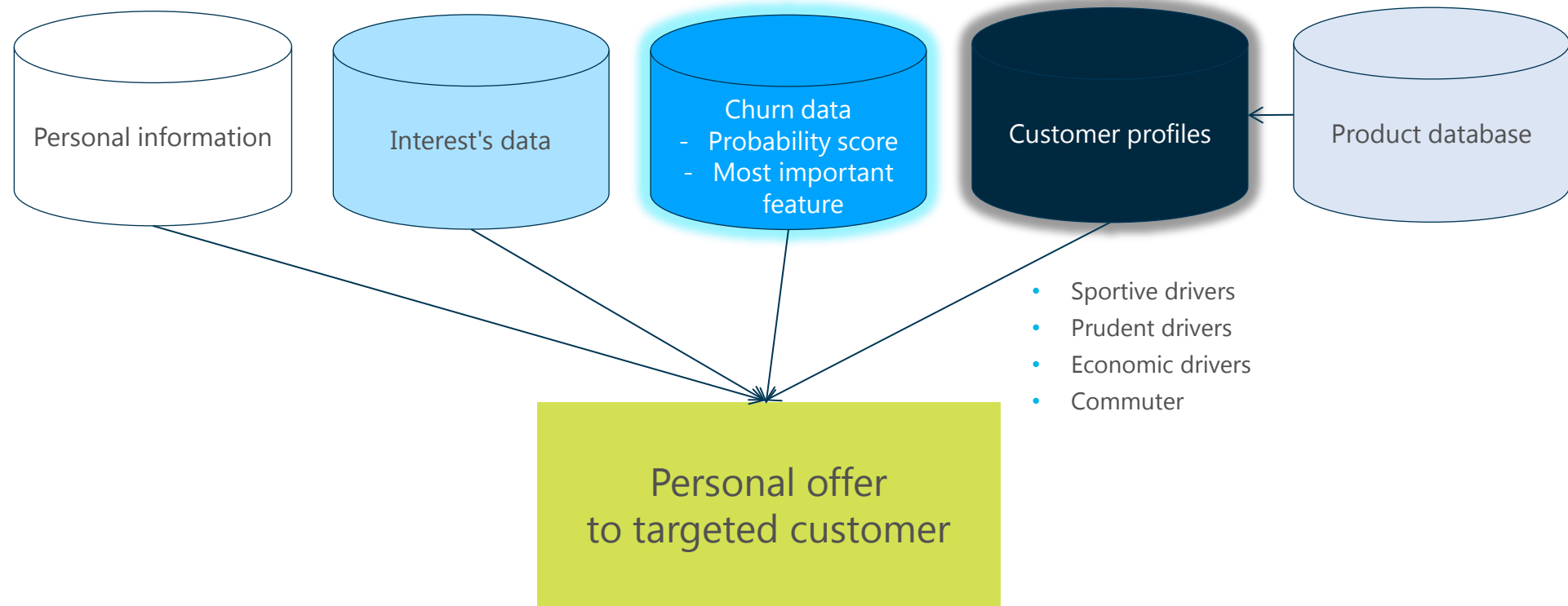
Numbers only taken into account when score $\geq 0,1$ on individual car

In combination with Customer profile information

- Sportive drivers
- Prudent drivers
- Economic drivers
- Commuter

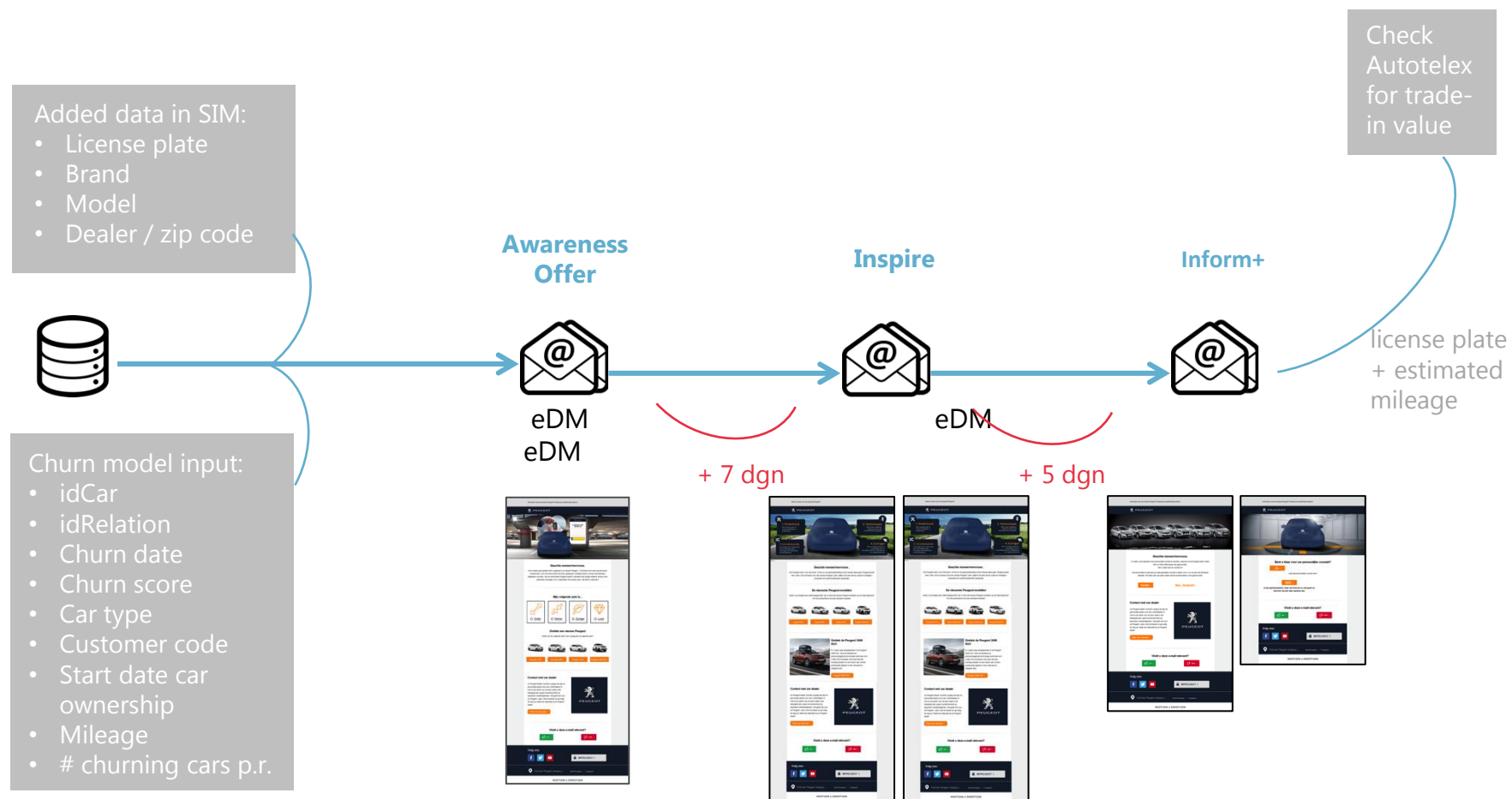
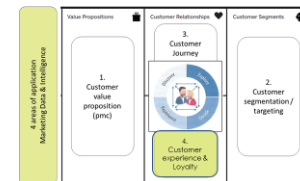
Activate & Implement

'Best offer' based on integrated data sets



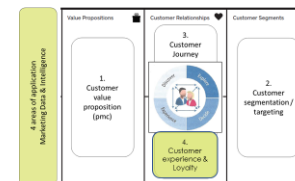
Loyalty - Churn Reduction case

The eDM Churn campaign



Loyalty - Churn Reduction case

Results Prescriptive marketing Churn (*)



IMPACT OF TARGETED CHURN CAMPAIGN			
	No specific campaign Actual churn after 3 months	Targeted churn campaign Actual churn after 3 months	Effect of targetted campaign on churn
Churn predction above treshold	62,2%	48,4%	-22,2%
	Included churn not predicted	6,9%	

Result : Improvement Customer Value by 22%

Data driven business defining the type of analysis

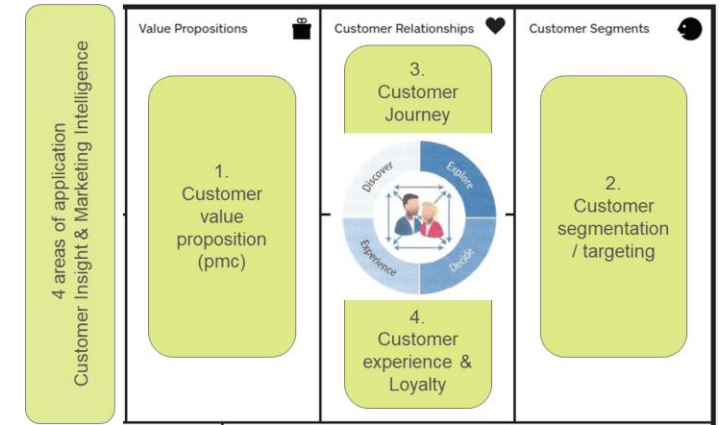
Onderzoek: verband tussen data
Voorspel een waarde
Classificeer: sorteren van items in groepen
Cluster data in homogene groepen
Cluster onderliggende waarden tot betekenisvolle waarden
Aanbeveling: wat moet ik doen
Methoden om met behulp van bovengenoemde predictive analyse technieken op een geautomatiseerde wijze Tot actiegericht output te komen

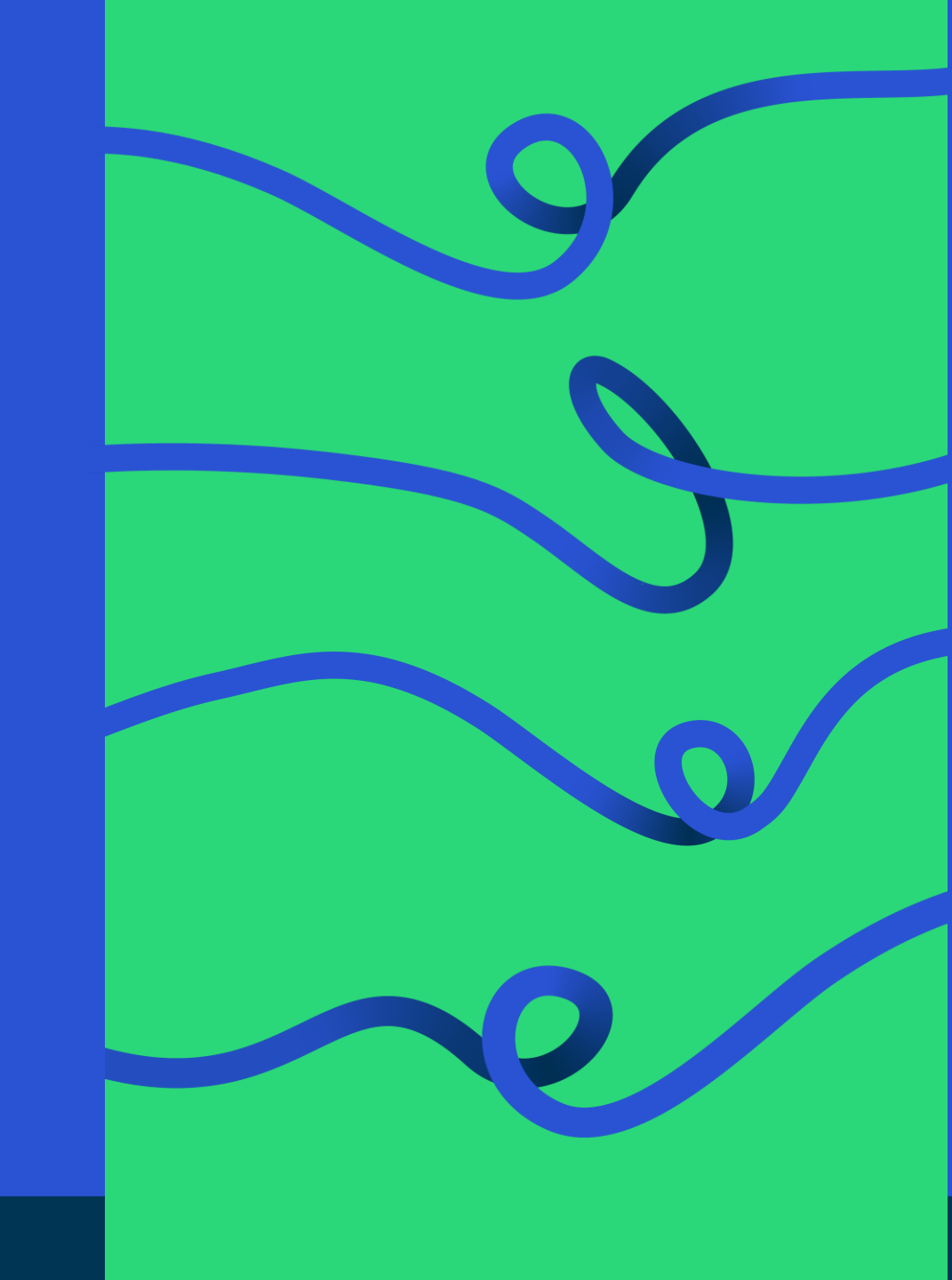
Data driven business defining the type of analysis

1 Descriptive Analytics	Onderzoek: verband tussen data
2 Predictive Analytics	
A Supervised	Voorspel een waarde
	Classificeer: sorteren van items in groepen
B Unsupervised	Cluster data in homogene groepen
	Cluster onderliggende waarden tot betekenisvolle waarden
C Reinforced	Aanbeveling: wat moet ik doen
3 Prescriptive Analytics	Methoden om met behulp van bovengenoemde predictive analyse technieken op een geautomatiseerde wijze Tot actiegerichte output te komen

Data driven business defining the type of analysis

	Analytic technique	
1 Descriptive Analytics	correlations	Onderzoek: verband tussen data
2 Predictive Analytics		
A Supervised	regression	Voorspel een waarde
	decision tree	Classificeer: sorteren van items in groepen
B Unsupervised	RFM	Cluster data in homogene groepen
	clustering	
	principal component analysis	Cluster onderliggende waarden tot betekenisvolle waarden
C Reinforced	collaborative filtering	Aanbeveling: wat moet ik doen
3 Prescriptive Analytics	artificial intelligence - testing	Methoden om met behulp van bovengenoemde predictive analyse technieken op een geautomatiseerde wijze Tot actiegerichte output te komen
	machine learning	





THANK YOU

CONTACTGEGEVENS

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☎ 06 53 62 84 70

in <https://www.linkedin.com/in/frank-van-delft-cim7>

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🌐 www.beeckestijn.org

VOLG ONS OOK OP SOCIAL MEDIA

